EE104 Fundamentals of Electric Circuits (Lesson 05)

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Review of Lesson 4 (a)

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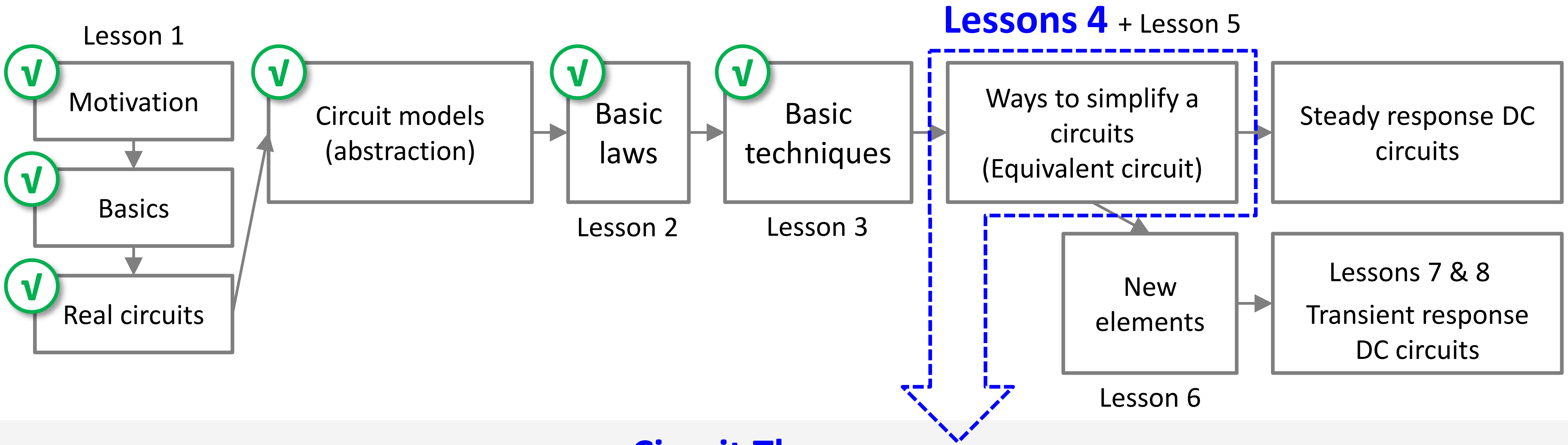
- EE104 roadmap (DC portion)



Circuit Theorems

1. Linear circuits

2. Superposition theorem

3. Source transformation theorem

4. Thévenin's theorem

5. Norton's theorem

6. Maximum power transfer theorem

Review of Lesson 4 (b)

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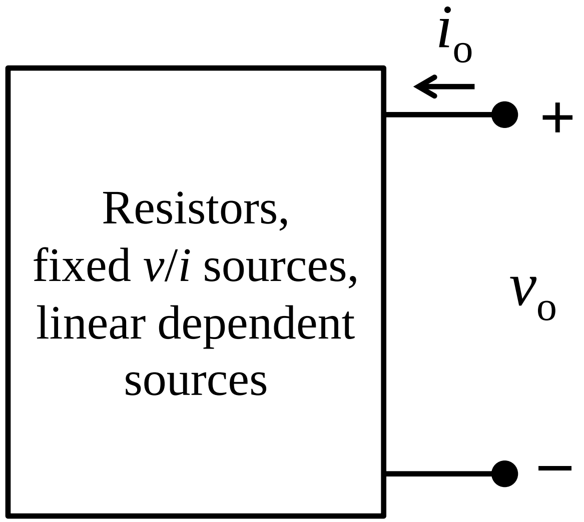
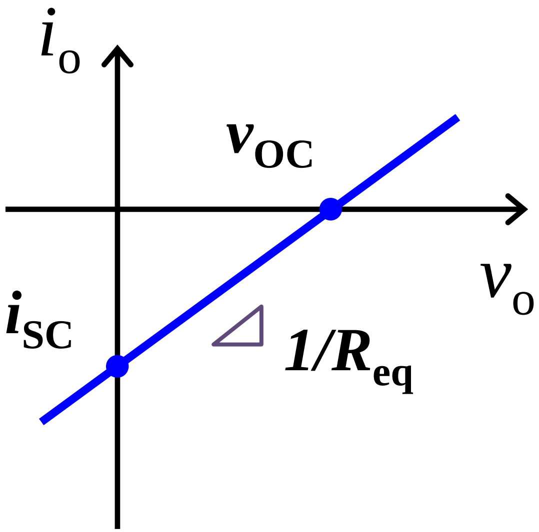
- Why v_{oc}, i_{sc} and R_{eq}? To determine the *i*-*v* characteristics at the port.





$$i_o = v_o/R_{eq} - i_{sc}$$

Or

$$v_o = R_{eq} \cdot i_o + V_{oc}$$

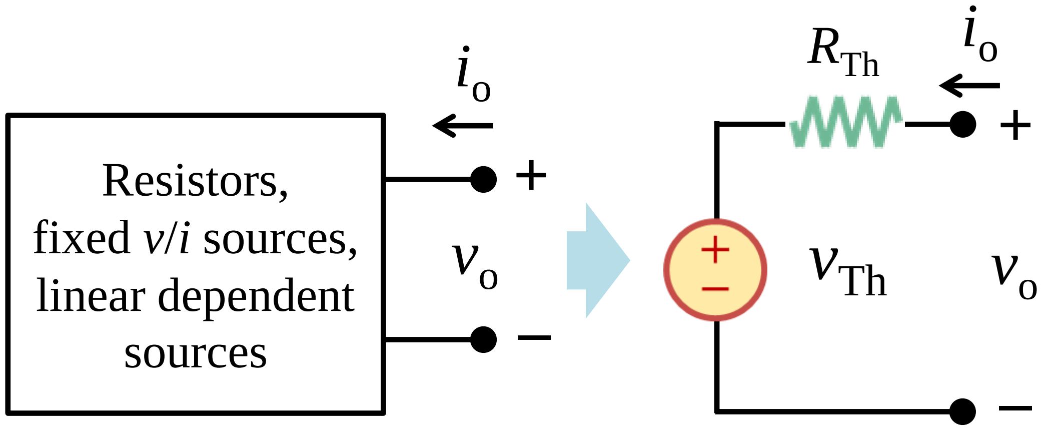
a) When you want to substitute a certain circuit by its equivalent circuit, in order to simplify a circuit, you need to first know need to know *i*-*v* characteristics of the original circuit.

b) For a circuit based on linear elements, independent sources and linear dependent sources, v_{oc}, i_{sc} and R_{eq} are parameters to determine the *i*-*v* characteristics at the port.

c) THIS IS WHY YOU NEED THEM IN THEVENIN AND NORTON THEOREMS !

Review of Lesson 4 (c)

Thévenin's theorem



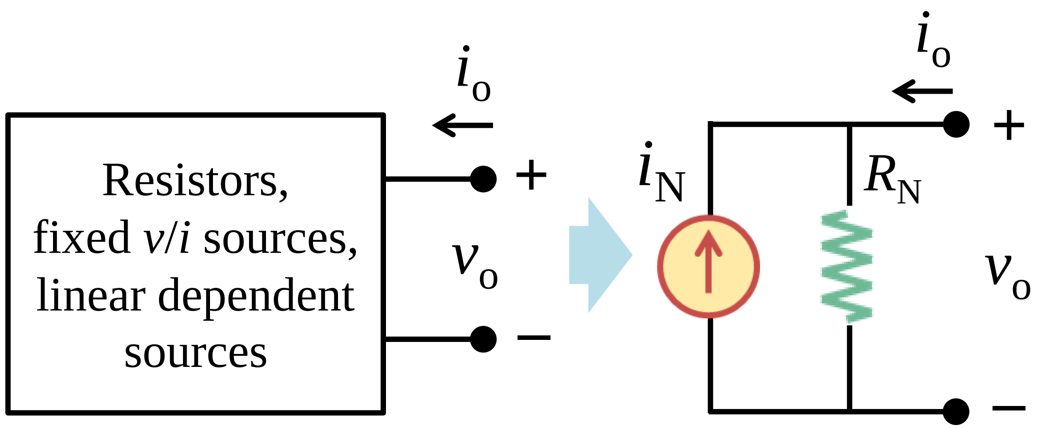
$v_o = R_{Th} \cdot i_o + V_{Th}$

Open-circuit voltage: $v_{OC} = v_{Th}$

Short-circuit current: $i_{SC} = v_{Th}/R_{Th} = -i_o$

Equivalent R: $R_{Th} = v_{OC}/i_{SC} = R_{eq}$

Norton's theorem



$i_o = v_o/R_N - i_N$

Open-circuit voltage: $v_{OC} = i_N \cdot R_N$

Short-circuit current: $i_{SC} = i_N = -i_o$

Equivalent R: $R_N = v_{OC}/i_{SC} = R_{eq}$

Thévenin-Norton transformation: $v_{Th} = i_N \cdot R_{Th}$

Review of Lesson 4 (d)

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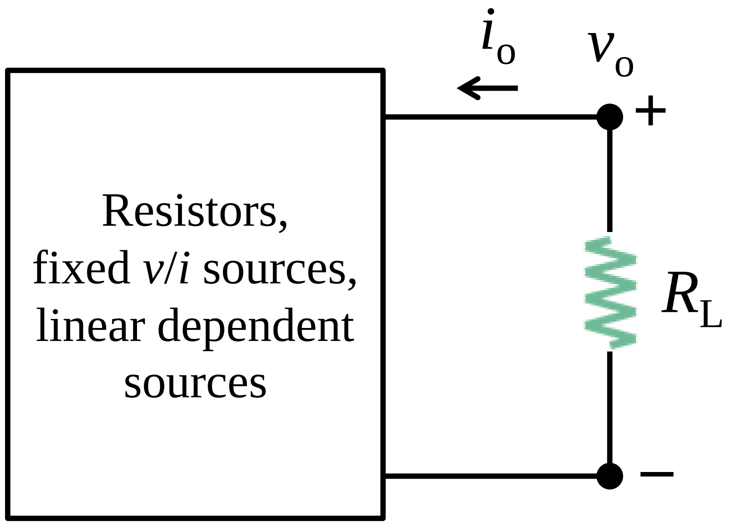
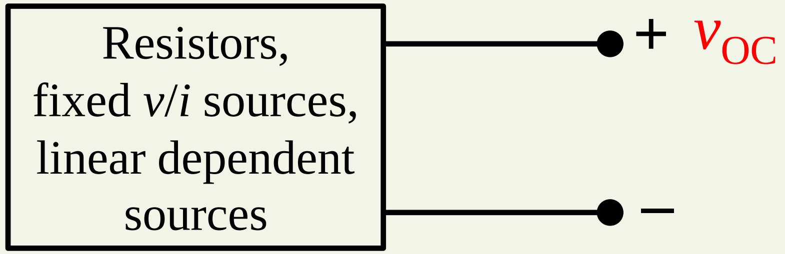
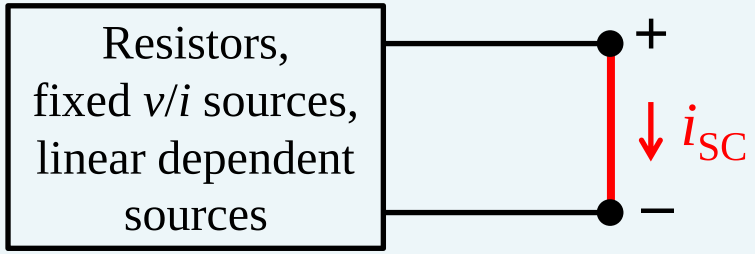
- To find v_{OC}, i_{SC}, R_{eq}

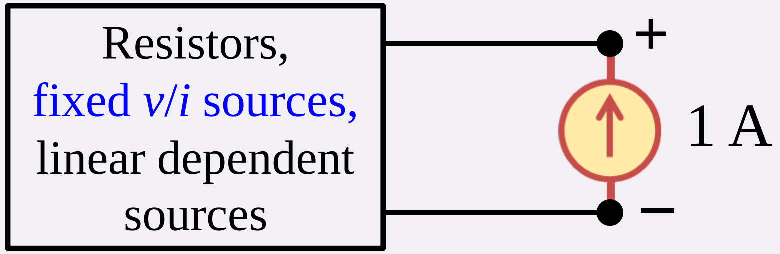




To find v_{OC}, remove the load and calculate the v_{OC} using nodal/mesh analysis.



To find i_{SC}, replace the load with an ideal wire and calculate the i_{SC} using nodal/mesh analysis.



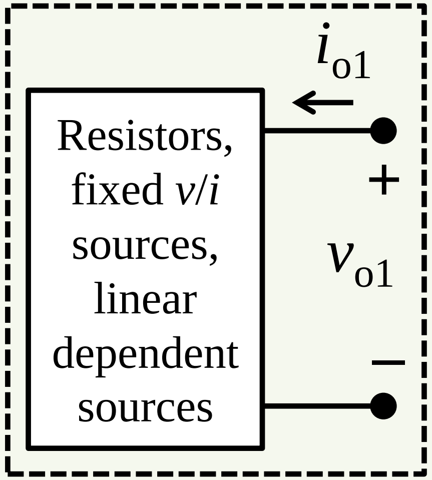
To find R_{eq}, replace the load with a 1 A current source (i_o = 1 A), turn off the independent sources, calculate the v_O, and $R_{eq} = v_O/i_O$.

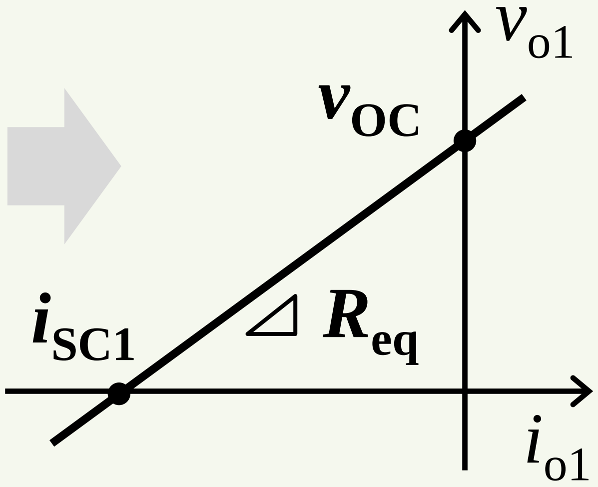
*You can also use a 1 V voltage source

Review of Lesson 4 (e)

- Thévenin's theorem

The original circuit





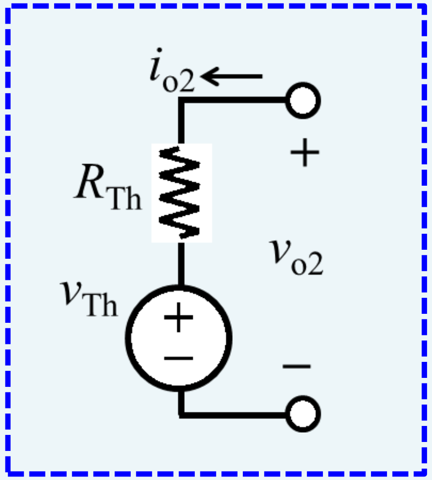
$$v_{o1} = i_{o1}R_{eq} + v_{oc}$$

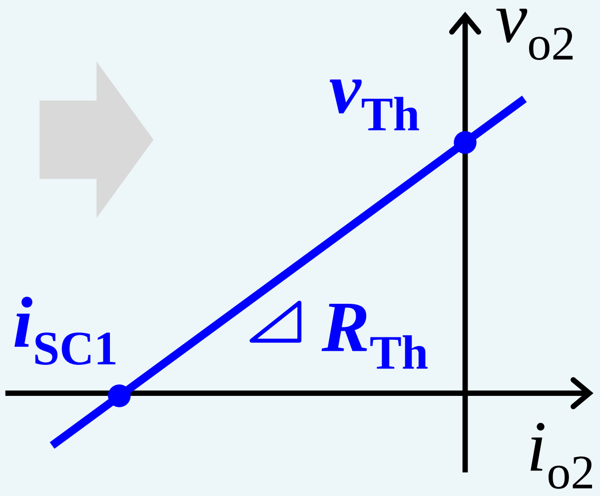
We make them equivalent:

$$v_{Th} = v_{oc} \;\&\; R_{Th} = R_{eq}$$

To substitute

The equivalent circuit



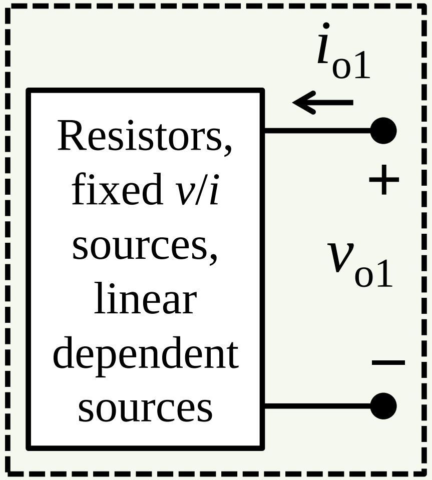


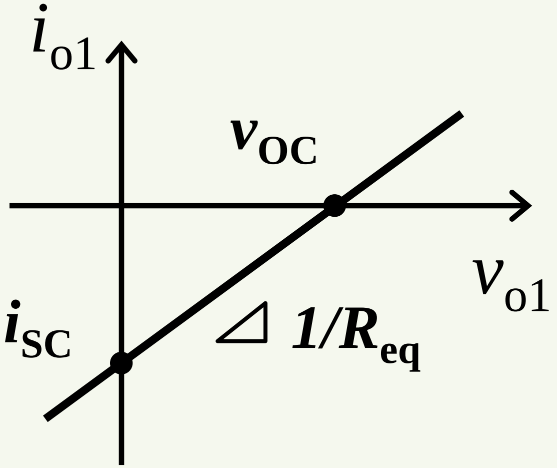
$$v_{o2} = i_{o2}R_{Th} + v_{Th}$$

Review of Lesson 4 (f)

- Norton's theorem

The original circuit





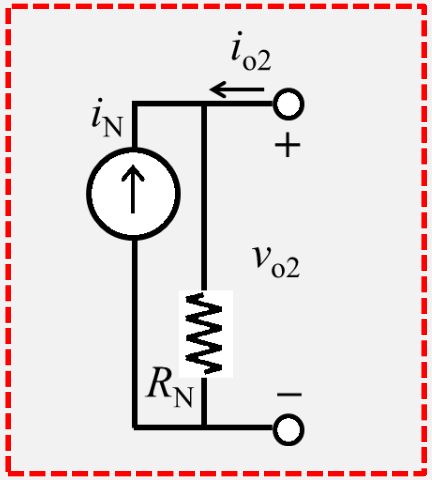
$$i_{o1} = v_{o1}R_{eq} - i_{sc}$$

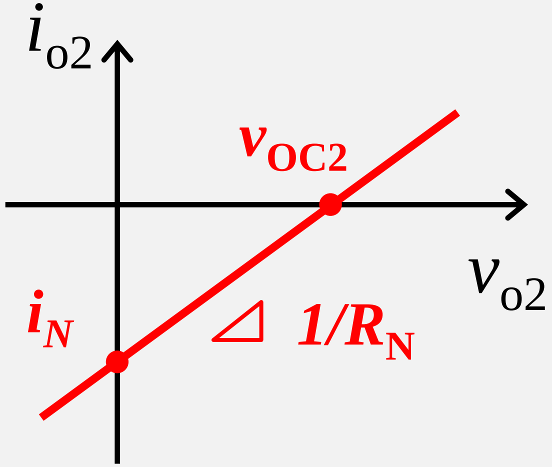
To substitute

We make them equivalent:

$$i_N = i_{SC} \ \&\ R_N = R_{eq}$$

The equivalent circuit





$$i_{o2} = v_{o2}R_N - i_N$$

Review of Lesson 4 (g)

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- You've learned a lot regarding how to simplify DC circuits !!!

(1) You know how to simplify a DC circuit based on resistors

Equivalent circuits: voltage division, current division, delta-wye transformation

(2) You know how to simplify a DC circuit based on independent sources

Equivalent circuits: voltage sources in series, current sources in parallel
voltage sources in series with a current source
current sources in parallel with a voltage source

(3) You know how to simplify a DC circuit based on transistors

Equivalent circuits: $I_c = I_b + I_c$, $I_c = \beta I_b$

(4) You also know how to simplify a DC circuit including sources and resistors

Equivalent circuits: source transformation + Thévenin's & Norton's theorems

Abstract of Lesson 5 (a)

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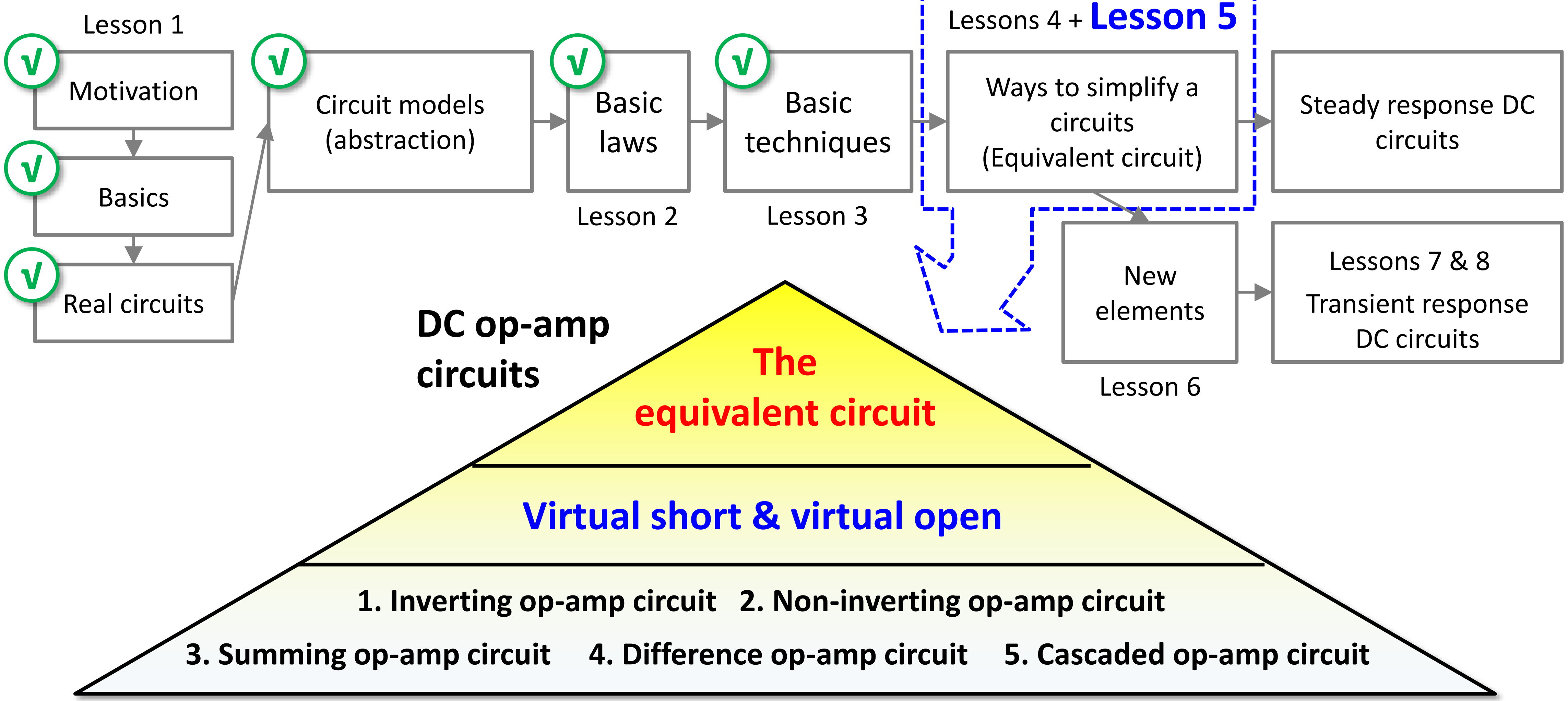
- EE104 roadmap (DC portion)



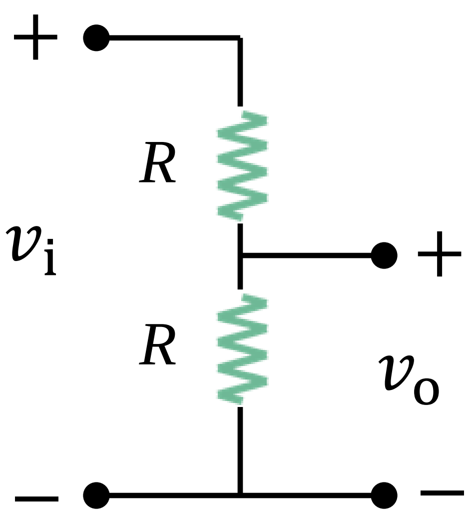
Lesson 5 Operational Amplifiers (Op Amp)

5.1	**Introduction**	**Slides 11-19**
5.2-5.4	Op amp basics and negative feedback	Slides 21-48
5.5-5.6	Inverting and noninverting amplifiers	Slides 50-60
5.7-5.9	Summing and difference amplifiers	Slides 62-72
5.10	Cascaded Op Amp circuits	Slides 74-82
5.11	Summary and assignments	Slides 84-85

5.1 Introduction (a)

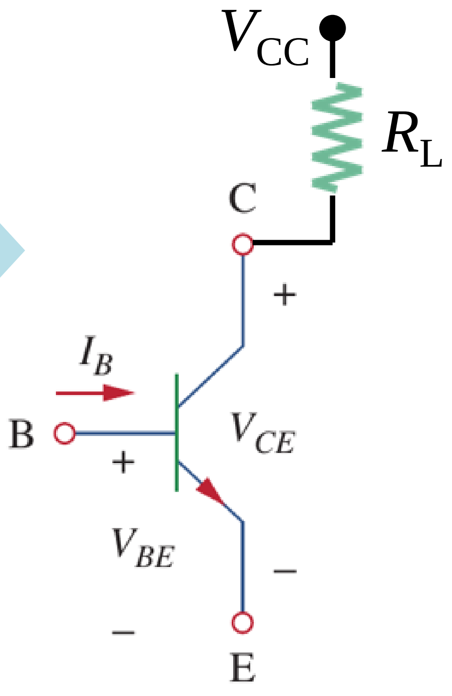
- attenuators and amplifiers

Voltage divider



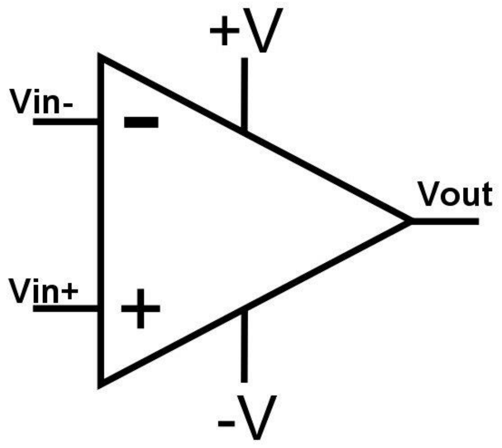
$$\frac{v_o}{v_i} = 0.5$$

BJT amplifier



$$i_c = \beta i_b$$

Op-amp



$$v_o = Av_i$$

5.1 Introduction (b)

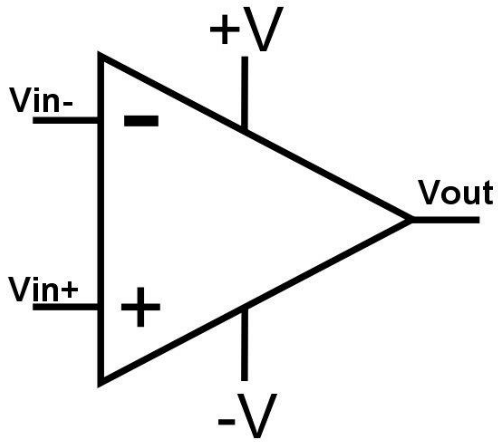
- Operational amplifier (Op-Amp) 运算放大器

a) An amplifier uses external voltage to amplify the input.

b) An operation amplifier (op-amp) can perform mathematical operations of addition, subtraction, multiplication, division, differentiation, and integration, in addition to amplification of the input through a very high gain.

c) The op-amp is an electronic unit that behaves like a voltage-controlled voltage source, can also be used in making a voltage- or current-controlled current source.

d) Op-amp are a mass-produced component found in countless electronics

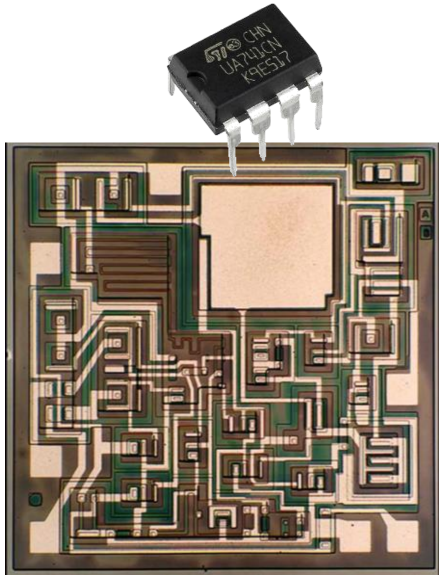


5.1 Introduction (c)

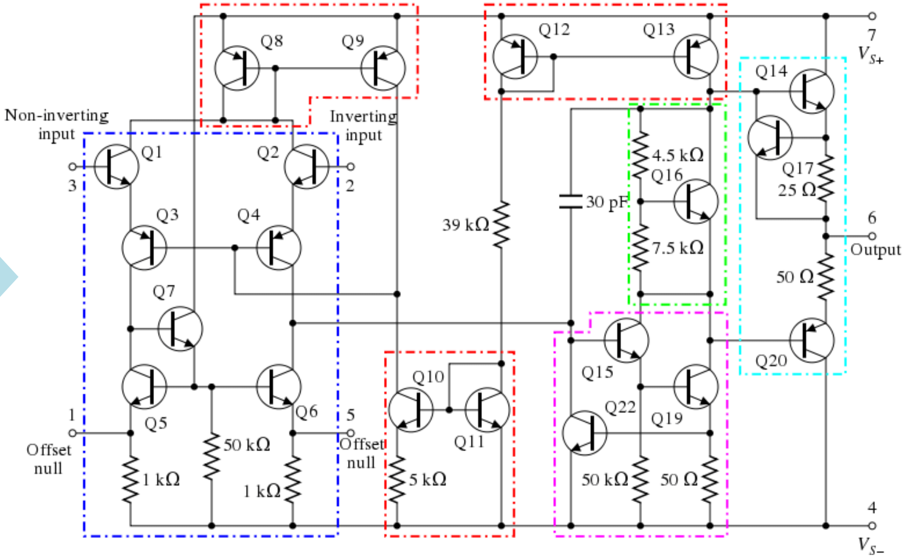
- µA741 Op Amp

a) An op amp consists of a complex arrangement of resistors, transistors, capacitors, and diodes.

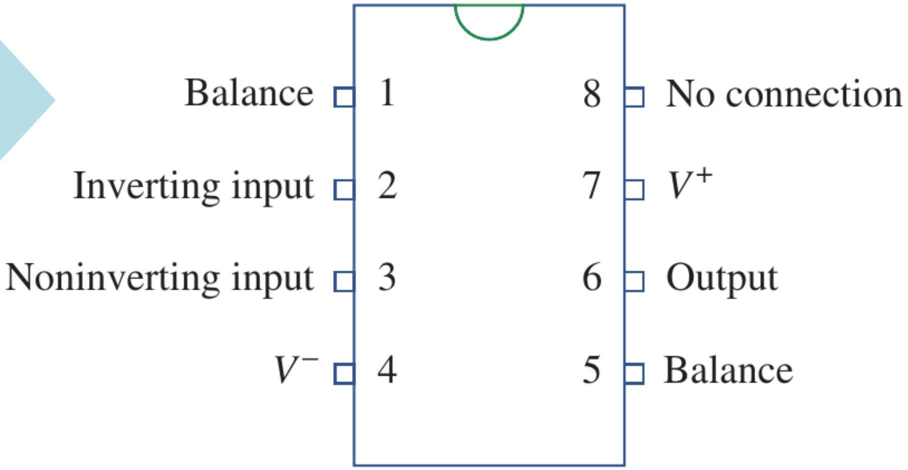
b) It will suffice to treat the op amp as a circuit building block and simply study what takes place at its terminals.



µ741 IC



Internal circuit of µ741



Pin configuration of µ741 (counterclockwise numbering)

5.1 Introduction (d)

- μA741 Op Amp

a) μA741 Op Amp has 7 terminals, including 2 inputs and 1 output.

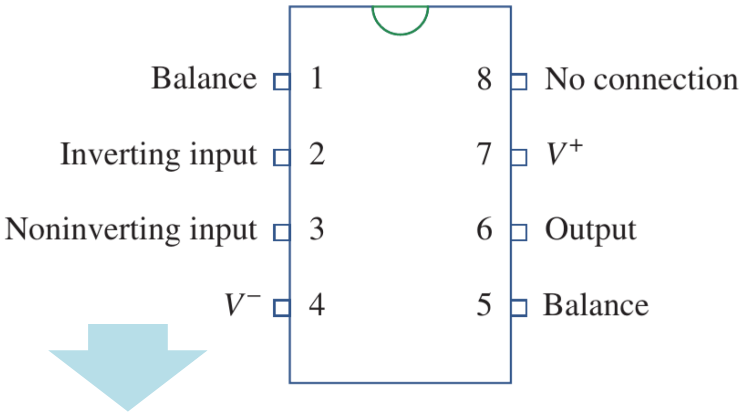
b) Noninverting input 同相输入端 (terminal 3): an input applied to this terminal will appear with the same polarity at the output.

c) Inverting input 反相输入端 (terminal 2): an input applied to the inverting terminal will appear with inverted polarity at the output.

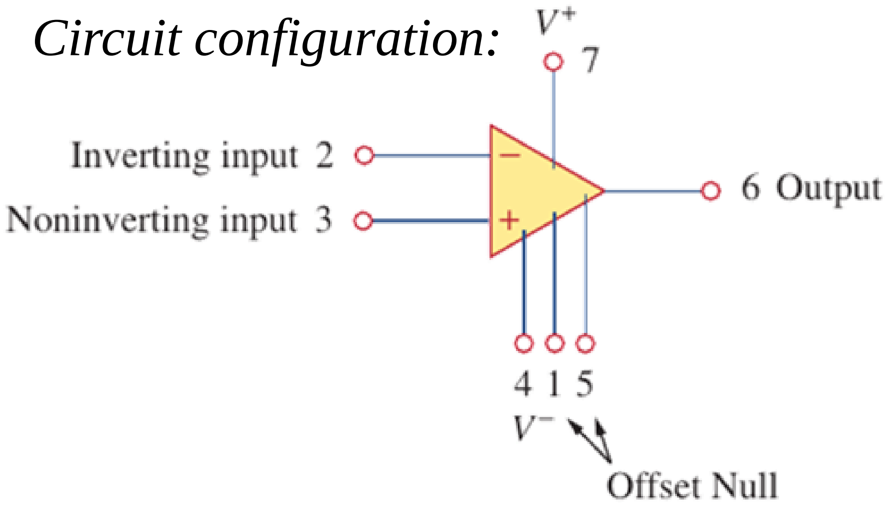
d) Power supply voltages (terminals 4 and 7): op amp is an active device and therefore needs extra power supply to work.

e) Offest null (terminals 1 and 5): The op amp offset null capability is used to null any small DC offsets at the output for DC amplifiers.

Pin configuration:

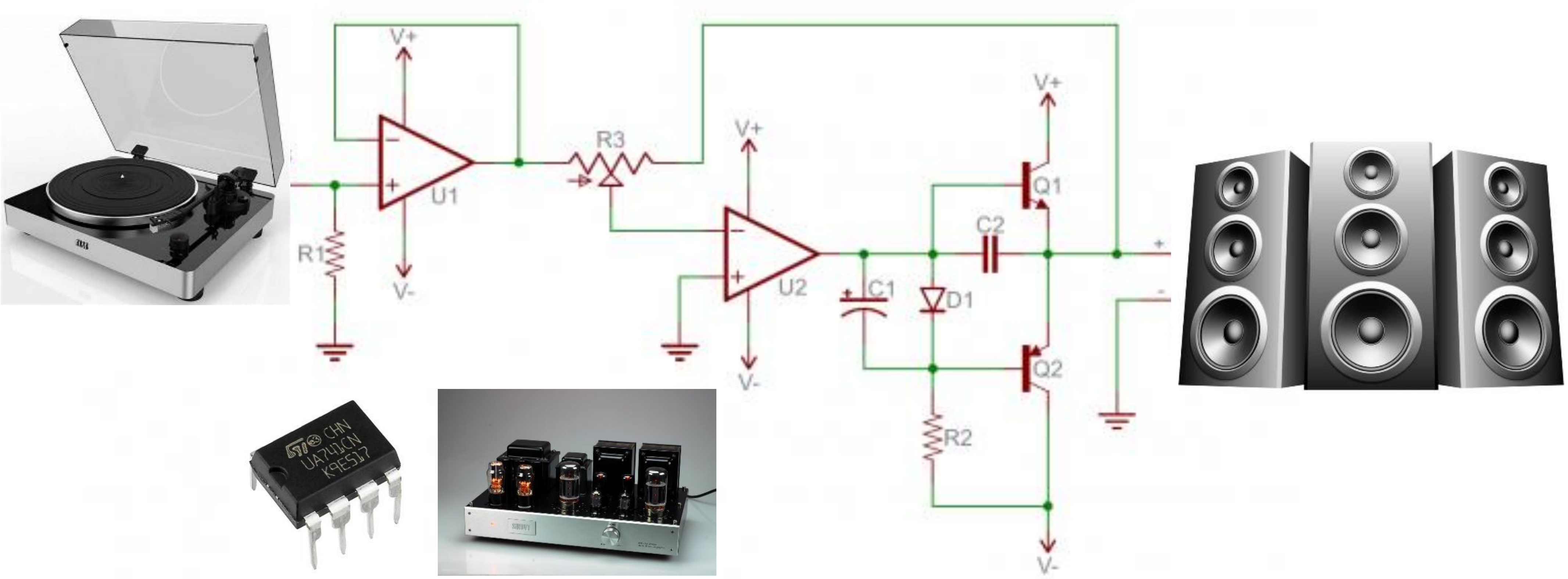


Circuit configuration:



5.1 Introduction (e)

- Op Amp is highly useful



5.1 Introduction (f)

- Applications of Op Amp

Applications of Op Amp

Linear	Nonlinear	Wave shaping
• Scale/sign changer • Adder • Substractor • Integrator • Differentiator • Log & antilog amplifier • instrumentation amplifier • etc.	• Comparator • Zero-cross detector • Schmitt trigger • Precision half and full wave rectifier • Sample and hold circuits • Clippers and clampers • etc.	• Sine wave generator • Tri-angular and rectangular wave generator • etc.

Reference: https://www.researchgate.net/publication/332574630_A_Novel_Approach_to_Test_Op-Amp_Ic-741_Using_Arduino

5.1 Introduction (g)

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- History – The dawn (interesting to know)

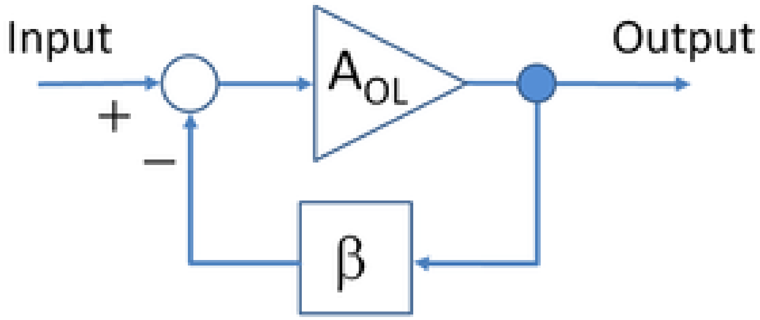
a) Before the Op-amp, Harold S. Black develops the negative feedback amplifier (1920–1930).

b) Many electronic devices used to provide gain are nonlinear and results in distortion and noise. In early 20th century, long-distance calls are highly noisy due to these issues.

c) A negative feedback amplifier is an amplifier in which a fraction of its output is combined with the signal at its input that opposes the signal in what is called negative feedback.

d) Negative feedback yields a circuit technique that trades gain for higher linearity (reducing distortion)

Negative feedback amplifiers



"The answer scrawled on a blank page in a daily newspaper, was conceived whilst aboard a ferry."
– Harold Stephen Black

5.1 Introduction (h)

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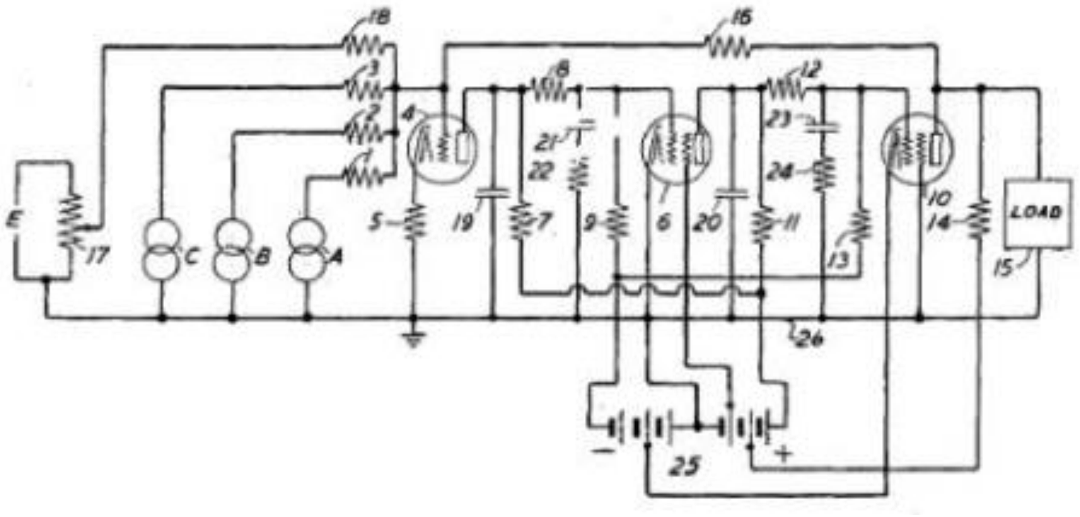
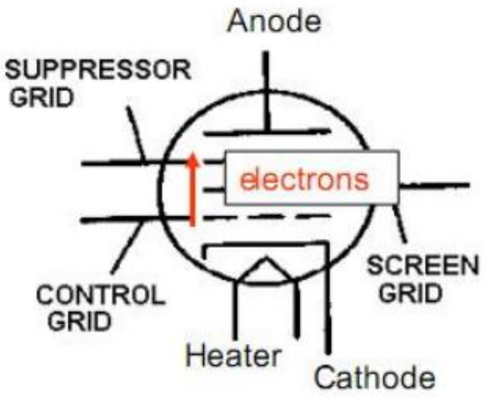
- History – the vacuum tube age (interesting to know)





a) The first op-amp designed by Karl Swartzel for Bell labs M9 gun director.
b) 3 vacuum tubes were used, with only one input, as well as +350 V and -350 V sources to attain a gain of 90 dB
c) Loebe Julie then develops an Op-amp with two inputs: inverting and non-inverting .

5.1 Introduction (i)

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- History – the semiconductor age (interesting to know)

Transistor + The integrated circuits + The planar process

- 1960s Beginning of the solid state Op-amp. Example: GAP/R P45 (1961-1971) $118
- 1963s Robert J. Wilder develops the first monolithic op-amp μA702 .
- Fairchild Semiconductor vs. National semiconductor
  - National: LM101 (1967) & LM101a (1968)
  - Fairchild: The "famous" μA741 (design by Dave Fullager in 1968)

Lesson 5 Operational Amplifiers (Op Amp)

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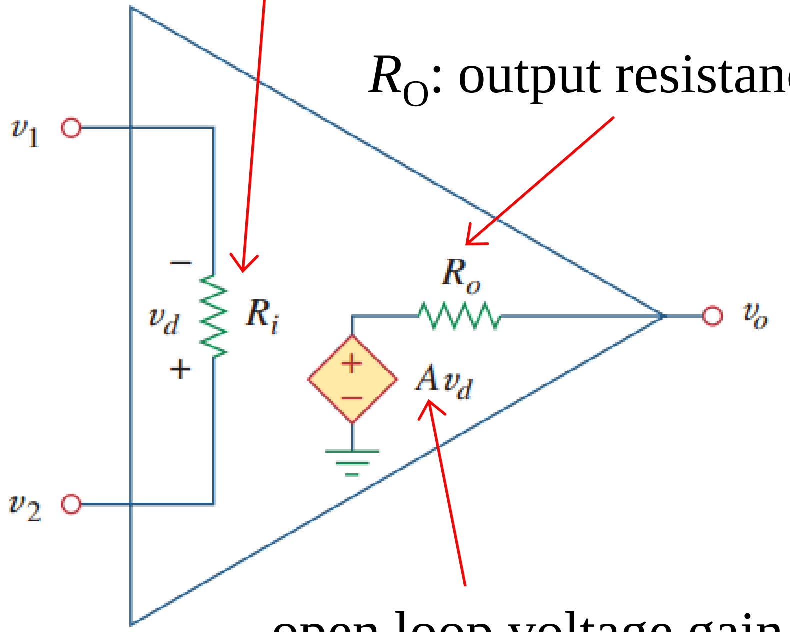
5.2 Op amp basics (a)

- The equivalent circuit of a op amp

R_i: input resistance

R_O: output resistance



open loop voltage gain
开环增益

a) The equivalent circuit of an op amp is shown in the left.

b) The input resistance R_i (输入电阻) is the Thevenin equivalent resistance seen from the input terminal.

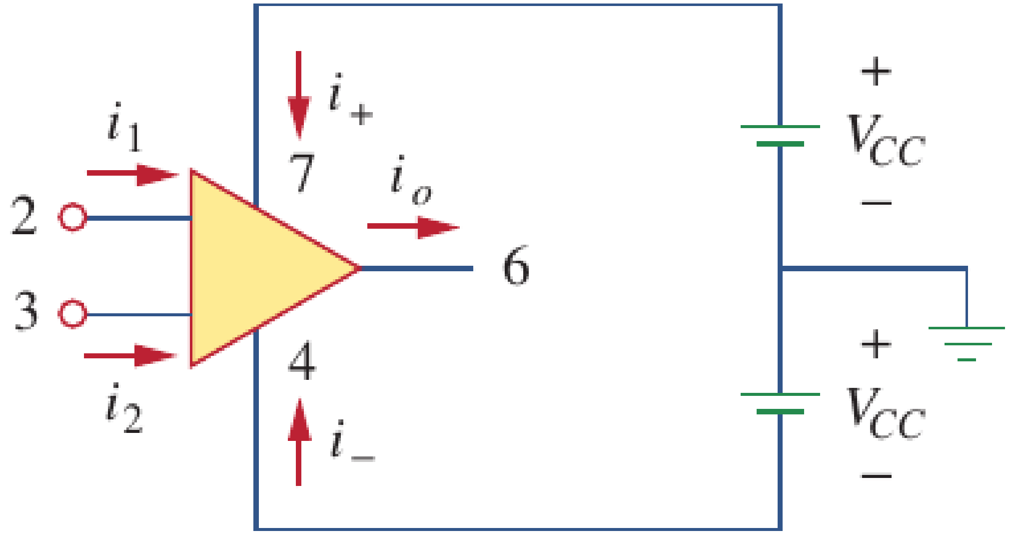
c) The differential input voltage $v_d = v_2 - v_1$

d) The output section consists of a voltage-controlled source in series with the output resistance R_o (输出电阻), which is the Thevenin equivalent resistance seen from the output terminal.

e) The output voltage $v_o = Av_d = A(v_2 - v_1)$, A is called open-loop voltage gain (开环增益) (the gain of the op amp without any external feedback from output to input).

5.2 Op amp basics (b)

- Always remember the power supplies



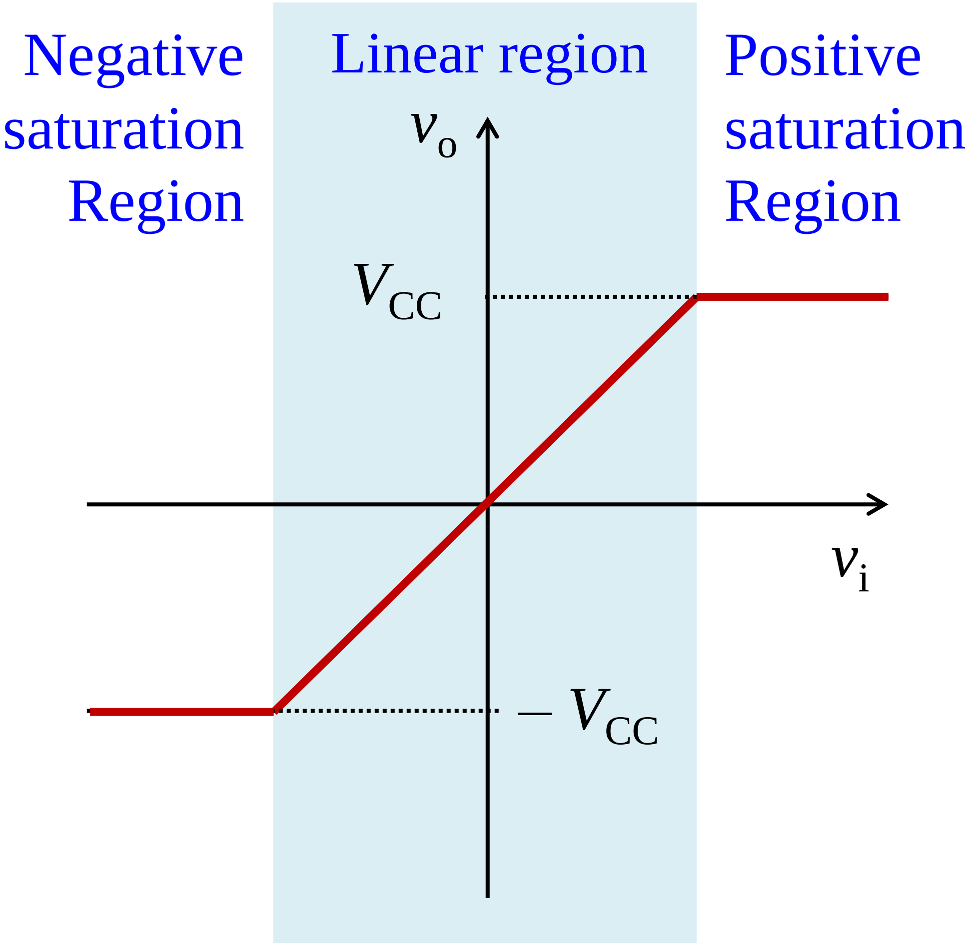
a) As active elements, op-amp must be powered by power supplies.

b) Although power supplies are often ignored in op-amp circuit diagrams for the sake of simplicity, their currents should not be overlooked.

c) The KCL only applies when power supplies are included.

5.2 Op amp basics (c)

- Three operating regions



a) Three operating regions:

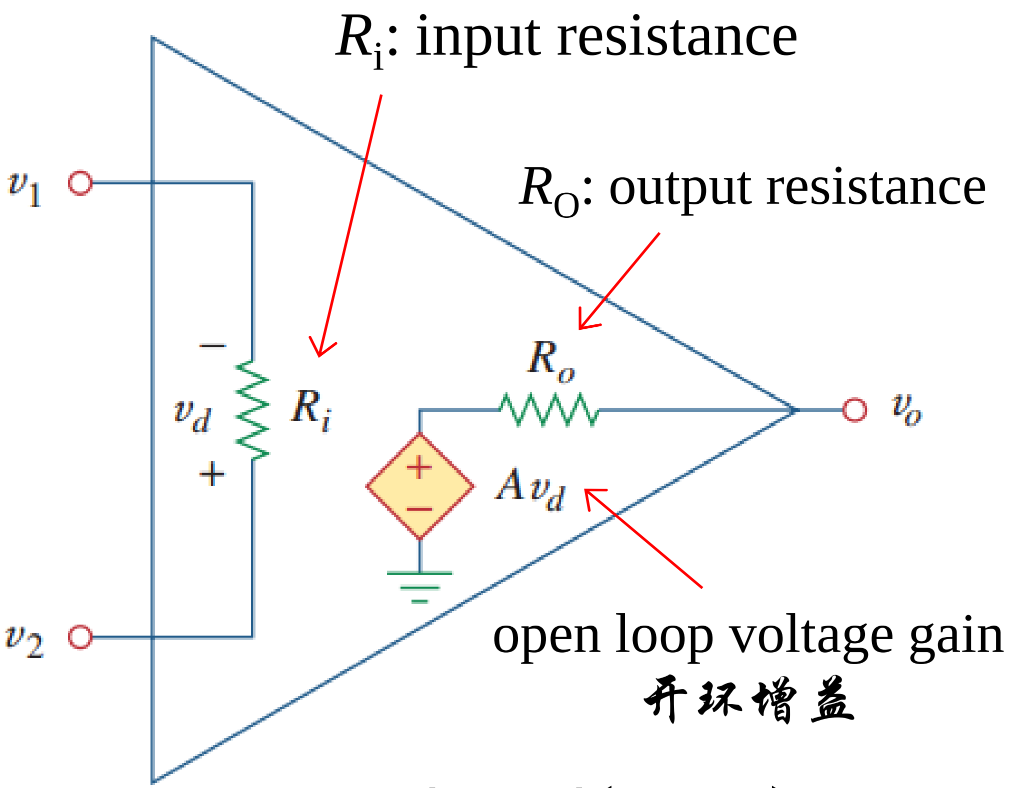
1. Positive saturation, $v_o = V_{CC}$.
2. Linear region, $-V_{CC} \leq v_o = Av_d \leq V_{CC}$.
3. Negative saturation, $v_o = -V_{CC}$.

b) The output can not exceed the maximum or minimum voltage of the power supply.

c) This is because the op-amp is really just a power converter that takes DC power from the supplies, converting and transferring it to amplify the input. So the maximum voltage and the power available for delivering depend on the power supplies.

5.2 Op amp basics (d)

- Ideal and non-ideal op-amps



$$v_o = Av_d = A(v_2 - v_1)$$

	Ideal	Non-ideal
A	∞	10^5 to 10^8
R_i	∞	10^5 to 10^{13} Ω
R_o	0	10 to 100 Ω

5.2 Op amp basics (e)

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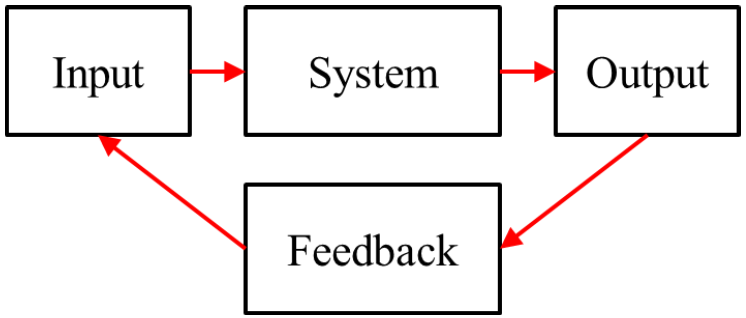
- Practical op-amps are not perfect (interesting to know).

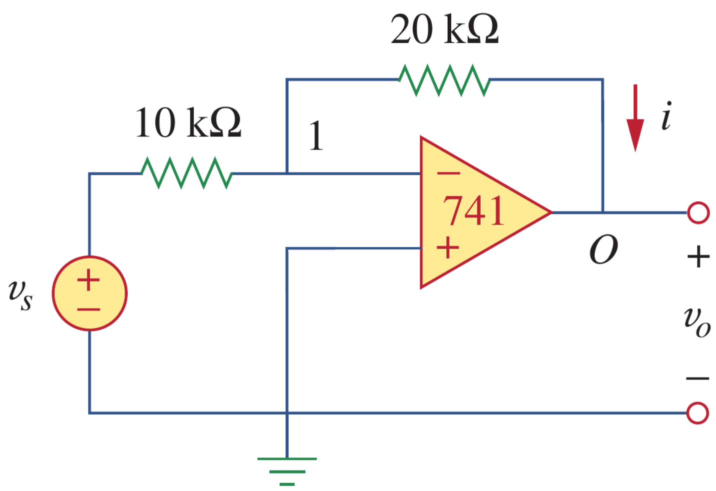
- Linear Imperfections:
  - Finite open-loop gain ($A_0 < \infty$)
  - Finite input resistance ($R_i < \infty$)
  - Non-zero output resistance ($R_o > 0$)
  - Finite bandwidth / Gain-BW Trade-Off
- Other (non-linear) imperfections:
  - Slew rate limitations
  - Finite swing
  - Offset voltage
  - Input bias and offset currents
  - Noise and distortion

A real device deviates from a perfect difference amplifier. It may have an offset which is not zeroed. The inputs may draw current. The characteristics may drift with age and temperature. Gain may be reduced at high frequencies, and phase may shift from input to output. These imperfection may cause no significant errors in some applications, unacceptable errors in others. So， choose your op amp carefully.

5.3 Negative feedback (a)

- Negative feedback (负反馈)





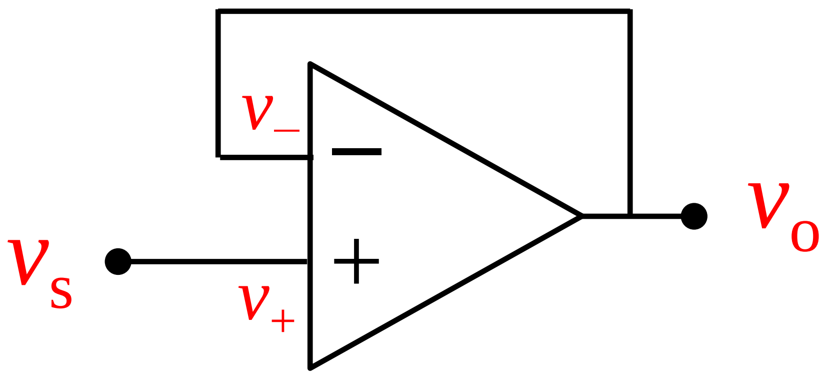
a) Feedback is an event that occurs when the output of a system is used as input back into the system as part of a chain of cause and effect. Negative feedback is when the occurrence of an event causes something to happen that counteracts the original event.

b) In electronics, feedback is defined as the process of returning part of the signal output from a circuit or device back to the input of that circuit or device. A feedback is negative if the output signal is opposite in value or phase to the input signal.

c) For example, in a central heating system, if the temperature falls too low the thermostat turns on the heating, when it rises the thermostat turns it off again.

5.3 Negative feedback (b)

- (a) The negative feedback decides the gain



(1) $v_i = v_+ - v_- = v_s - v_o$

(2) $v_o = Av_i = A(v_s - v_o)$

(3) $v_o = v_s/(1+1/A) \approx v_s$

Where is the again/amplification?

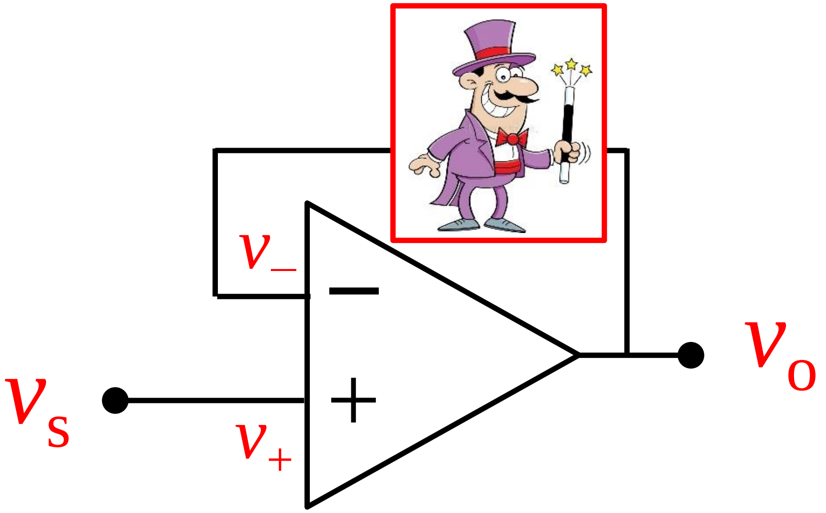
a) Negative feedback stabilizes the gain: $v_s\uparrow$, $v_s - v_o\uparrow$, $v_o\uparrow$, $v_s - v_o\downarrow$, $v_o\downarrow$

b) Since A is practically large (10^5 to 10^8), so $v_i = v_+ - v_- = v_o/A$ is normally very small.

c) Negative feedback adjusts the output to make $v_+ - v_- \approx 0$.

5.3 Negative feedback (c)

- (a) The negative feedback decides the gain



Now we have a magician who makes $v_- = v_o/2$, in which ½ is defined as F.

(1) $v_i = v_+ - v_- = v_s - v_o/2$

(2) $v_o = Av_i = A(v_s - v_o/2)$

(3) $v_o = v_s/(1/2+1/A) \approx 2v_s = v_s/F$

a) Now you have the gain, or namely the amplification!

b) You take a portion (F) of the v_o as the negative feedback, and you get a gain of $1/F$

c) Think about: how can you design/build this magician by circuits ?

5.3 Negative feedback (d)

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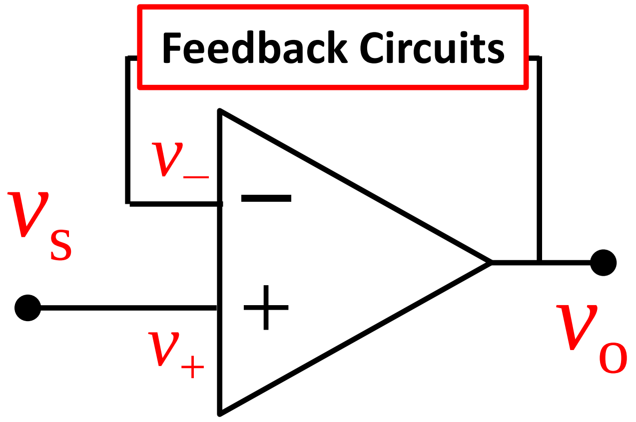
- (a) The negative feedback decides the gain

a) Route the output to the inverting input and you get negative feedback

b) As a negative feedback is applied, the closed-loop gain is determined mainly by the feedback factor. In this scenario, it is the feedback circuit that determines the gain, rather than the op-amp.

c) The feedback depth shows the difference in gains with and without the negative feedback.



Now the magician makes $v_- = Fv_o$.

(1) $v_i = v_+ - v_- = v_s - Fv_o$

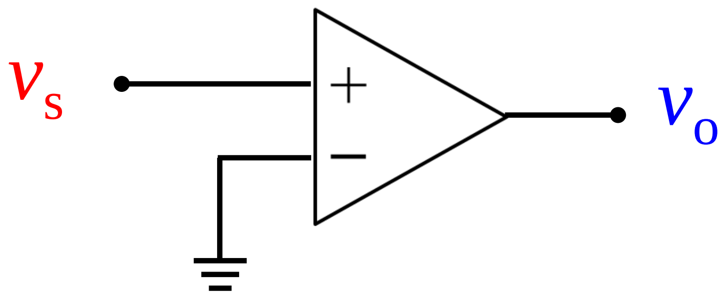
(2) $v_o = Av_i = A(v_s - Fv_o)$

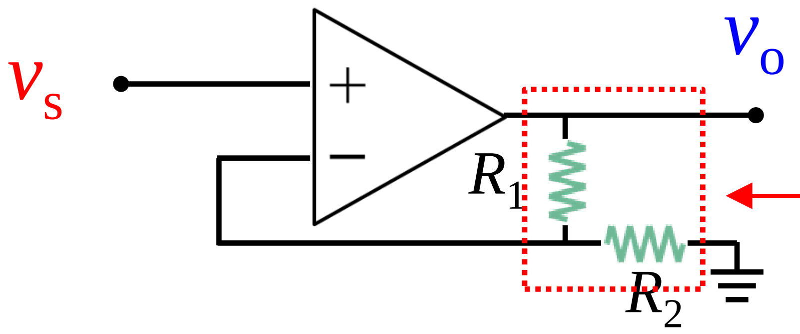
(3) $\dfrac{v_o}{v_s} = \dfrac{A}{1+AF} = \dfrac{1}{1/A+F} \approx \dfrac{1}{F} = A_F$

Item	Name	名称	Determined by
A	Open-loop Gain	开环增益	Semiconductor ICs
F	Feedback Factor	反馈系数	Feedback circuits
A_F	Closed-loop Gain	闭环增益	Feedback circuits
1+AF	Feedback Depth	反馈深度	Mainly feedback circuits

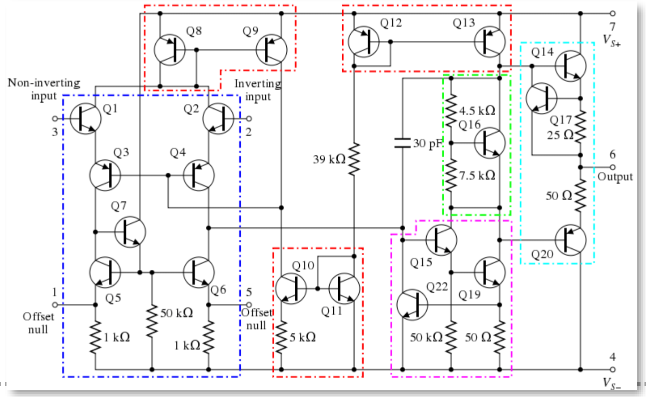
5.3 Negative feedback (d)

- (b) Negative feedback simplifies and stabilizes the gain





Without negative feedback	With negative feedback
Open-loop gain: G = v_o/v_s = *A*, *A* is the gain of the op amp	$F = R_2/(R_1 + R_2)$, gain of the feedback path, loop gain: G $= v_o/v_s = [A(v_s - Fv_o)]/v_s$ $= A/(1 + AF)$ $= 1/(A^{-1} + F)$ So the **Closed-loop gain $A_F \approx 1/F$** (if $A^{-1} << F$)



5.3 Negative feedback (e)

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- (b) Negative feedback simplifies and stabilizes the gain

$$A_F = \frac{A}{1+AF} \Rightarrow \frac{dA_F}{A_F} = \frac{1}{1+AF} \cdot \frac{dA}{A}$$

a) This equation shows how much the closed-loop gain would vary versus the variation of the open-loop gain, namely how the closed-loop gain is affected by the non-stability of the semiconductor ICs.

b) When A is reduced by 20%, the A_F is only reduced by 0.002%, assuming $A = 10^6$ and $F = 10^{-4}$.

c) The results show that the closed-loop gain can be far more stabilized with a negative feedback if *F* can be delivered using stable passive components such as resistors, which can be more stable versus temperature, frequency, etc.

5.3 Negative feedback (f)

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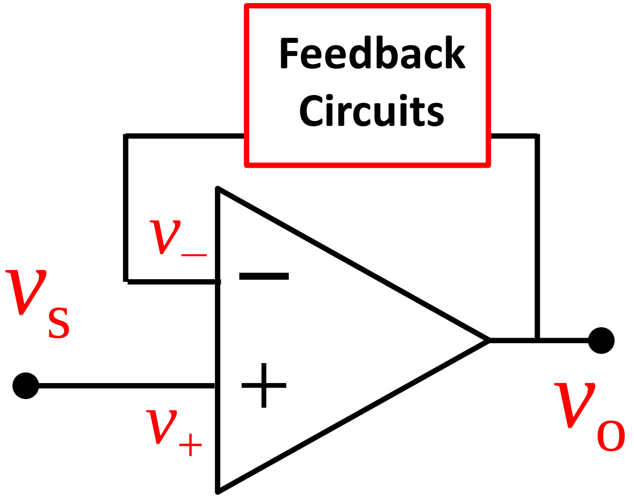
- (c) Negative feedback shorts the inverting and non-inverting inputs



$$\begin{cases} v_i = v_+ - v_- = v_s - F v_o \\ \dfrac{v_o}{v_s} = A_F \Rightarrow v_o = v_s A_F = \dfrac{v_s}{F} \end{cases}$$

$$\Rightarrow v_+ - v_- = v_s - F v_o = v_s - F\frac{v_s}{F} = 0$$

$$\Rightarrow v_+ - v_- = 0$$ **(virtual short, 虚短)**

a) An op-amp with a negative feedback will yield zero voltage between the inverting and non-inverting inputs.

b) This property is called virtual short.

5.3 Negative feedback (g)

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- (Interesting to know) Negative feedback is wonderful

Negative feedback features advantages including:

a) Gain stabilization: as we have explained.

b) Distortion reduction: High power amplifiers are often non-linear, e.g. their gain decreases at high signal amplitudes. Since the gain of a feedback system does not depend much on the gain of the amplification path, the non-linearity has little effect.

c) External disturbances: have little effect on the output of a feedback system because the feedback adjusts to compensate for them.

d) Other benefits: extend the bandwidth, control input and output impedance, etc.

Other examples of negative feedback :

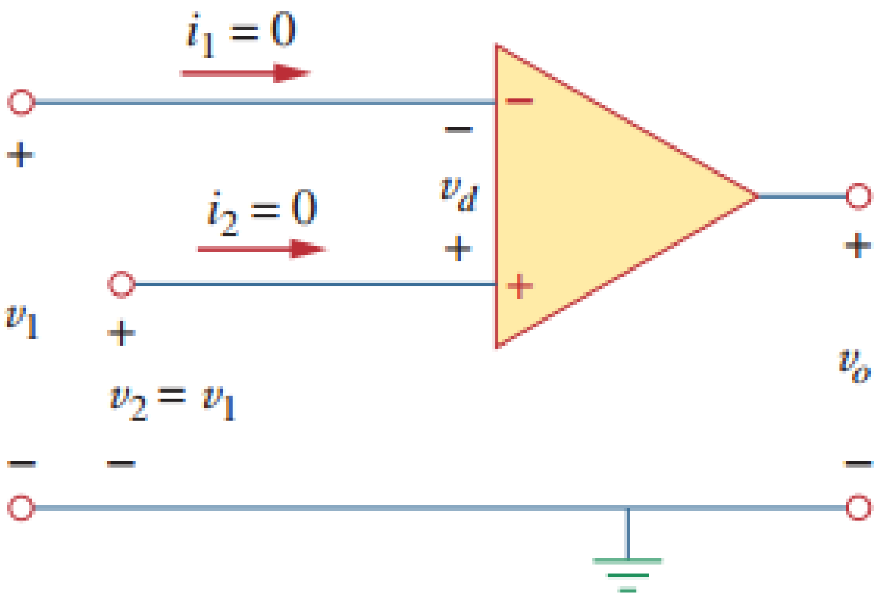
a) Economics: Supply↓ ⇒ Demand↑ ⇒ Price↑ ⇒ Supply↑ ⇒ Supply=Demand

b) Biology: Enough food ⇒ More rabbits ⇒ Not enough food ⇒ Less rabbits ⇒ Enough food

5.4 Op-amps with negative feedbacks (a)

- **THE GOLDEN RULES:**

Ideal op-amps	A	R_i	R_o
	∞	∞	0



Golden rules for analyzing ideal op-amp:

(1) Currents into both input terminals are 0:

$$i_1 = i_2 = 0$$

(2) The voltage across the input terminals is 0:

$$v_2 = v_1$$

a) To calculate voltages, the two input terminals can be considered as "shorted". ($A \rightarrow \infty$, 虚短)

($v_2 - v_1 = v_o/A$, $A \rightarrow \infty$, $v_2 - v_1 = 0$)

a) To calculate currents, the two input terminals and the internal portion of the op-amp can be considered as "open". ($R_i \rightarrow \infty$, 虚断)。

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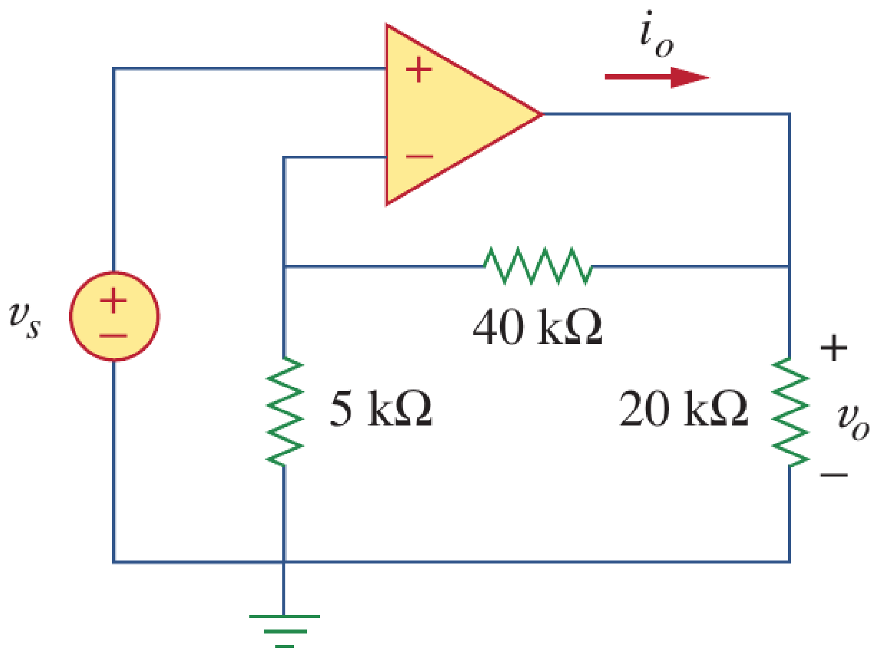
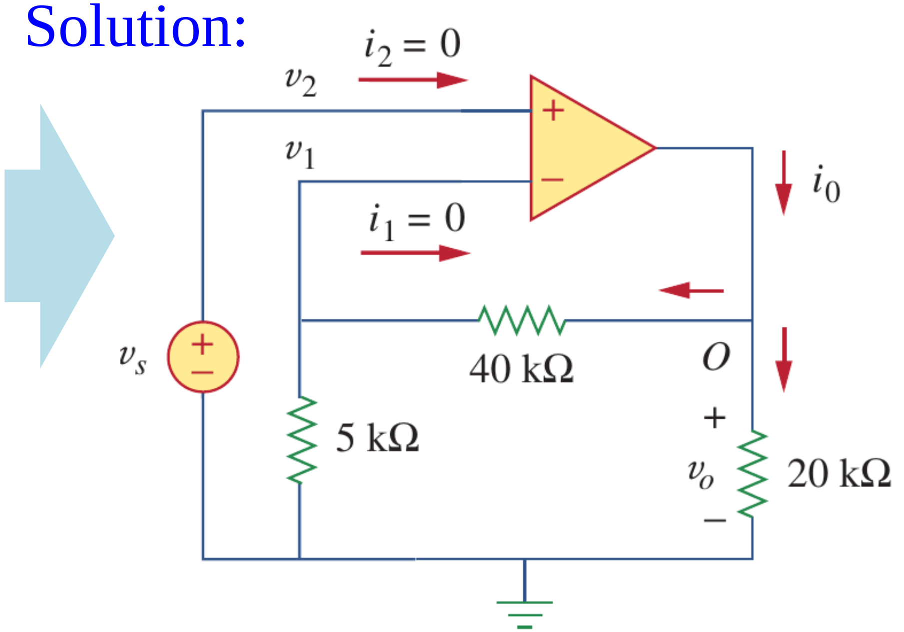
- Example 5.01

Example 5.01: Using the ideal op amp model to calculate the closed-loop gain v_o/v_s, and find i_o when $v_s = 1$ V



Solution:



$$i_1 = i_2 = 0$$

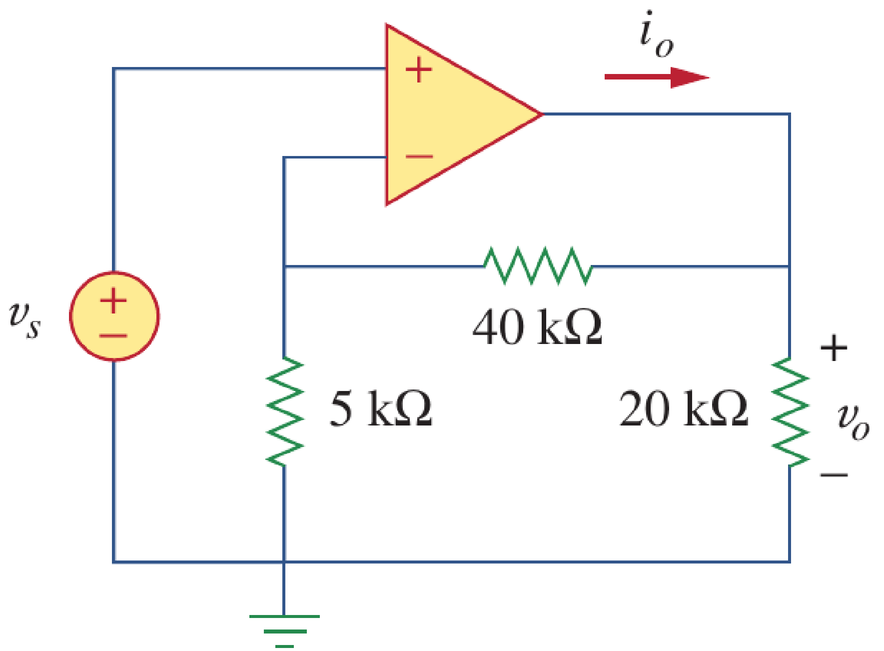
$$v_2 - v_1 = 0$$

$$\begin{cases} v_2 = v_s \\ v_1 = \dfrac{5\text{k}}{5\text{k} + 40\text{k}} v_o \end{cases}$$

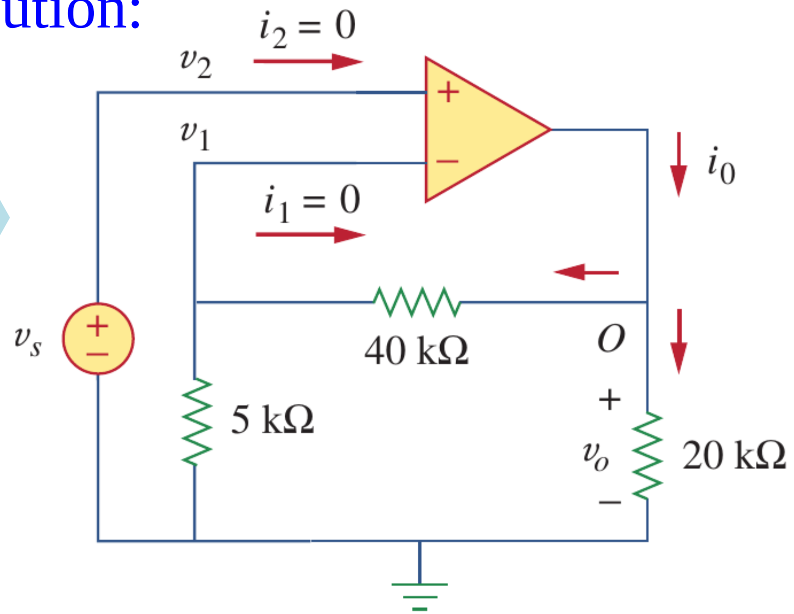
$$\rightarrow \frac{v_o}{v_s} = 9$$

- Example 5.01

Example 5.01: Using the ideal op amp model to calculate the closed-loop gain v_o/v_s, and find i_o when $v_s = 1$ V



Solution:



$$i_o = \frac{v_o}{40+5} + \frac{v_o}{20} \text{mA}$$

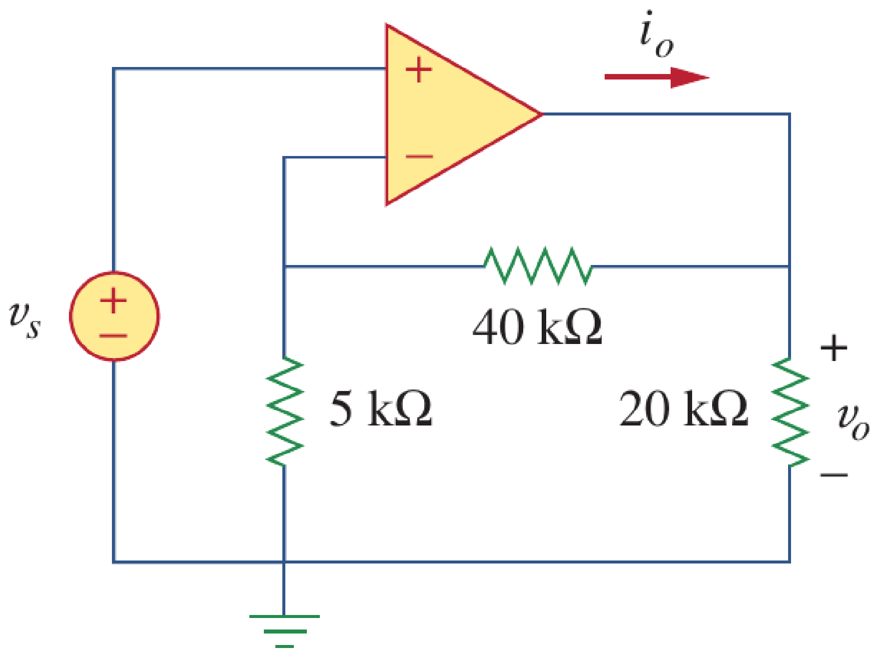
when $v_s = 1$ V, $v_o = 9$ V.

$$i_o = 0.2 + 0.45 = 0.65 \text{ mA}$$

5.4 Op-amps with negative feedbacks (d)

- Example 5.01

Example 5.01: Using the ideal op amp model to calculate the closed-loop gain v_o/v_s, and find i_o when $v_s = 1$ V



Solution:

	Ideal op-amp	μA741
Closed-loop gain	9	9.00041
i_o @ v_s = 1V	0.65 mA	0.647 mA

The difference is very very very small!

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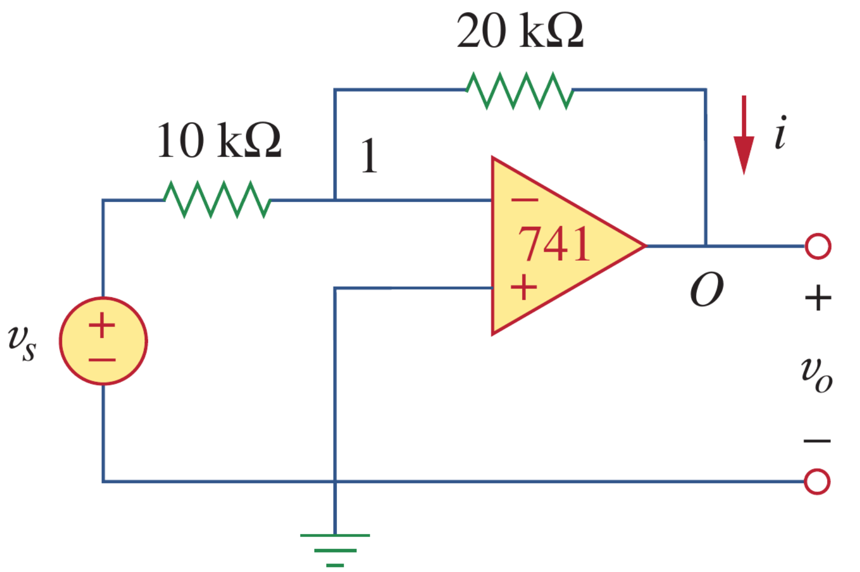
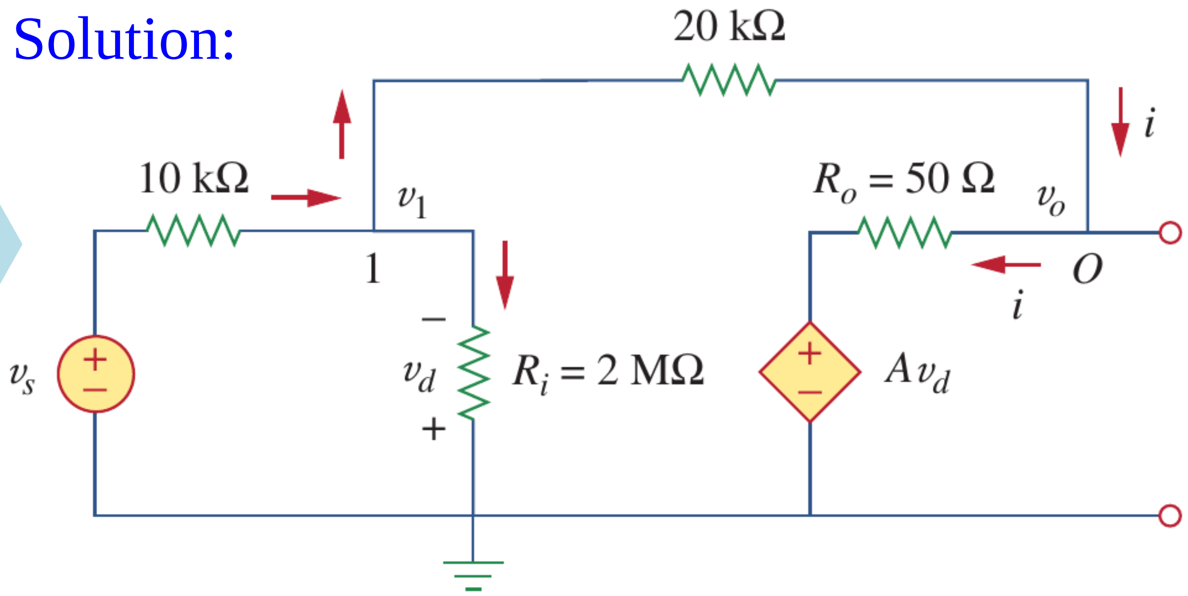
- Example 5.02

Example 5.02: A 741 op-amp has an open-loop voltage gain of 2×10^5, input resistance of 2 MΩ, and output resistance of 50 Ω. The op amp is used in the following circuit. Find the closed-loop gain v_o/v_s. Determine current i when v_s = 2 V.



Solution:



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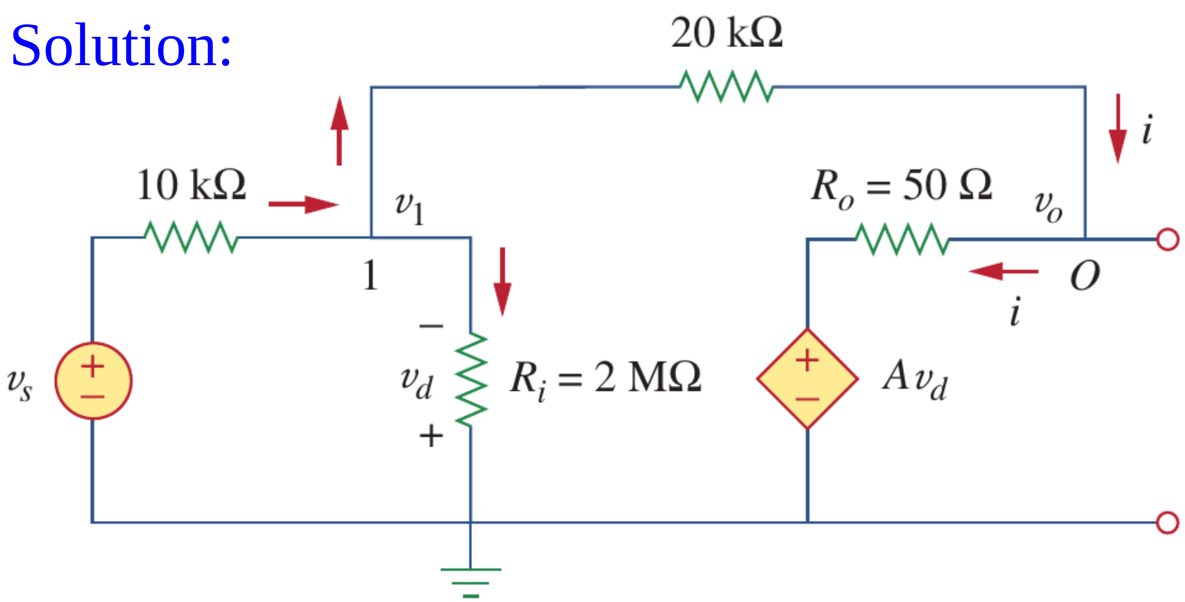
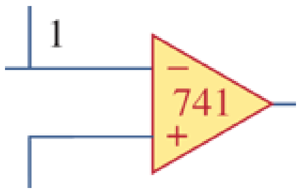
- Example 5.02

Example 5.02: A 741 op amp has an open-loop voltage gain of 2×10^5, input resistance of 2 MΩ, and output resistance of 50 Ω. The op amp is used in the following circuit. Find the closed-loop gain v_o/v_s. Determine current i when v_s = 2 V.

Solution:



(a) $v_d = -v_1$:



Then we use nodal analysis:

(b) KCL at node 1:

$$\frac{v_s - v_1}{10\text{k}} - \frac{v_1 - v_o}{20\text{k}} - \frac{v_1}{2000\text{k}} = 0$$

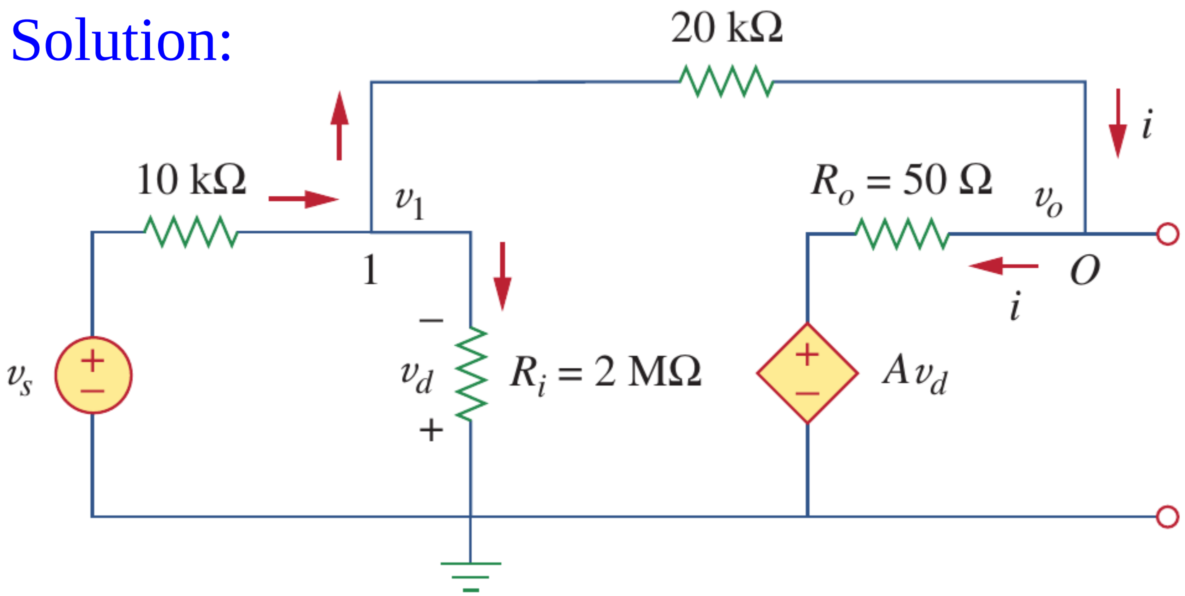
(c) KCL at node O:

$$\frac{v_1 - v_o}{20\text{k}} - \frac{v_o - Av_d}{50} = 0$$

- Example 5.02

Example 5.02: A 741 op amp has an open-loop voltage gain of 2×10^5, input resistance of 2 MΩ, and output resistance of 50 Ω. The op amp is used in the following circuit. Find the closed-loop gain v_o/v_s. Determine current i when $v_s = 2$ V.

Solution:



Thus we have: $v_o/v_s = -1.9999699\ (\approx -2)$

When $v_s = 2V$,

$$v_o = -3.99994\ V, v_1 = 20.066667\ \mu V,$$

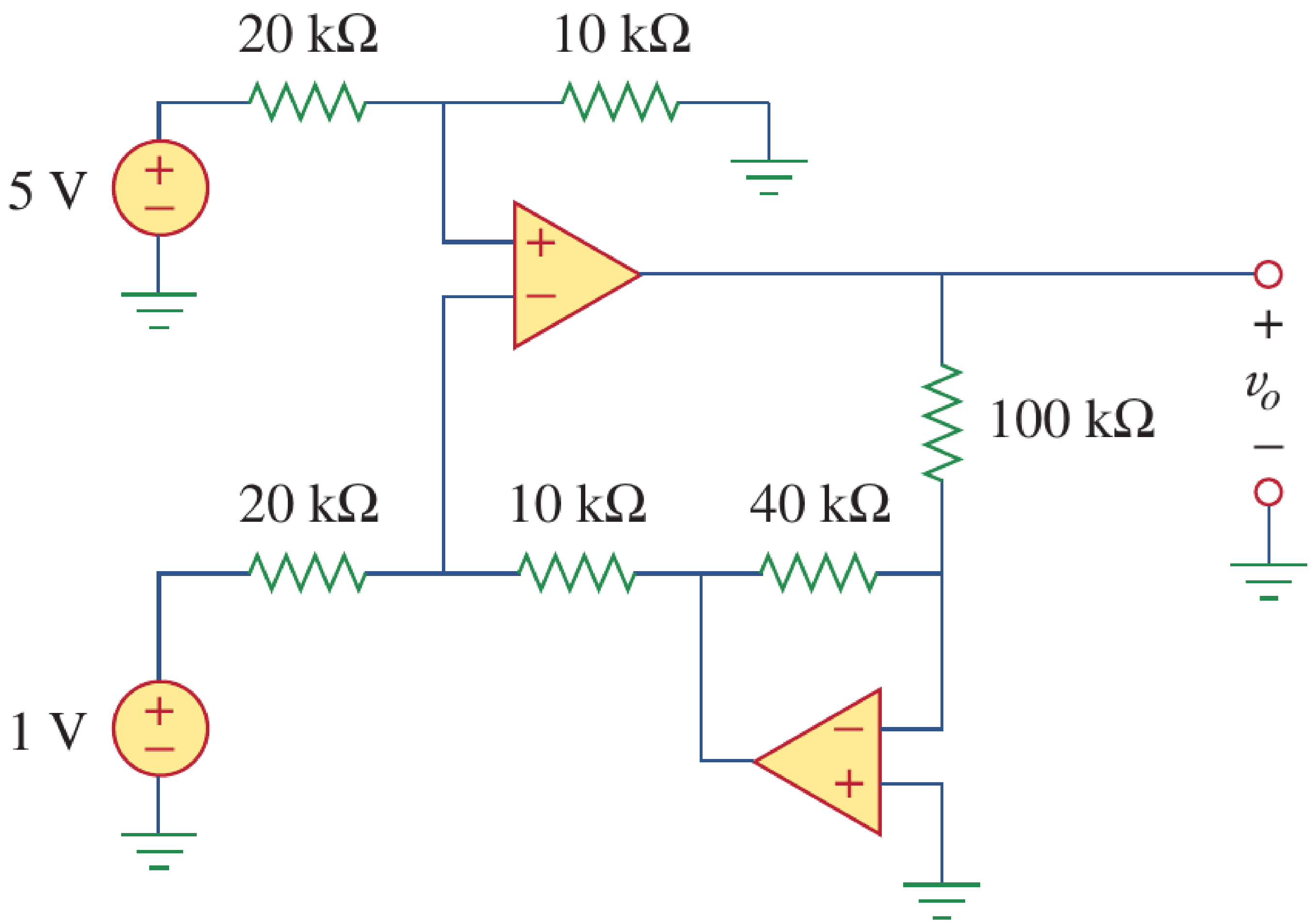
so $i = (v_1 - v_o)/20\,k = 0.1999\ mA.$

The voltage across and current through R_i is very small, and negligible.

5.4 Op-amps with negative feedbacks (g)

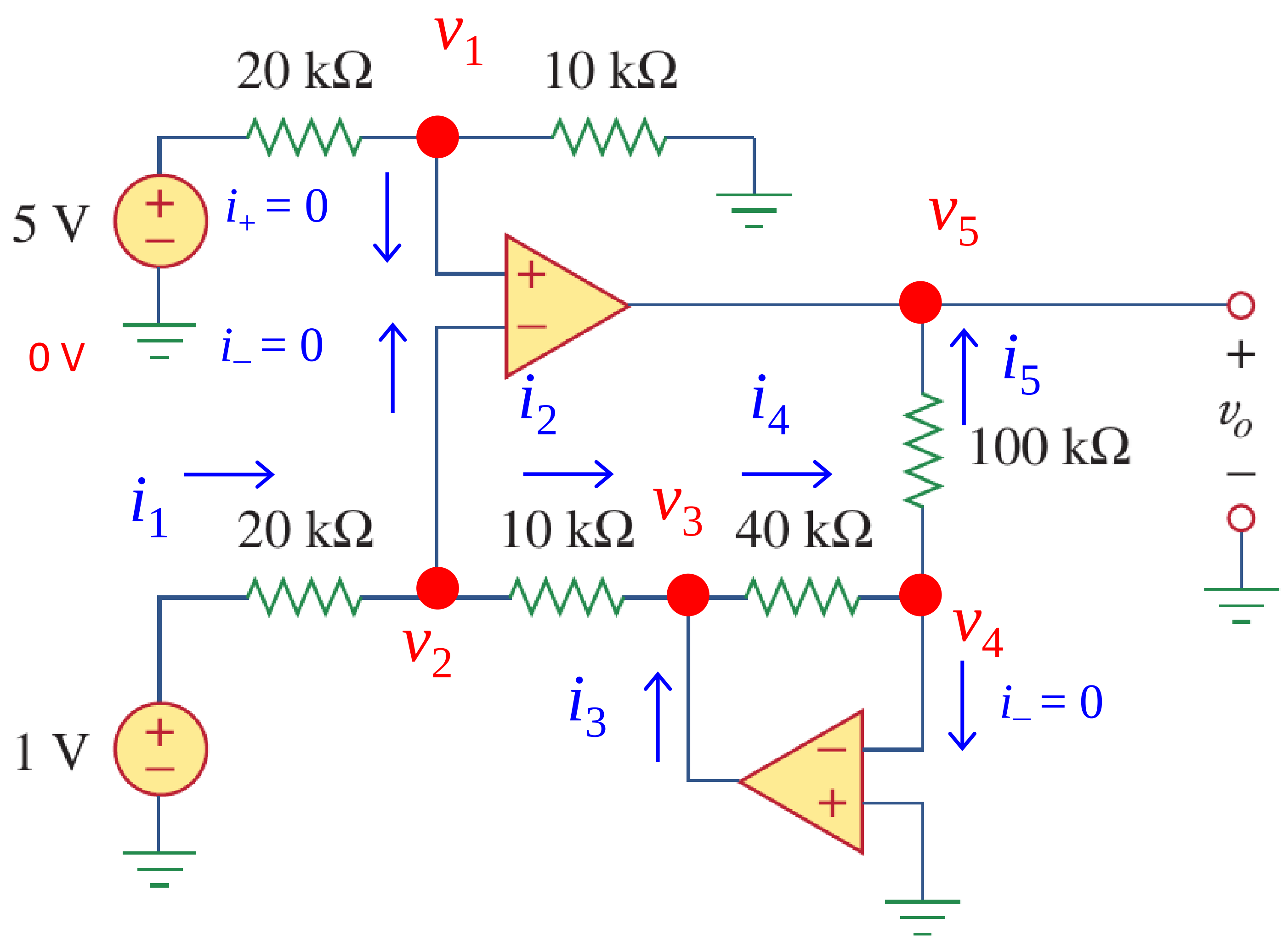
- Example 5.03

Example 5.02: determine Vo.



5.4 Op-amps with negative feedbacks (g)

- Example 5.03, solution I



(1) $v_1 = 5/3$ V

(2) Since $v_+ - v_- = 0$

$$v_2 = v_1 = 5/3 \text{ V}, \quad v_4 = 0$$

(3) Since $i_+ = i_- = 0$

$$i_1 = i_2 \Rightarrow \frac{1 - \frac{5}{3}}{20\,k} = \frac{\frac{5}{3} - v_3}{10\,k} \Rightarrow v_3 = 2 \text{ V}$$

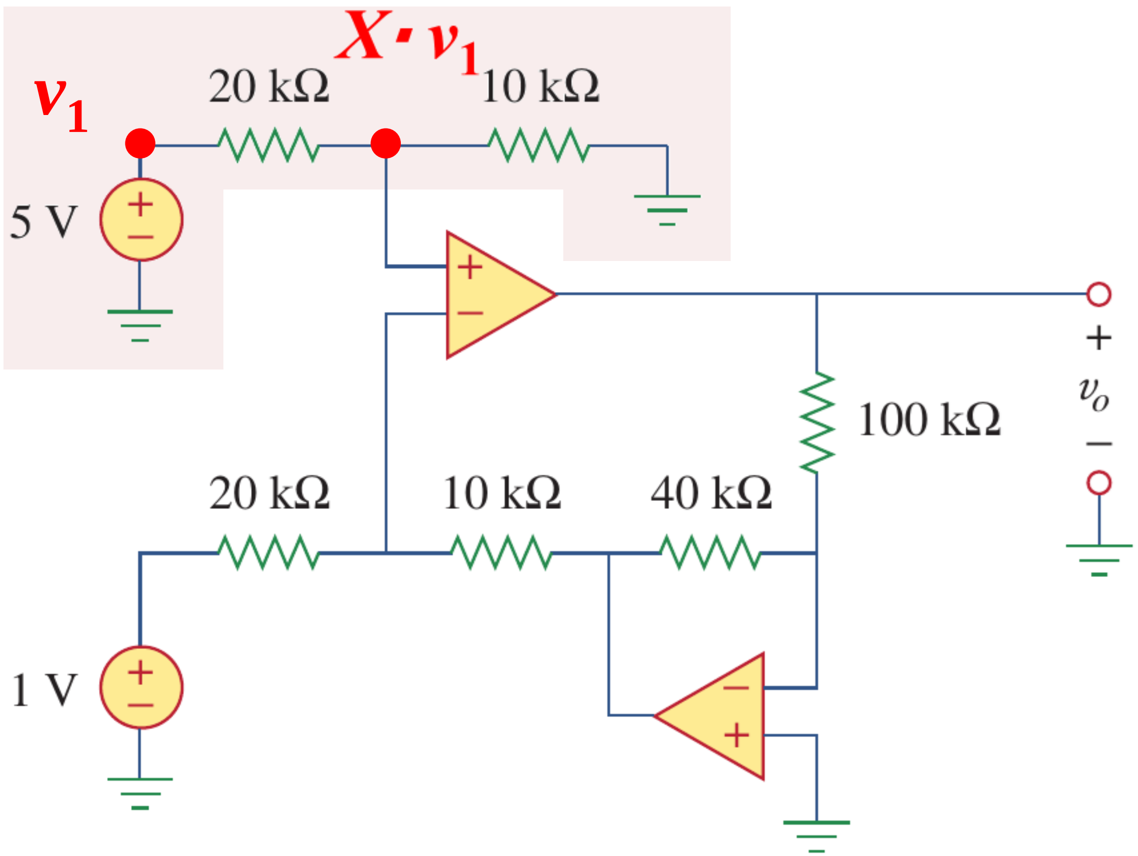
(4) at node v_4

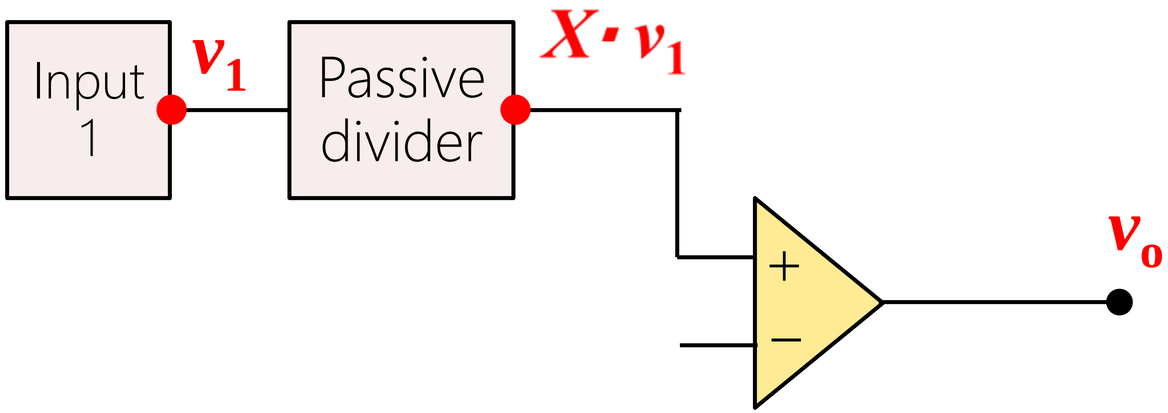
$$i_4 = i_5 \Rightarrow \frac{2 - 0}{40\,k} = \frac{0 - v_5}{100\,k} \Rightarrow v_5 = -5 \text{ V}$$

5.4 Op-amps with negative feedbacks (g)

- Example 5.03, solution II

$$X = \frac{1}{3}$$





5.4 Op-amps with negative feedbacks (g)

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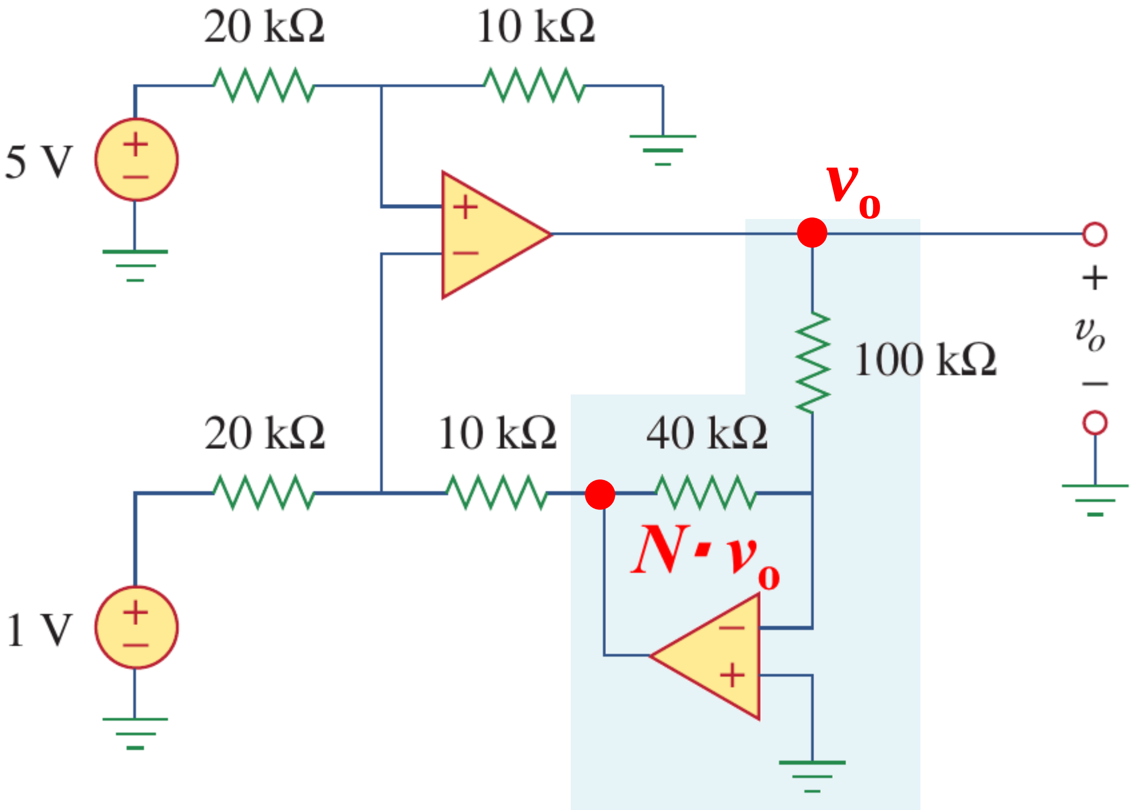
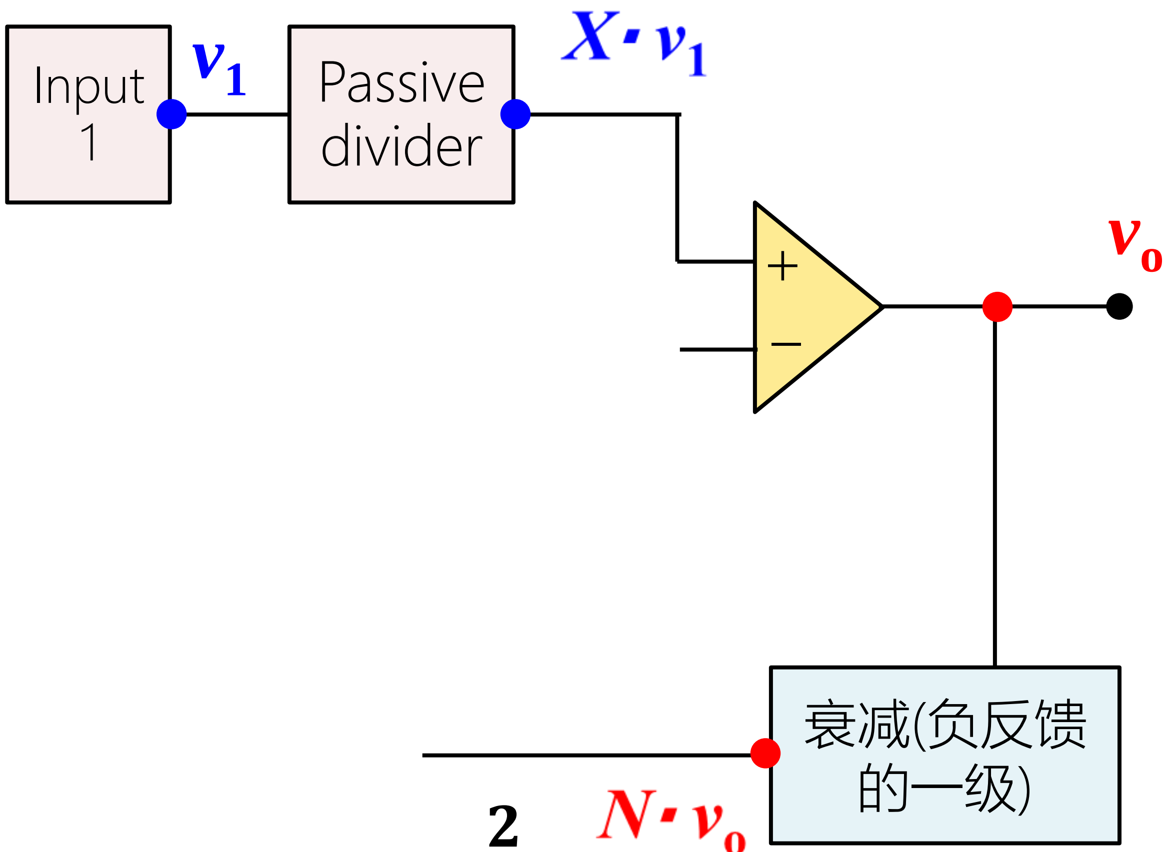
- Example 5.03, solution II

$$X = \frac{1}{3}$$

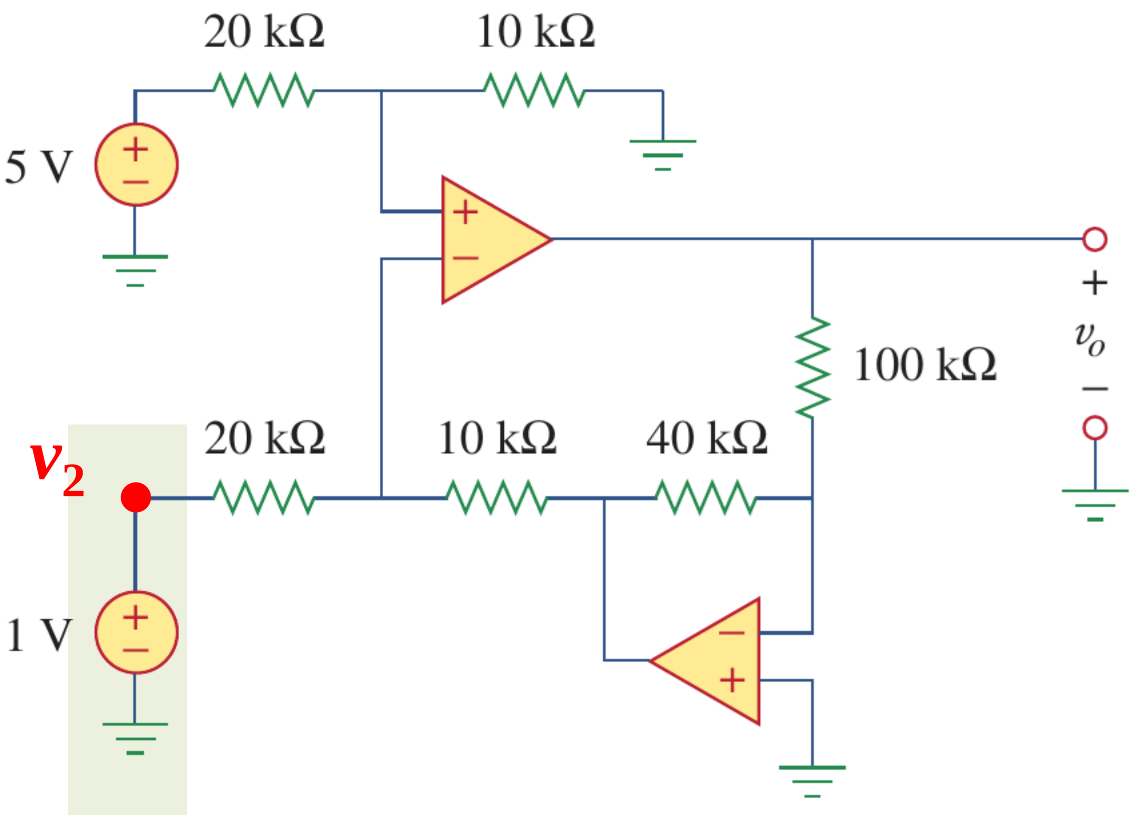




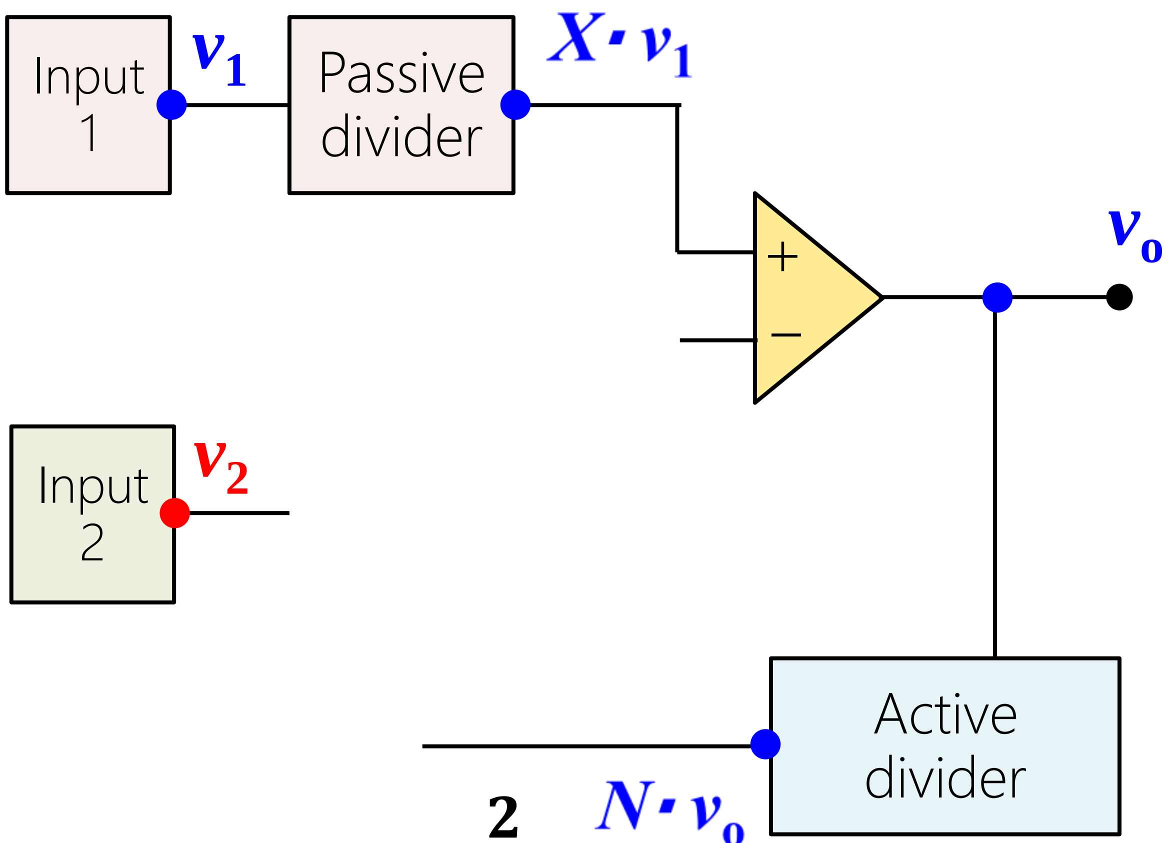
$$N = -\frac{2}{5}$$

5.4 Op-amps with negative feedbacks (g)

- Example 5.03, solution II



$$X = \frac{1}{3}$$



$$N = -\frac{2}{5}$$

5.4 Op-amps with negative feedbacks (g)

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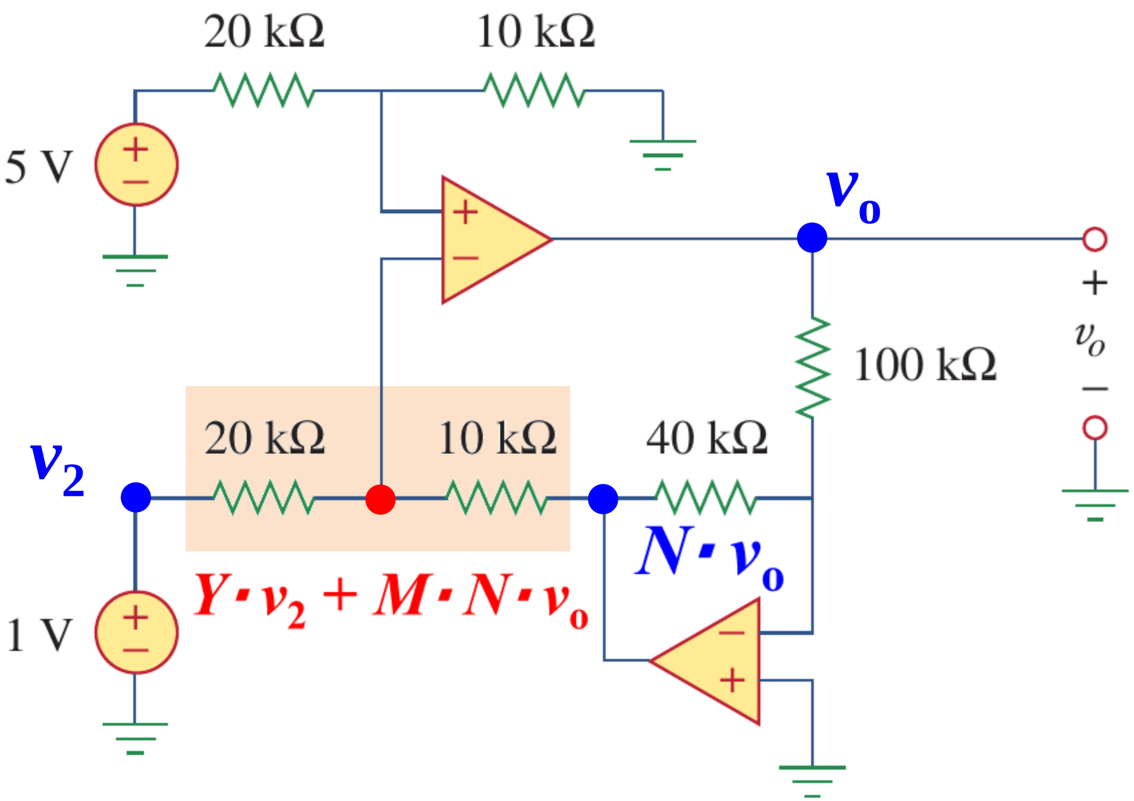
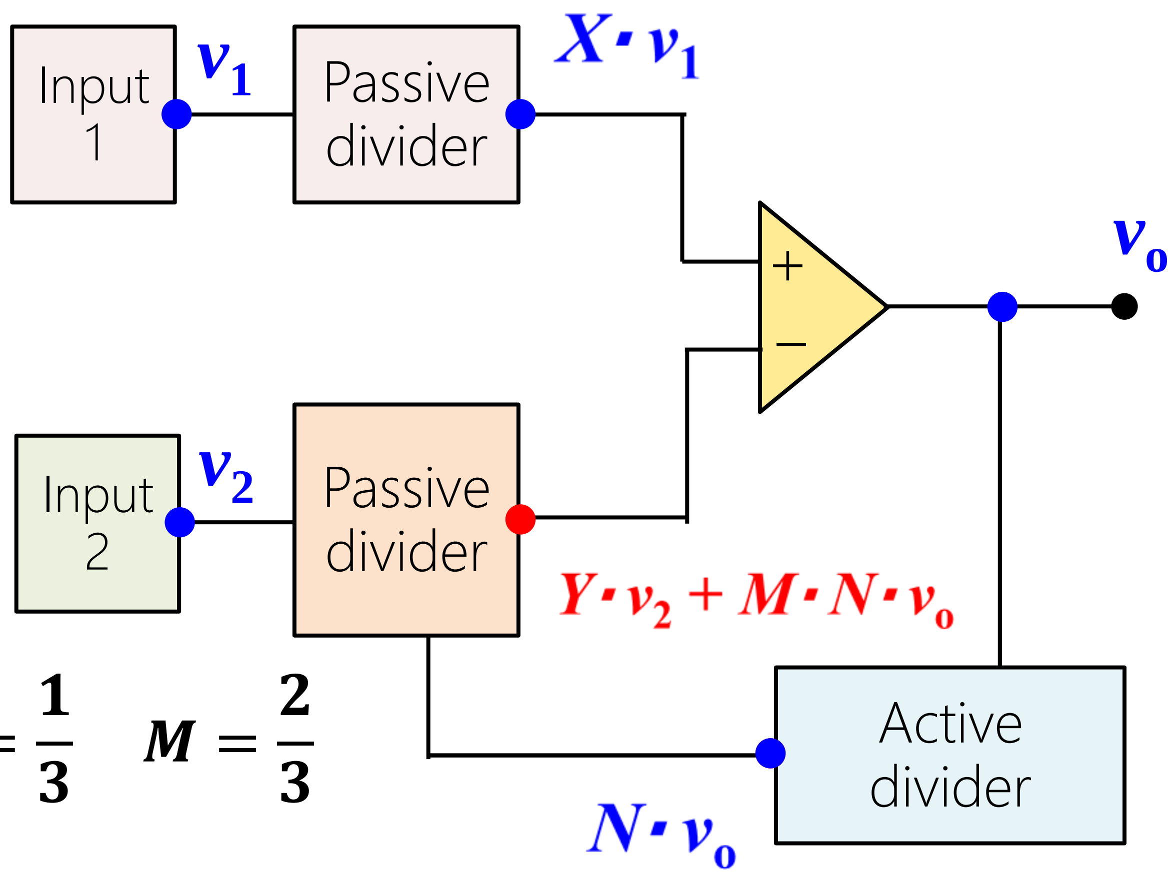
- Example 5.03, solution II





$$Y = \frac{1}{3} \quad M = \frac{2}{3}$$

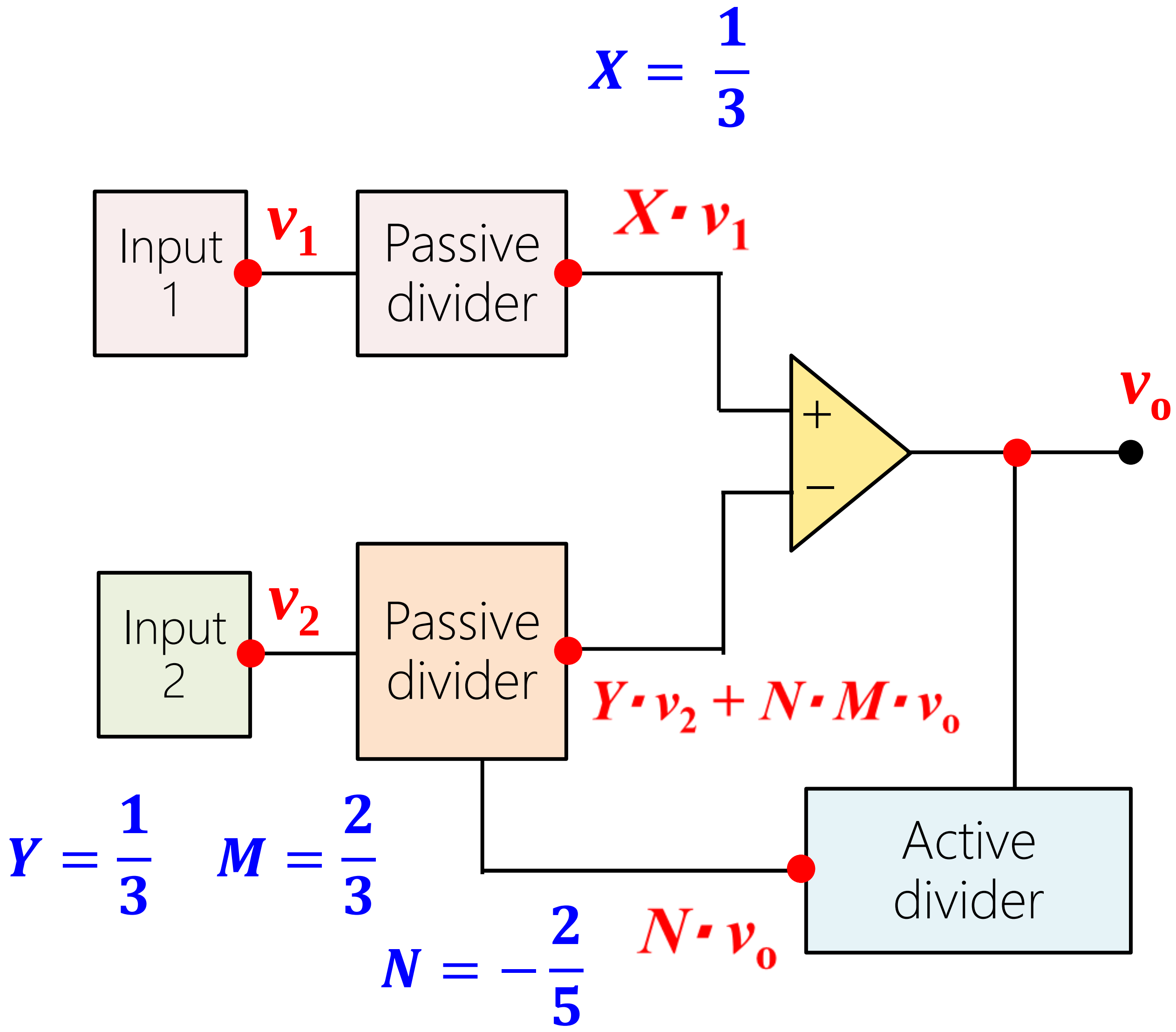
- Example 5.03, solution II

$$X \cdot v_1 = Y \cdot v_2 + N \cdot M \cdot v_o$$

$$5 \times \frac{1}{3} = 1 \times \frac{1}{3} - \frac{2}{3} \times \frac{2}{5} \times v_o$$

$$\frac{4}{3} = -\frac{4}{15} \times v_o$$

$$v_o = -5\ \mathrm{V}$$



- Example 5.03, solution III

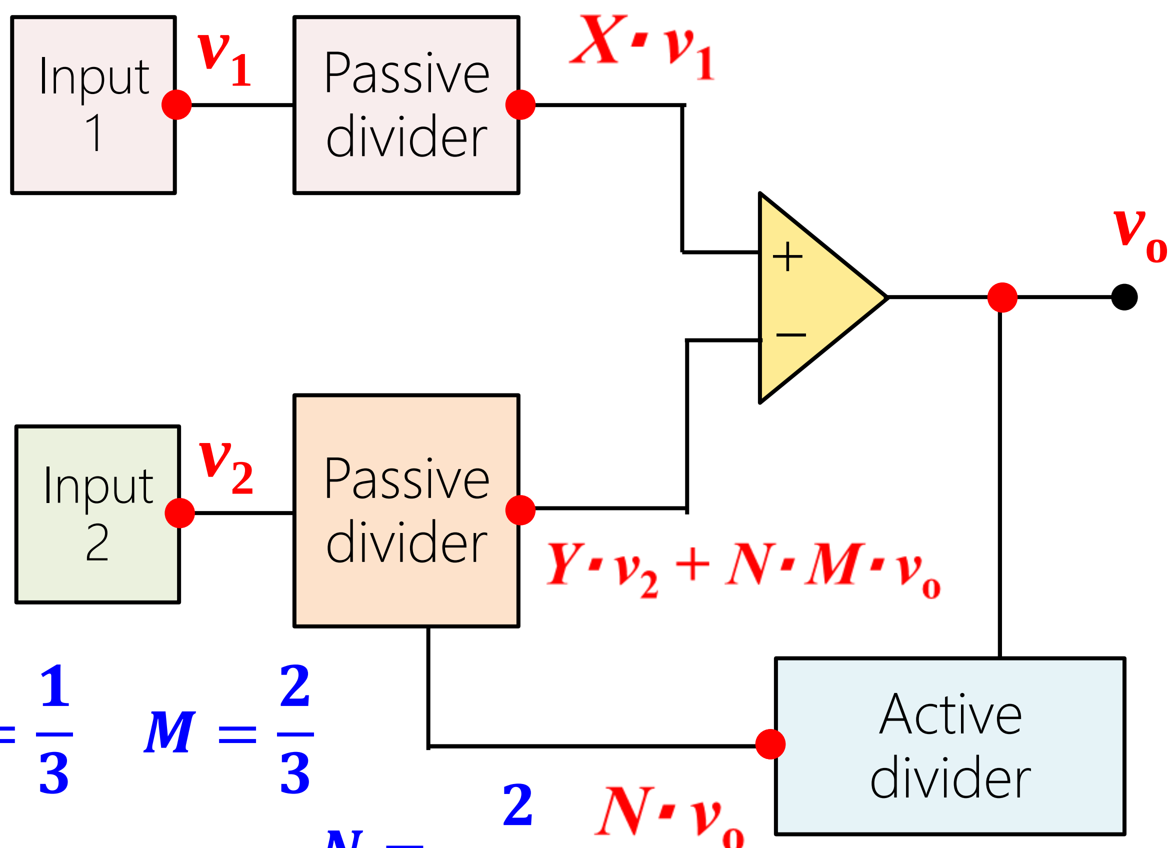
$$Feedback\ factor$$

$$F=\frac{2}{3}\times\left(-\frac{2}{5}\right)=-\frac{4}{15}$$

$$Closed-loop\ gain\ A_F=\frac{1}{F}=-\frac{4}{15}$$

$$v_o=\frac{1}{3}(5-1)\left(-\frac{4}{15}\right)=-5\ V$$

$X=\frac{1}{3}$



$Y=\frac{1}{3}$ $M=\frac{2}{3}$

$N=-\frac{2}{5}$

Lesson 5 Operational Amplifiers (Op Amp)

5.1	Introduction	Slides 11-19
5.2-5.4	Op amp basics and negative feedback	Slides 21-48
5.5-5.6	**Inverting and noninverting amplifiers**	**Slides 50-60**
5.7-5.9	Summing and difference amplifiers	Slides 62-72
5.10	Cascaded Op Amp circuits	Slides 74-82
5.11	Summary and assignments	Slides 84-85

5.5 Inverting amplifiers (a)

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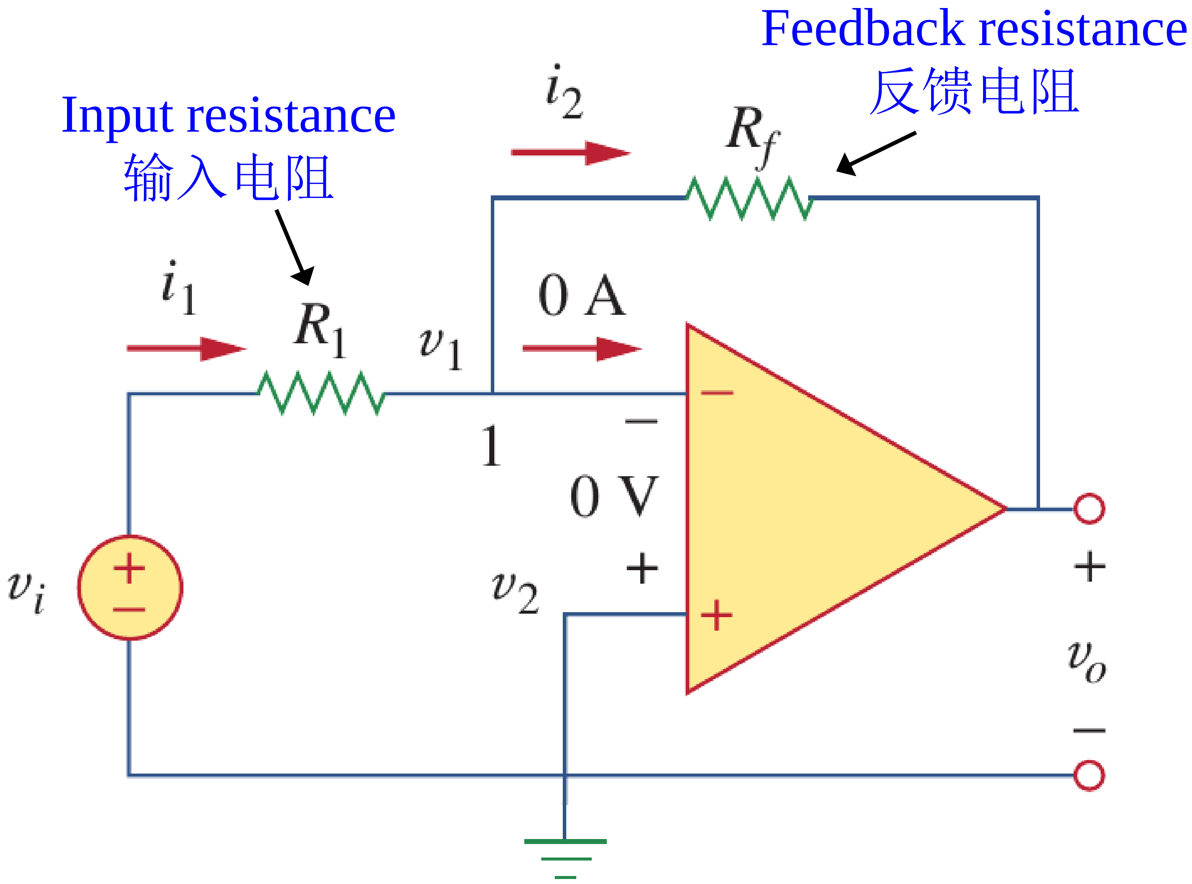
- Inverting amplifiers (反向放大器)



a) An inverting op-amp inverts the polarity of the signal while amplifying it (a negative voltage gain).

b) The noninverting input (同相输入) is grounded, v_i is connected to the inverting input (反相输入) through R_1, and the feedback resistor R_f is connected between the inverting input and output.

c) Notice the negative feedback is added to the input v_i in the left circuit.

d) Notice the input resistance here is the input resistance of the loop.

5.5 Inverting amplifiers (b)

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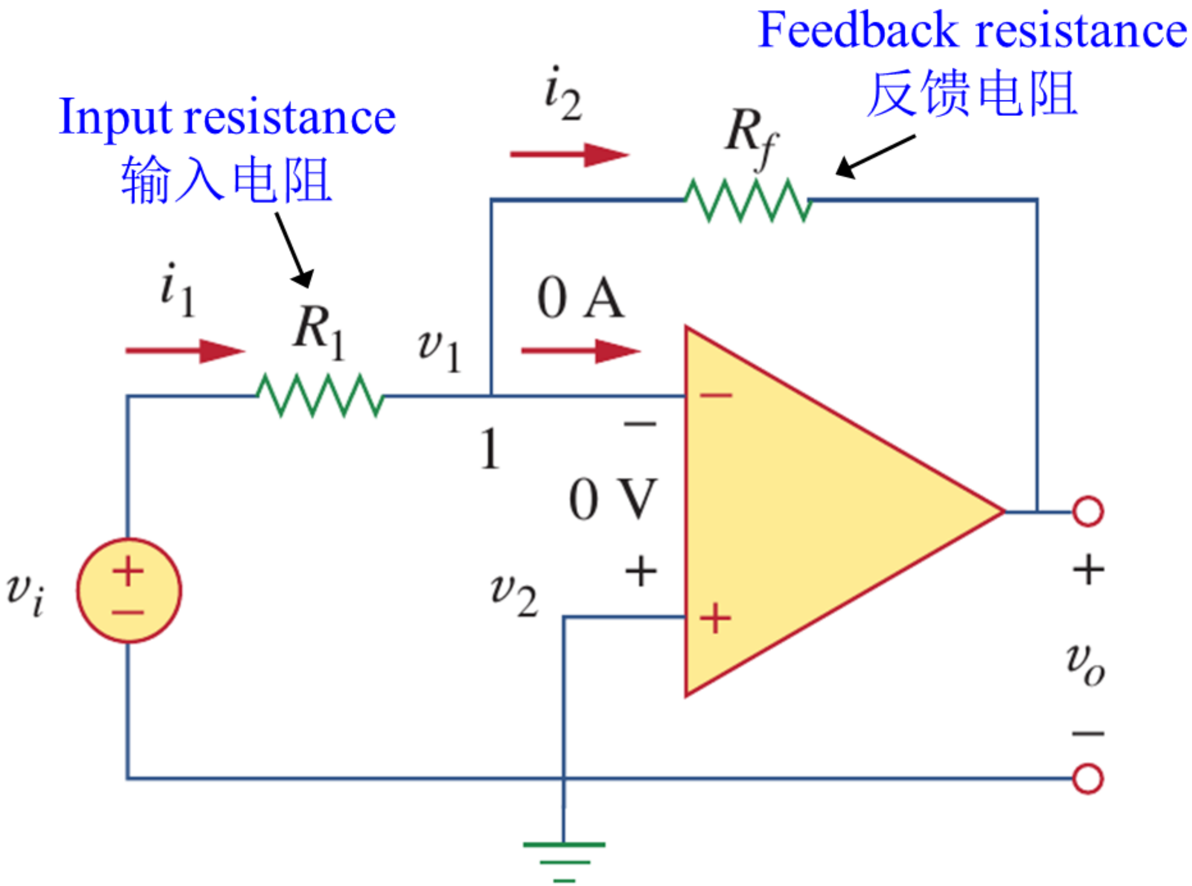
- The closed-loop gain (闭环增益)



- Golden rules for analyzing ideal op-amp:

  (1) Currents into the two inputs are zero:

$$i_1 = i_2 \rightarrow \frac{v_i - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$$

  (2) v_1 and v_2 are equal (zero here):

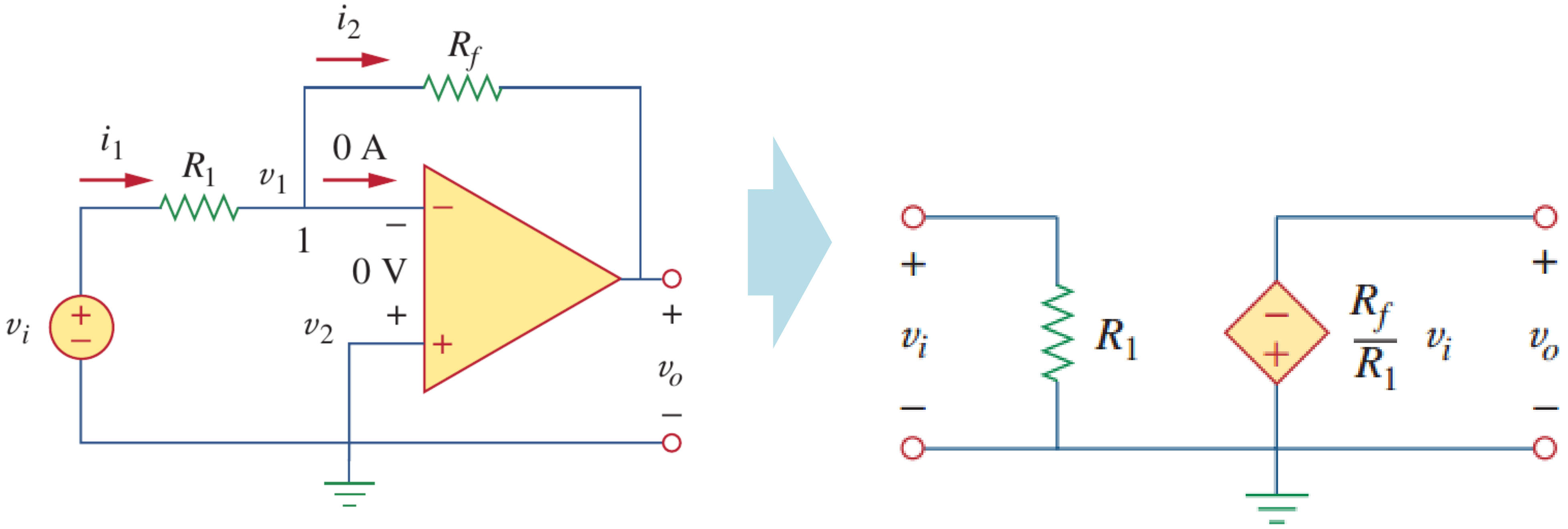
$$v_1 = v_2 = 0 \rightarrow \frac{v_i}{R_1} = \frac{-v_o}{R_f}$$

  (3) Therefore we have:

$$v_o = -\frac{R_f}{R_1} v_i$$

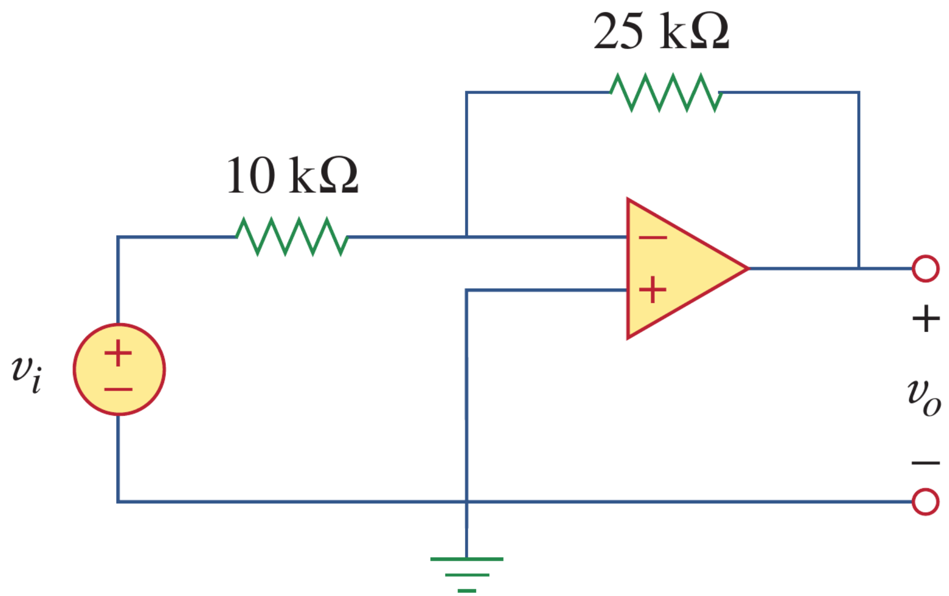
5.5 Inverting amplifiers (c)

- Equivalent circuits of ideal inverting amplifiers



- Example 5.04

Example 5.04: Refer to the op-amp in the following circuit. If $v_i = 0.5$ V, calculate: (a) the output voltage v_o and (b) the current in the 10 kΩ resistor.



Solution:

(a)
$$\frac{v_o}{v_i} = -\frac{R_f}{R_1} = -\frac{25}{10} = -2.5$$

$$v_o = -2.5v_i = -2.5(0.5) = -1.25 \text{ V}$$

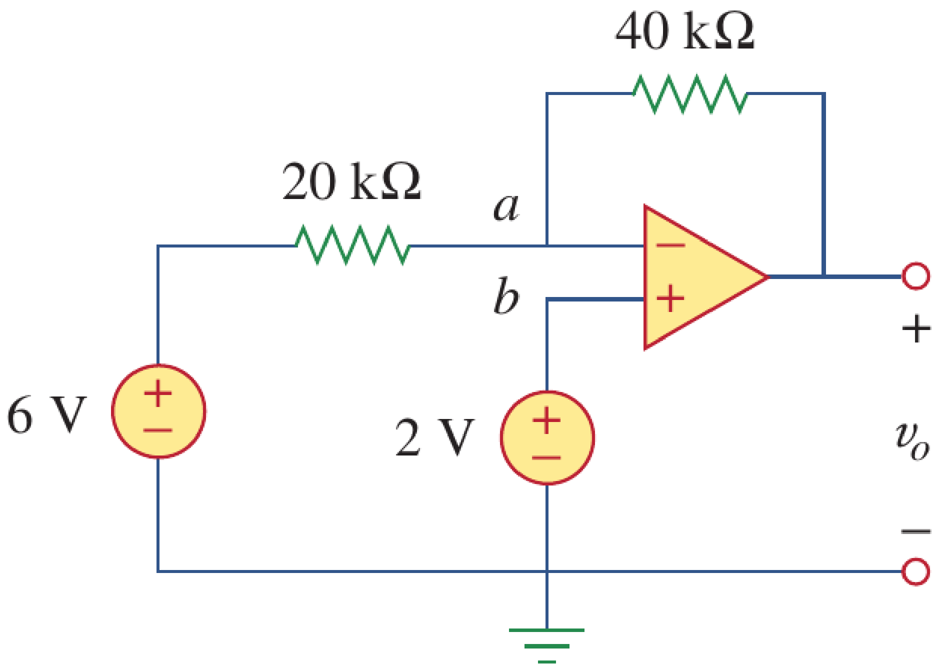
(b) $v_+ = v_-$ and $v_+ = 0$

$$i = \frac{v_i - 0}{R_1} = \frac{0.5 - 0}{10 \times 10^3} = 50\ \mu\text{A}$$

5.5 Inverting amplifiers (e)

- Example 5.05 (what if $v_+ = v_- \neq 0$?)

Example 5.05: Determine the v_o in the op-amp circuit shown below.



Solution:

Applying KCL at node a:

$$\begin{cases} \dfrac{v_a - v_o}{40\ \mathrm{k\Omega}} = \dfrac{6 - v_a}{20\ \mathrm{k\Omega}} \\ v_a = v_b = 2\ \mathrm{V} \end{cases}$$

$$\begin{aligned} v_o &= 3v_a - 12 \\ &= 6 - 12 = -6\ \mathrm{V} \end{aligned}$$

5.6 Noninverting amplifiers (a)

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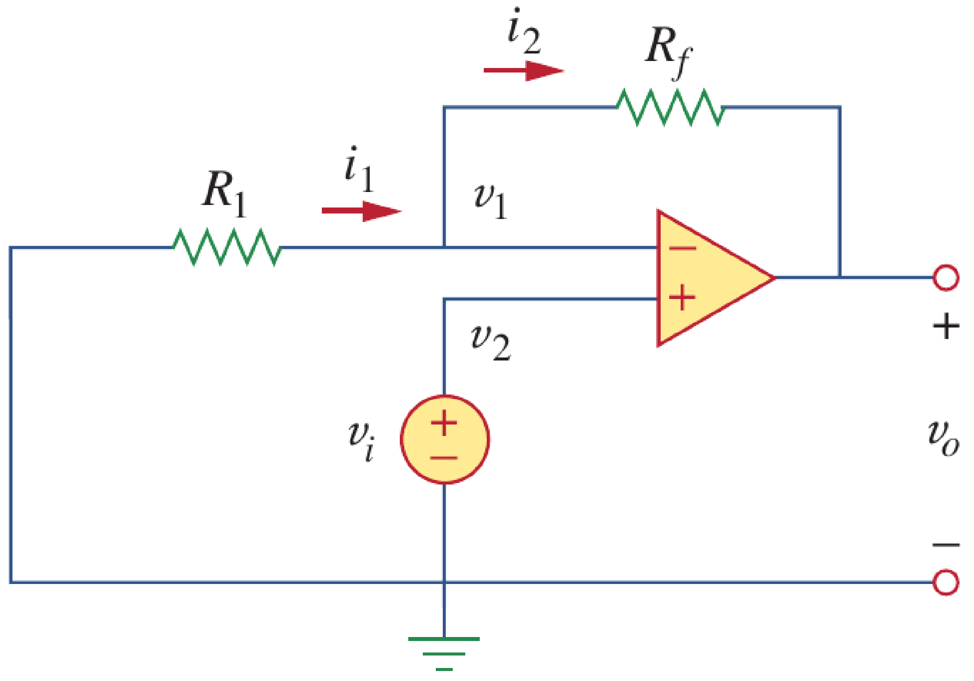
- Noninverting amplifiers (同向放大器)

a) Noninverting amplifiers are very similar to inverting amplifiers, but the input voltage is directly applied at the noninverting input terminal, which provide a positive voltage gain.

b) The negative feedback is still connected to the inverting input terminal.

c) One application of noninverting amplifiers is the voltage follower, which will be introduced later.



5.6 Noninverting amplifiers (b)

- Noninverting amplifiers

- Golden rules for analyzing ideal op-amp:

  (1) Currents into the two inputs are zero:

$$i_1 = i_2 \rightarrow \frac{0 - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$$

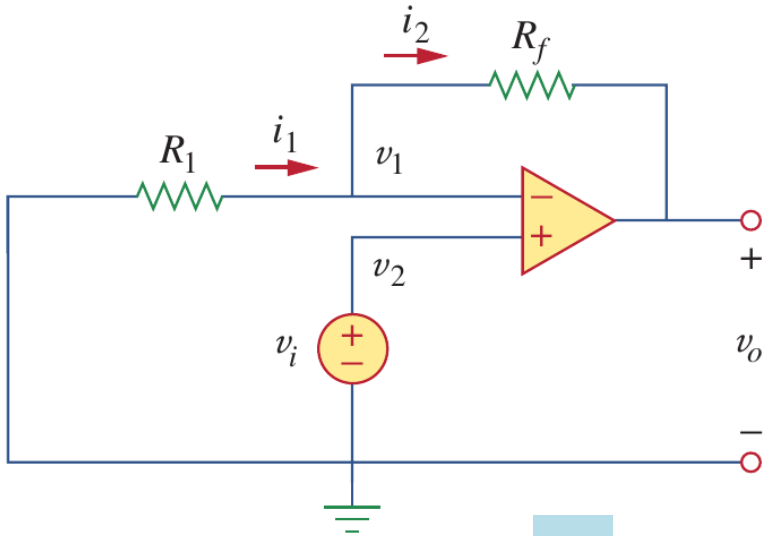
  (2) v_1 and v_2 are equal (zero here):

$$v_1 = v_2 = v_i \rightarrow \frac{-v_i}{R_1} = \frac{v_i - v_o}{R_f}$$

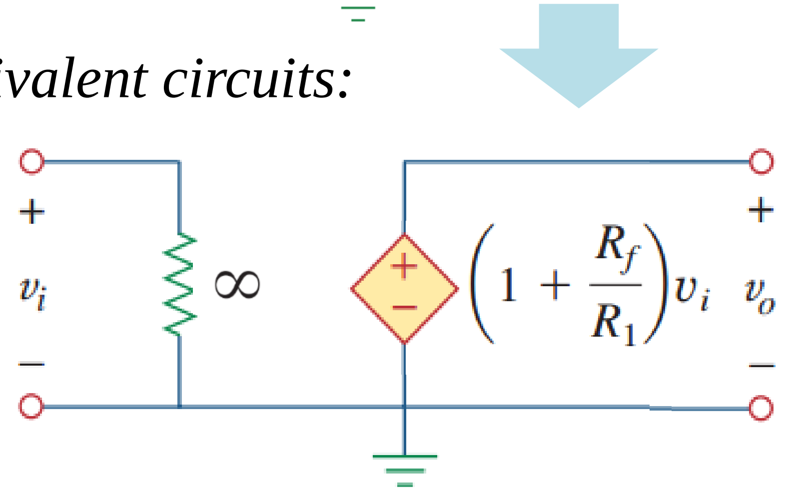
  (3) Therefore we have:

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

Noninverting amplifiers:



Equivalent circuits:



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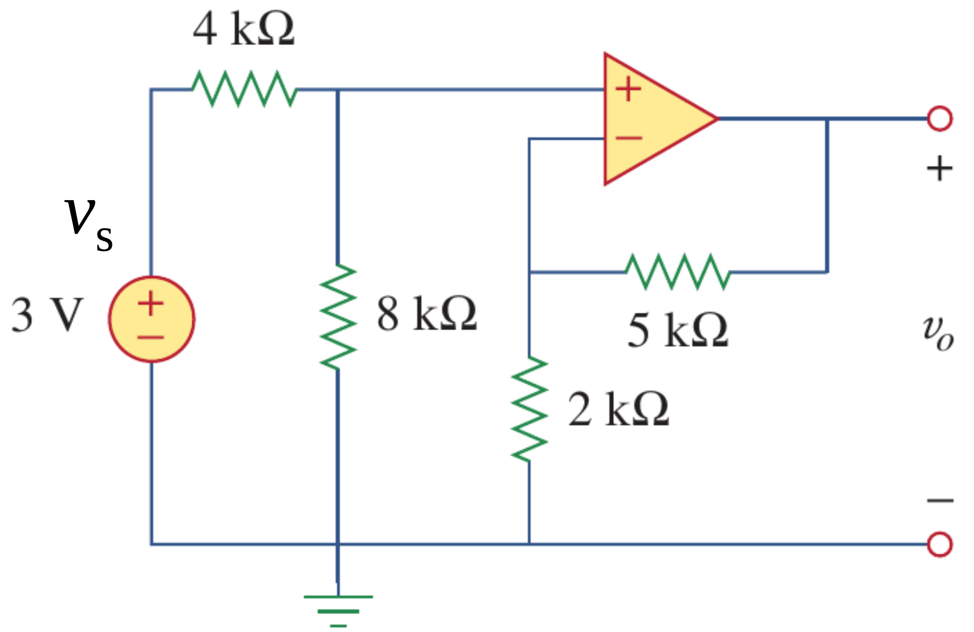
- Example 5.06

Example 5.06: Calculate the output voltage v_o for the op-amp circuit shown below.



Solution:

(b) Find the gain:

$$\frac{v_o - v_i}{5\text{k}} = \frac{v_i - 0}{2\text{k}} \Rightarrow v_o = 3.5vi$$

$$\boxed{\left(1+\frac{R_f}{R_1}\right) = 1+\frac{5}{2} = 3.5}$$

(a) Find the effective input voltage:

$$v_i = \frac{8\text{k}}{4\text{k}+8\text{k}} v_s = 2\text{ V}$$

(c) Then $v_o = 3.5v_i = 7$ V

5.6 Noninverting amplifiers (d)

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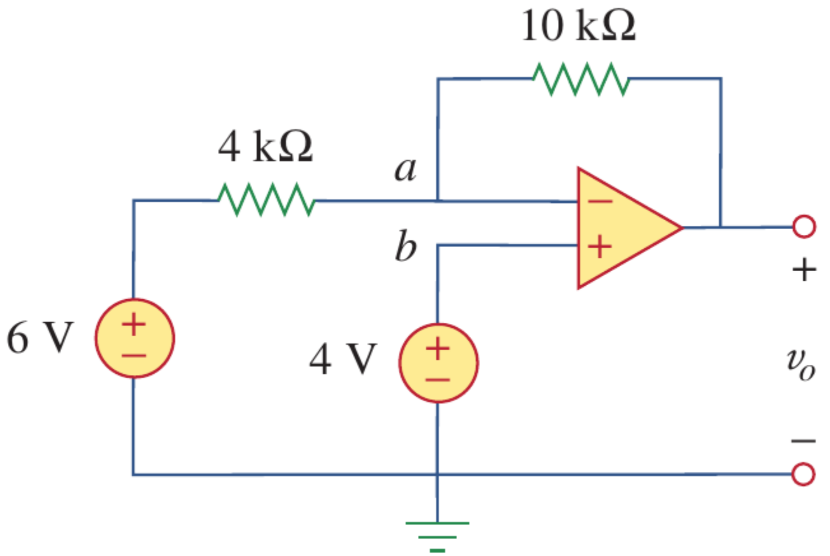
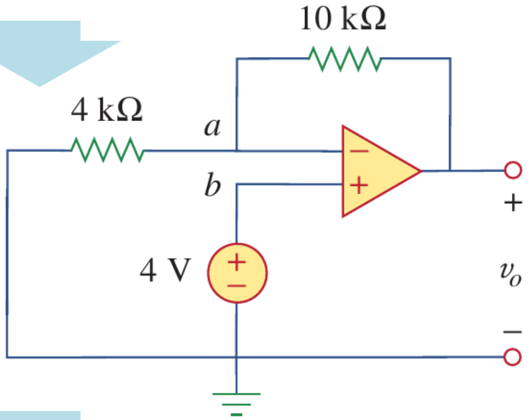
- Example 5.07

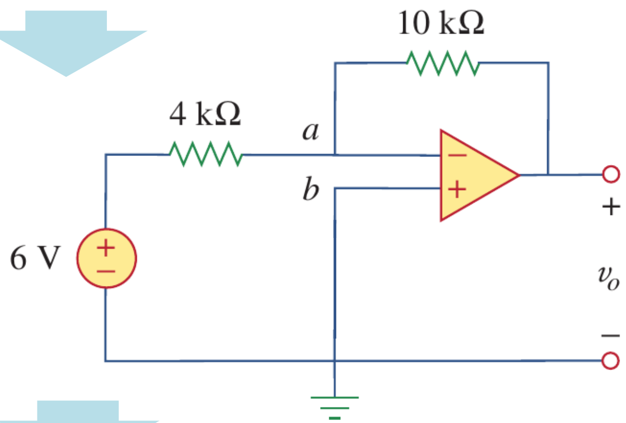
Example 5.07: Calculate the output voltage v_o for the op-amp circuit shown below.



Solution (based superposition theorem):

$v_o = v_{o1}$ with only the 4 V source + v_{o2} with only the 6 V source





$$v_{o1} = \left(1 + \frac{R_f}{R_1}\right) v_{s1} = \left(1 + \frac{10}{4}\right) \times 4 = 14\text{V}$$

$$v_{o2} = -\frac{R_f}{R_1} v_{s2} = -\frac{10}{4} \times 6 = -15\text{ V}$$

$$v_o = v_{o1} + v_{o2} = -1\text{ V}$$

5.6 Noninverting amplifiers (e)

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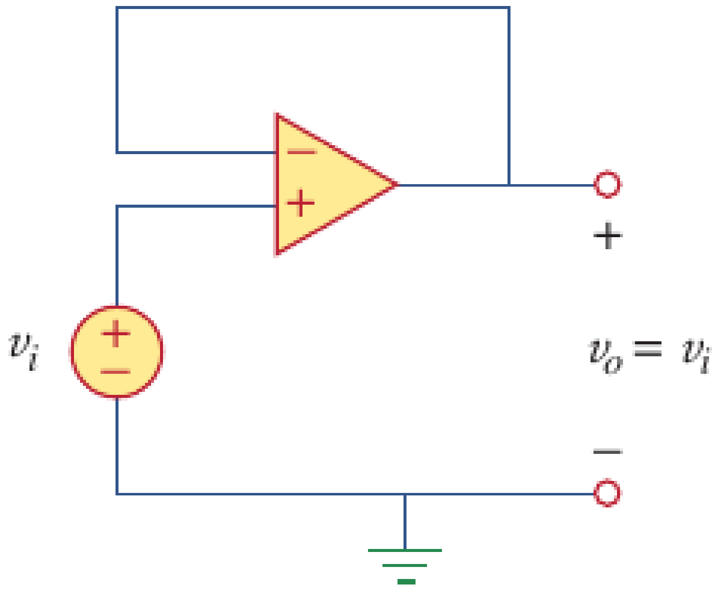
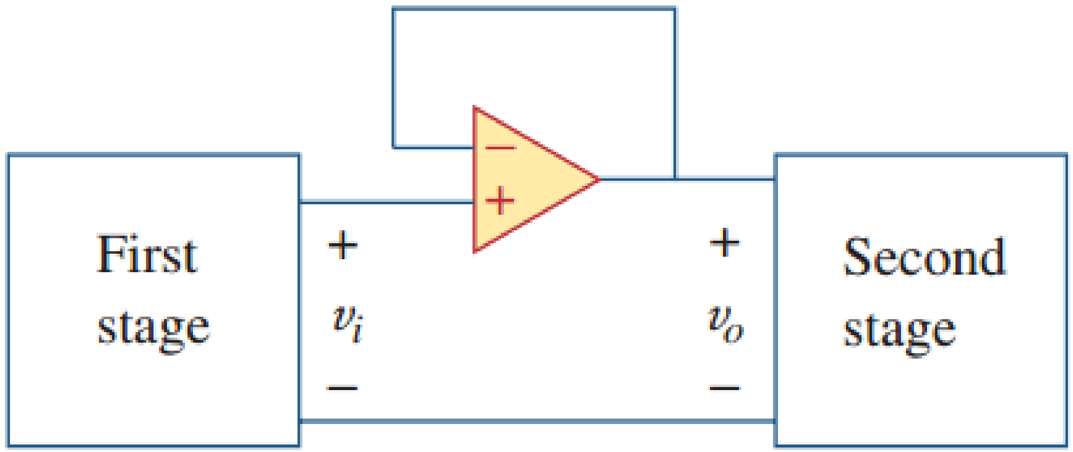
- Voltage followers

a) A noninverting amplifier functions as a voltage follower ($v_o = v_i$) if (a) $R_f = 0$ and (b) $R_i = \infty$, which is also called a unity gain amplifier.

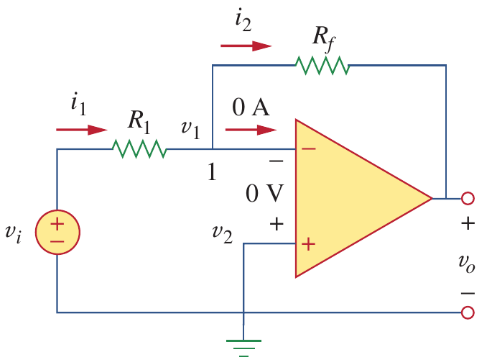
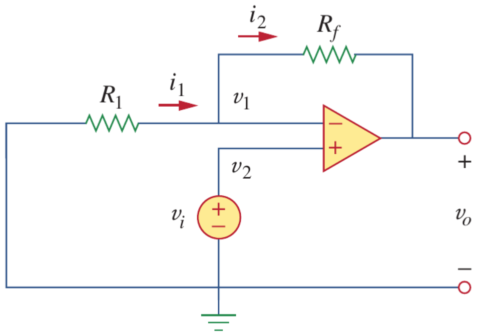
b) It is good for isolating (隔离) two circuits while allowing a signal to pass through.





5.6 Noninverting amplifiers (f)

- Brief summary: inverting and noninverting amplifiers

	Topology	Closed-loop gain
Inverting op-amp:		**Negative** $$v_o = -\frac{R_f}{R_1} v_i$$
Noninverting op-amp:		**Positive** $$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

Lesson 5 Operational Amplifiers (Op Amp)

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5.7-5.9	**Summing and difference amplifiers**	**Slides 62-72**
5.10	Cascaded Op Amp circuits	Slides 74-82
5.11	Summary and assignments	Slides 84-85

5.7 Summing amplifiers (a)

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- Summing amplifiers

a) The summing amplifier is a variation of the inverting amplifier, taking advantage of the fact that the inverting configuration can handle many inputs at the same time.

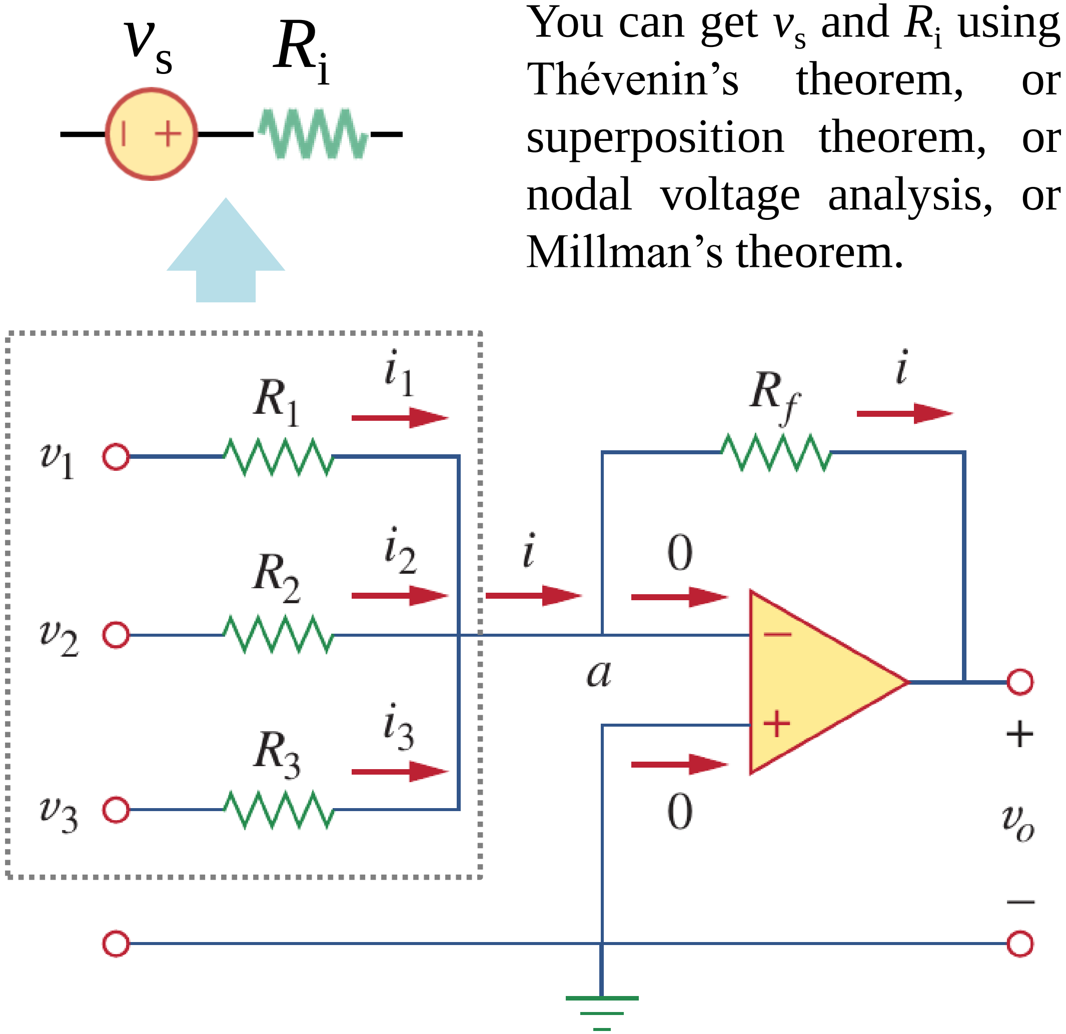
b) Based on Millman's theorem:

The effective input voltage of the loop:

$$v_i = \frac{G_1 v_1 + G_2 V_2 + G_3 V_3}{G_1 + G_2 + G_3}$$

The effective input resistance of the loop:

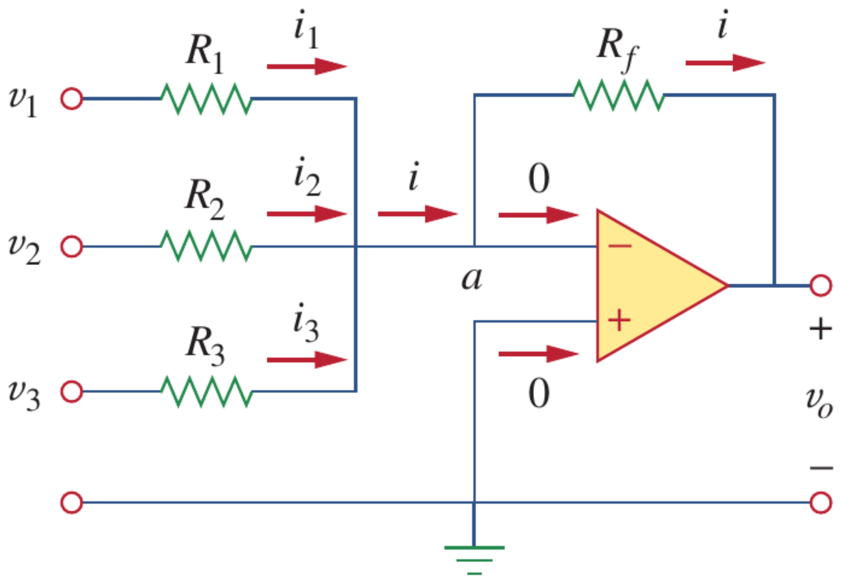
$$\frac{1}{R_i} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$



You can get v_s and R_i using Thévenin's theorem, or superposition theorem, or nodal voltage analysis, or Millman's theorem.

5.7 Summing amplifiers (b)

- Summing amplifiers



- $i_1 + i_2 + i_3 - i = 0$ & $v_+ = v_- = 0$

$$\frac{v_1 - 0}{R_1} + \frac{v_2 - 0}{R_2} + \frac{v_3 - 0}{R_3} - \frac{0 - v_o}{R_f} = 0$$

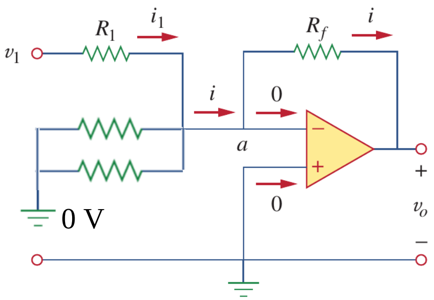
$$v_o = -\frac{R_f}{R_1} v_1 - \frac{R_f}{R_2} v_2 - \frac{R_f}{R_3} v_3$$

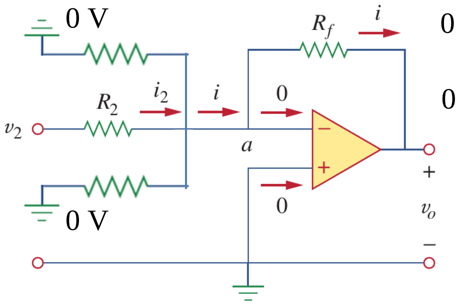
- You can also use superposition to find this, since the equivalent circuit of an op-amp contains only resistors and a linear voltage-controlled voltage source.

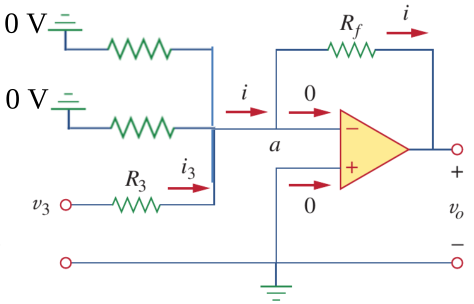
v_{o1} due to v_1

v_{o2} due to v_2

v_{o3} due to v_3



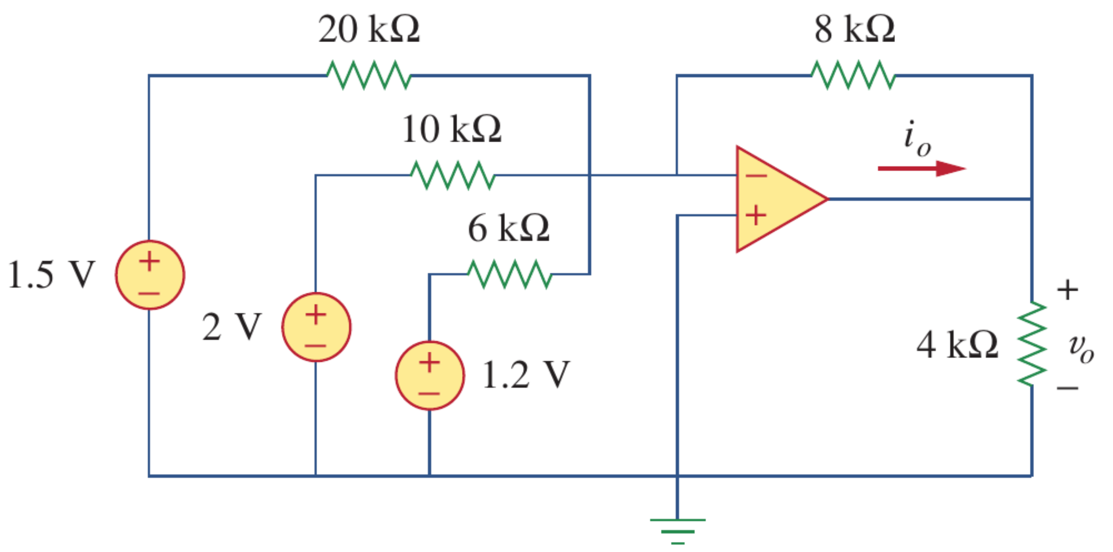




5.7 Summing amplifiers (c)

- Example 5.08

Example 5.08: Find the v_o and i_o in the following op-amp circuit.



Solution I:

$$v_o = -\frac{8}{20} \times 1.5 - \frac{8}{10} \times 2 - \frac{8}{6} \times 1.2 = -3.8\text{ V}$$

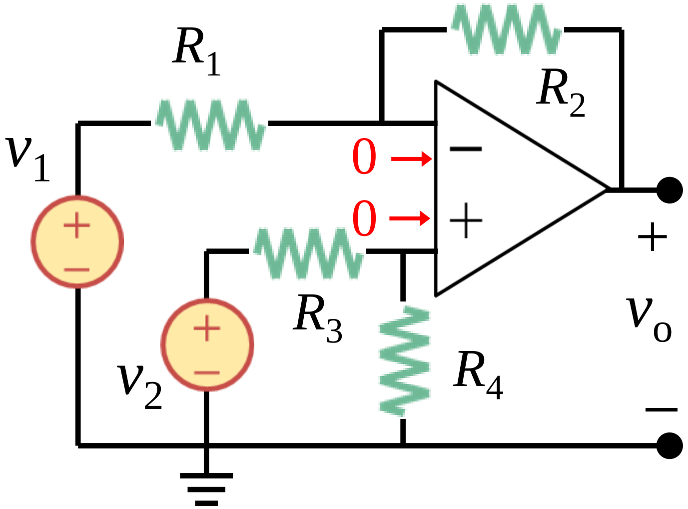
Using nodal analysis:

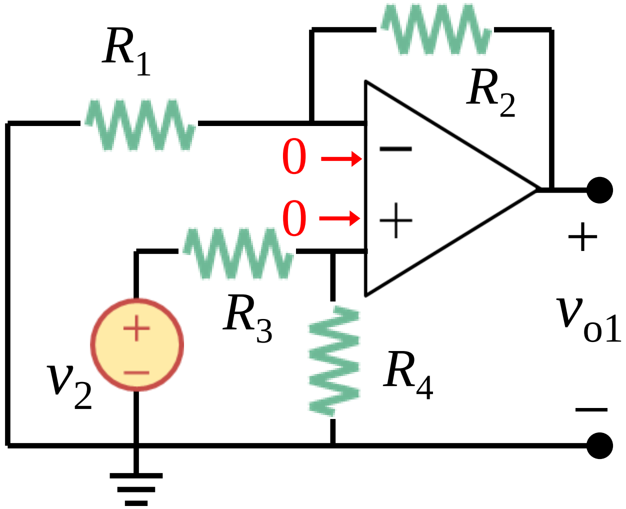
$$i_o - \frac{v_o - 0}{8\text{k}} - \frac{v_o - 0}{4\text{k}} = 0, and\ v_o = -3.8\text{ V}$$

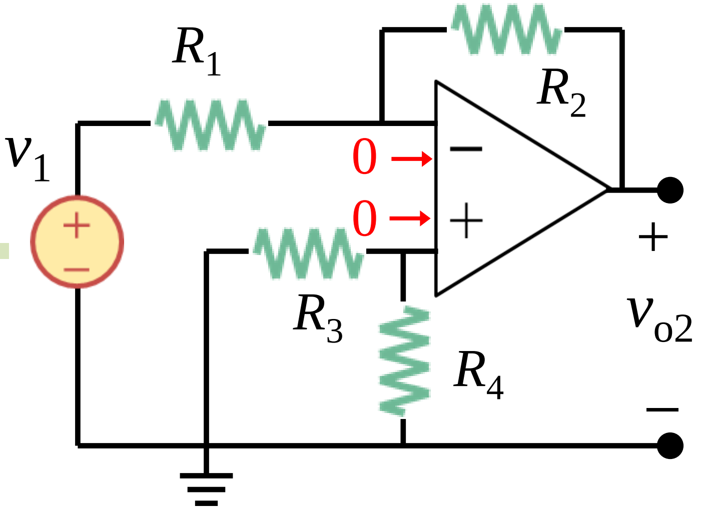
Therefore $i_o = -1.425$ mA

5.8 Difference amplifiers (a)

- Difference amplifiers







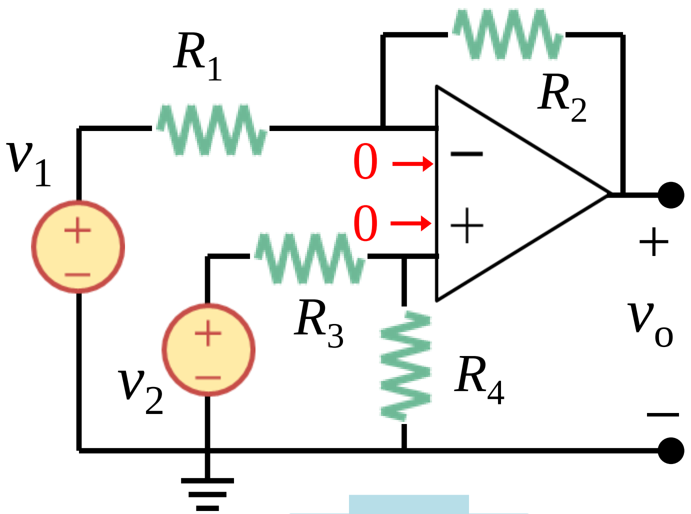
$$v_{o1} = \left(1 + \frac{R_2}{R_1}\right) \frac{R_4}{R_3 + R_4} v_2$$

$$v_{o2} = -\frac{R_2}{R_1} v_1$$

$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

5.8 Difference amplifiers (b)

- Difference amplifiers



$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

Difference amplifiers (差分放大器):

a) For difference amplifiers, when $v_1 = v_2$, $v_o = 0$ V.

b) When : $\dfrac{R_1}{R_2} = \dfrac{R_3}{R_4}$ $\quad v_o = \dfrac{R_2}{R_1}(v_2 - v_1)$

Substractors (减法器):

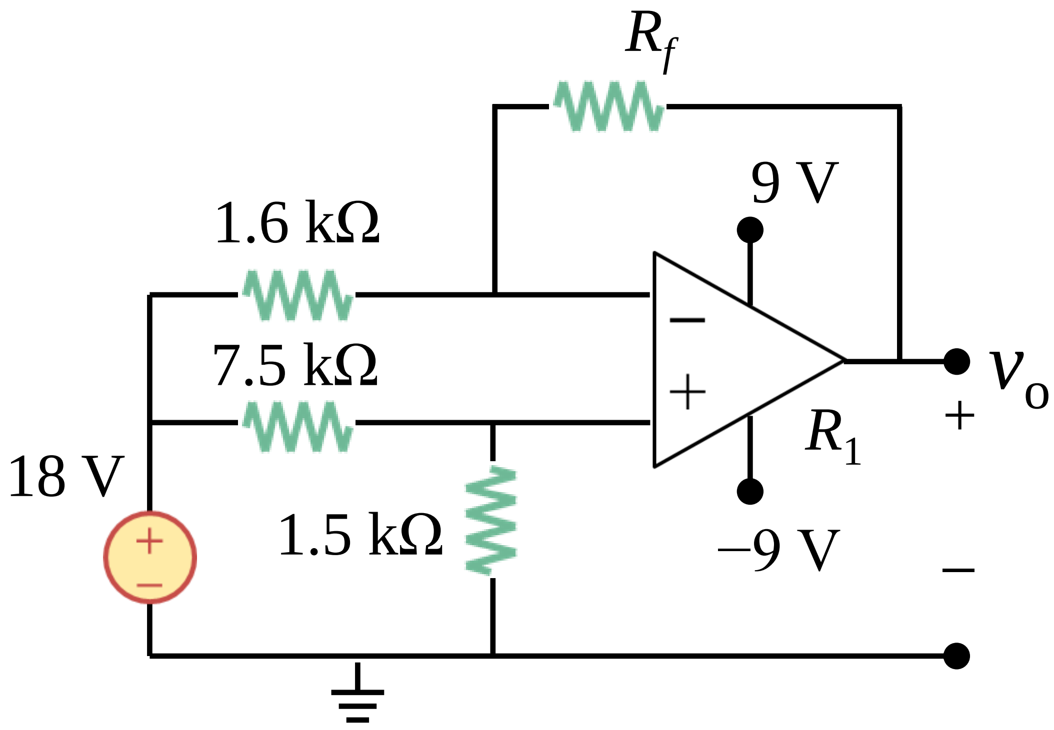
a) When $R_1 = R_2$ and $R_3 = R_4$:

$$v_o = v_2 - v_1$$

5.8 Difference amplifiers (c)

- Example 5.09

Example 5.09: Find the value of R_f that the ideal op-amp starts to operate in saturation region.



Solution I: $v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$

$$v_o = 18 \times \frac{x}{1.6} \times \left(\frac{1 + \frac{1.6}{x}}{1 + \frac{7.5}{1.5}} - 1\right) \Rightarrow v_o = 3 - 9.375x$$

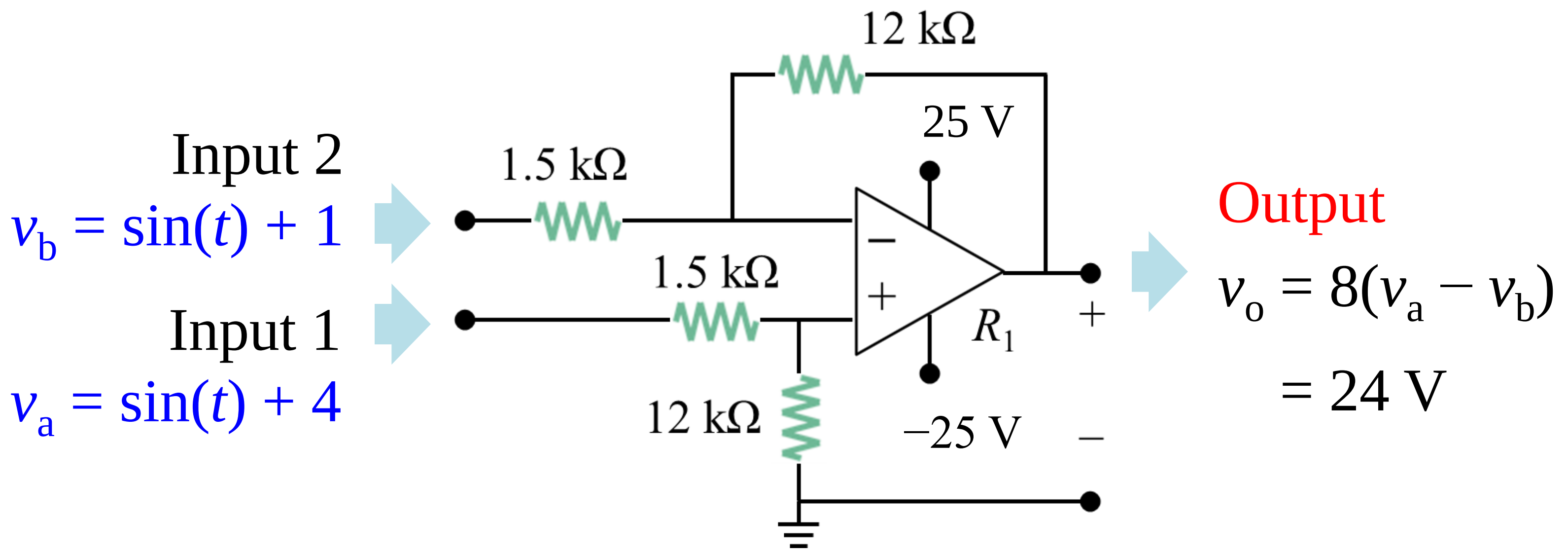
When $v_o = 9$ V, $R_f = 1.28$ kΩ

When $v_o = -9$ V, $R_f = -0.64$ kΩ (Is it possible?)

So when R_f reaches 1.28 kΩ the op-amp starts to work in saturation region.

- Common mode rejection in difference amplifiers

a) The following circuit has been analyzed in Example 5.11, which is difference amplifier with a closed loop gain of 8, $v_o = 8(v_a - v_b)$.
b) Assuming v_a and v_b are now time-dependent input of signals.



5.9 Common mode rejection (b)

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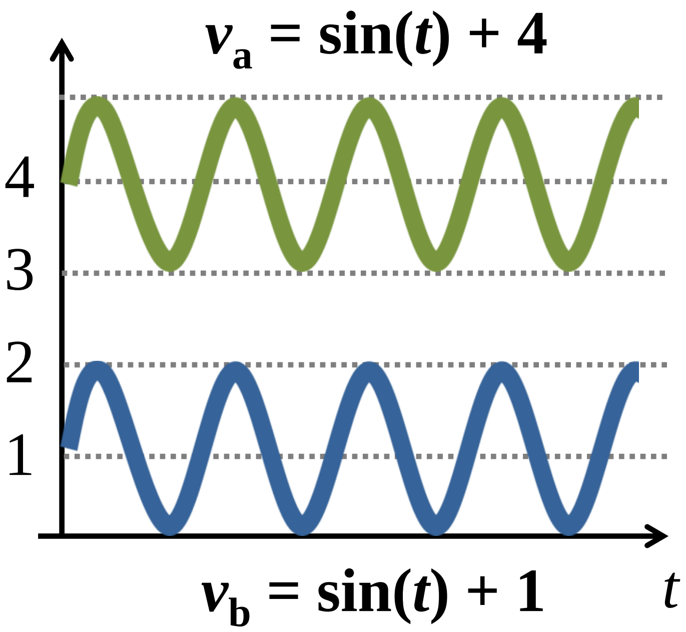
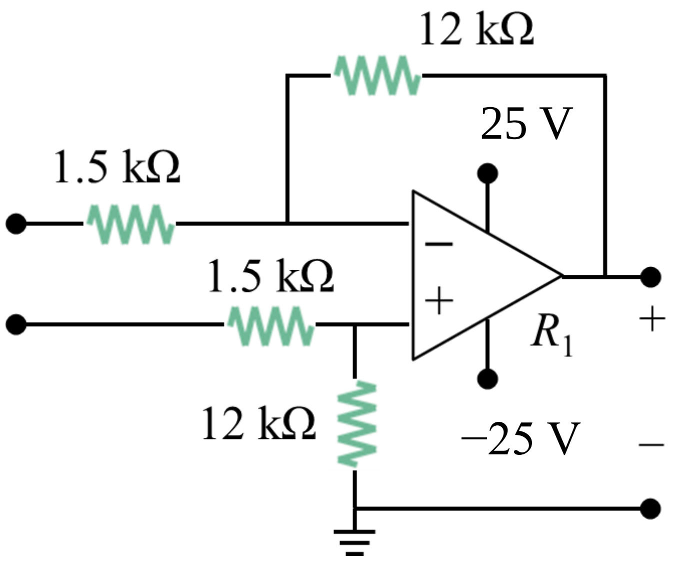
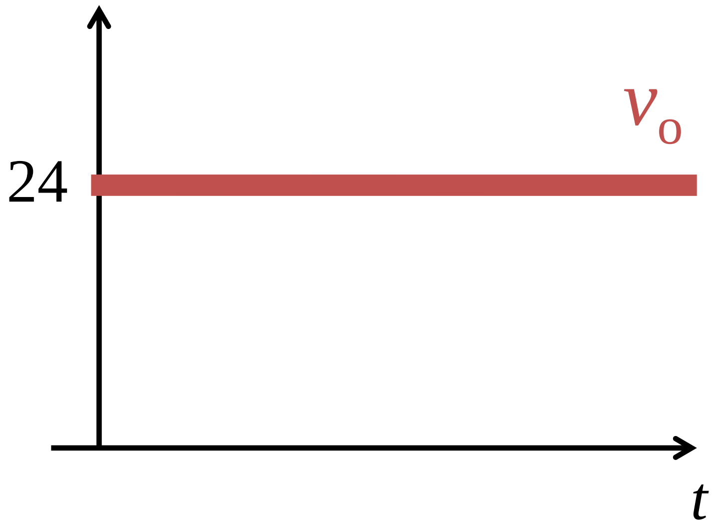
- Common mode rejection in difference amplifiers

a) The following circuit has been analyzed in Example 5.11, which is difference amplifier with a closed loop gain of 8, $v_o = 8(v_a - v_b)$.
b) Assuming v_a and v_b are now time-dependent input of signals.







5.9 Common mode rejection (c)

- Common mode rejection in difference amplifiers

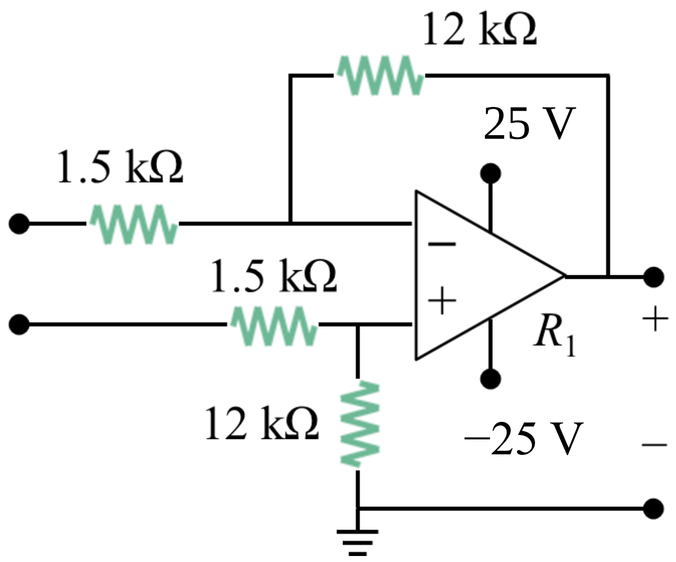
a) The following circuit has been analyzed in Example 5.11, which is difference amplifier with a closed loop gain of 8, $v_o = 8(v_a - v_b)$.
b) Assuming v_a and v_b are now time-dependent input of signals.

$$v_b = \sin(t) + 1 = [\sin(t) + 2.5] - 1.5$$

$$\mathbf{v_b = v_{cm} - v_{dm}/2}$$

$$v_a = \sin(t) + 4 = [\sin(t) + 2.5] + 1.5$$

$$\mathbf{v_a = v_{cm} + v_{dm}/2}$$

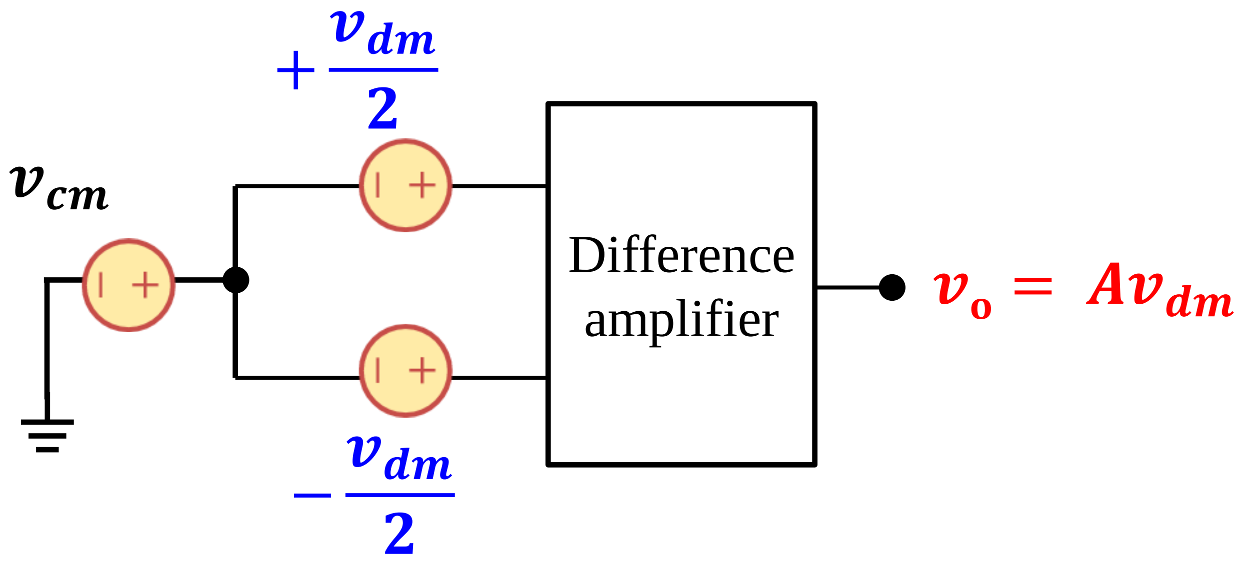


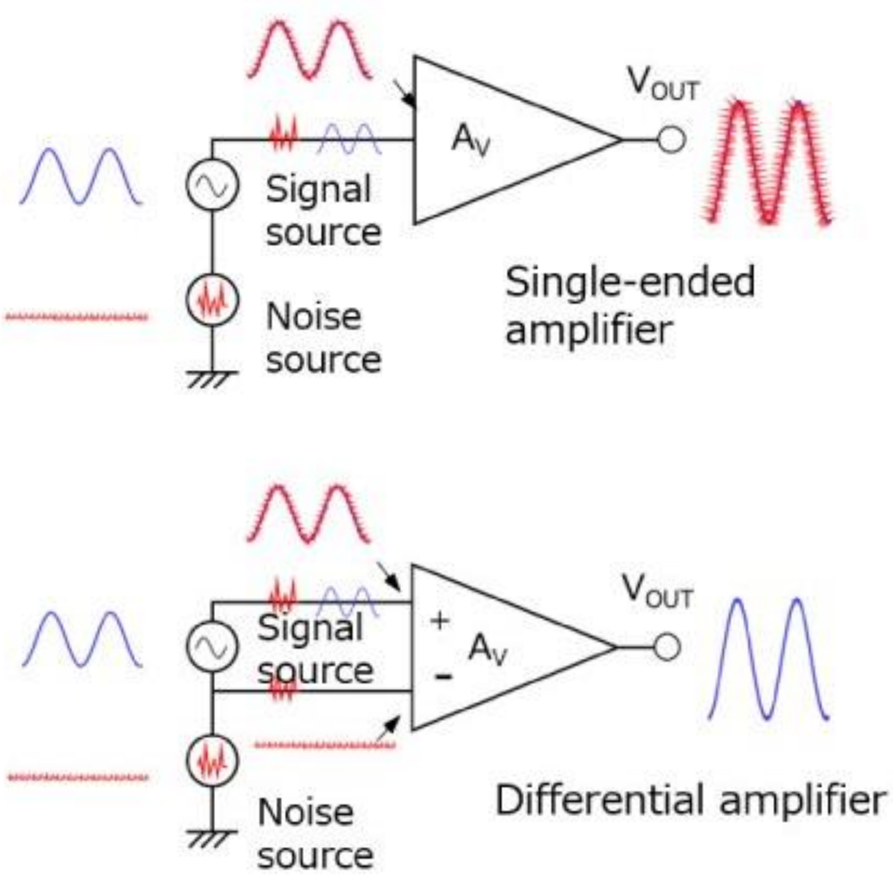
$$v_o = 8(v_a - v_b)$$
$$= 8[(v_{cm} + v_{dm}/2) - (v_{cm} - v_{dm}/2)]$$
$$= 8(v_{cm} - v_{cm}) + 8(v_{dm}/2 + v_{dm}/2)$$
$$\mathbf{v_o = 8v_{dm}}$$

5.9 Common mode rejection (d)

- Common mode rejection in difference amplifiers

An ideal difference amplifier rejects the common mode (v_{cm}) and amplifies only the difference mode (v_{dm}), which is of great value in suppressing noise.





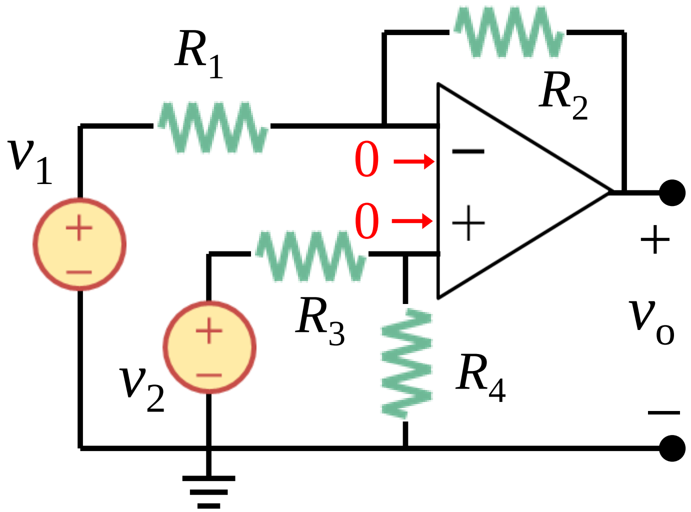
5.9 Common mode rejection (e)

- Common Mode Rejection Ratio (CMRR, 共模抑制比)

a) Interesting to know

b) Difference amplifiers in real life cannot fully reject the common mode (v_{cm}), therefore CMRR is used to evaluate how good a difference amplifier is.

c) The larger CMRR, the better the difference amplifier.



1. For ideal difference amplifiers:

$$v_o = A v_{dm}$$

2. For practial difference amplifiers:

$$v_o = A_{cm} v_{cm} + A_{dm} v_{dm}$$

$$CMRR = |A_{dm}/A_{cm}|$$

Lesson 5 Operational Amplifiers (Op Amp)

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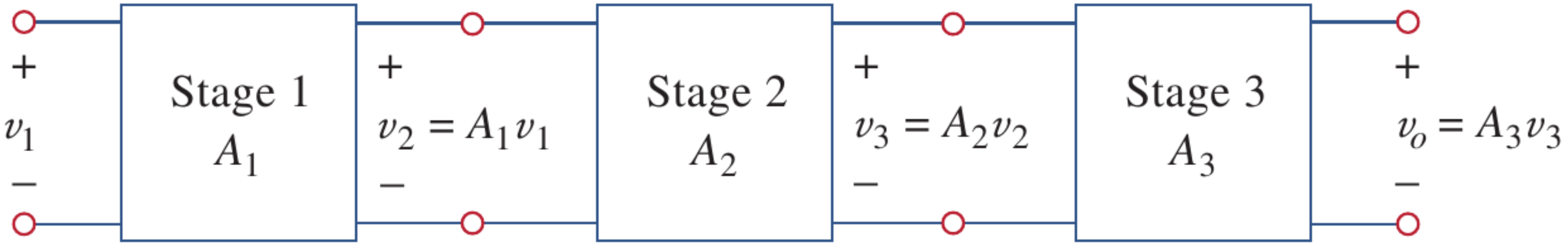
5.7-5.9	Summing and difference amplifiers	Slides 62-72
5.10	**Cascaded Op Amp circuits**	**Slides 74-82**
5.11	Summary and assignments	Slides 84-85

5.10 Cascaded Op-Amp Circuits (a)

- Cascaded Op-Amp Circuits (级联运放电路)



a) Op amps are modules or building blocks in the design of a more complex circuit. In practical applications, it is often necessary to connect op amp circuits in cascade (i.e., head to tail) to achieve higher gain or functionalities.

b) This head to tail configuration is called “cascading” (级联), and each amplifier is called a “stage” (级).

c) Due to the fact that ideal op amp has an infinite input resistance and zero output resistance, stages can be chained together without altering the input-output relationship of any one stage.

d) The overall gain of a cascade connection is the product of the gains of each individual op amps $A_{tot} = A_1 A_2 A_3$

5.10 Cascaded Op-Amp Circuits (b)

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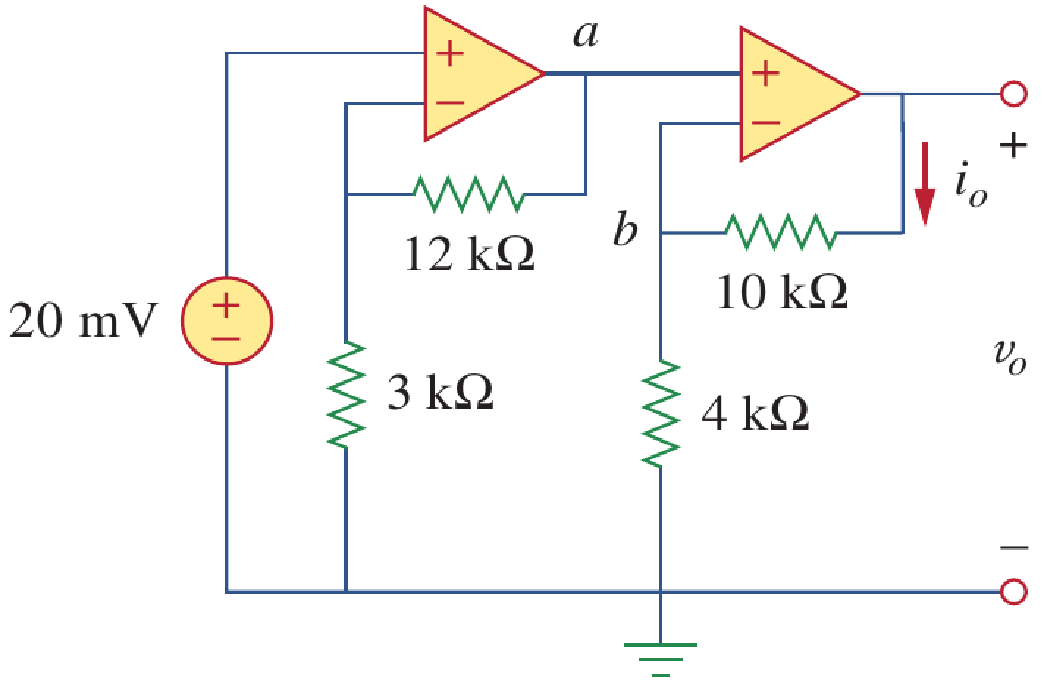
- Example 5.10

Example 5.10: Find the v_o and the i_o in the following circuit.



Solution: to find v_o

(1) Noninverting amplifiers, the closed-loop gain of one noninverting amplifier is $1 + R_f/R_i$

(2) For the 1st stage, the closed-loop gain:

$$G_1 = (1 + 12/3) = 5$$

(3) For the 2nd stage, the closed-loop gain:

$$G_2 = (1 + 10/4) = 3.5$$

(4) The total of gain: $G_{TOT} = 5 \times 3.5 = 17.5$

(5) $v_o = G_{TOT} v_i = 20 \times 17.5 = 350$ mV

5.10 Cascaded Op-Amp Circuits (c)

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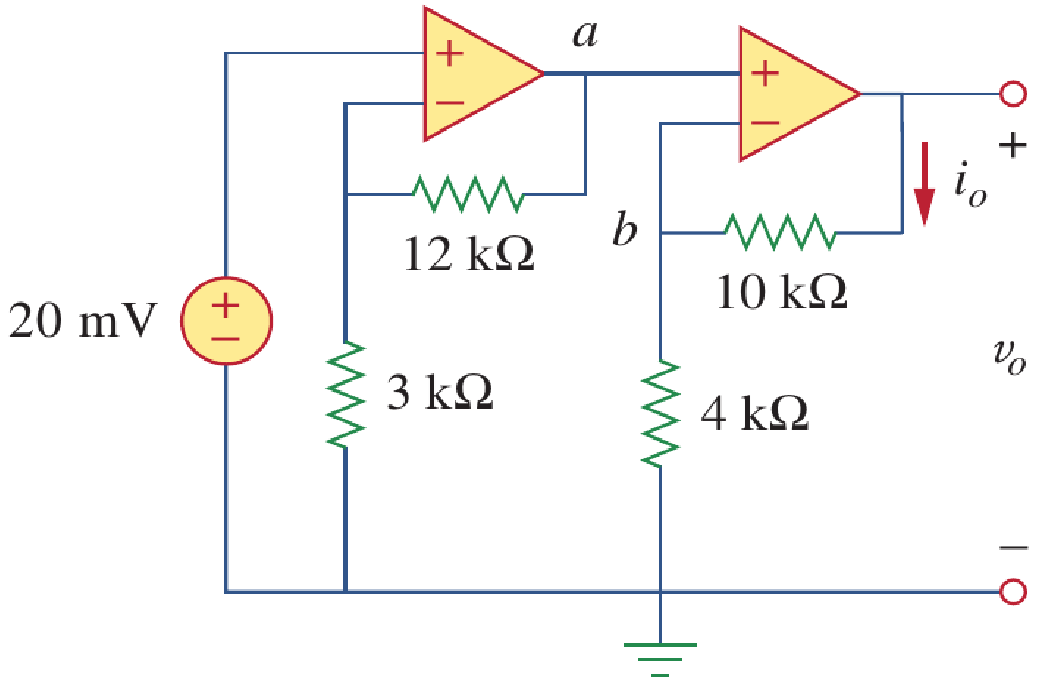
- Example 5.10

Example 5.10: Find the v_o and the i_o in the following circuit.



Solution: to find i_o

$i_o = (v_o - v_b) / 10\ \text{k}\Omega$

$v_b = v_a = G_1 v_i = 100\ \text{mV}$

So $i_o = (350 - 100)\ \text{mV} / 10\ \text{k}\Omega$

$= 25\ \mu\text{A}$

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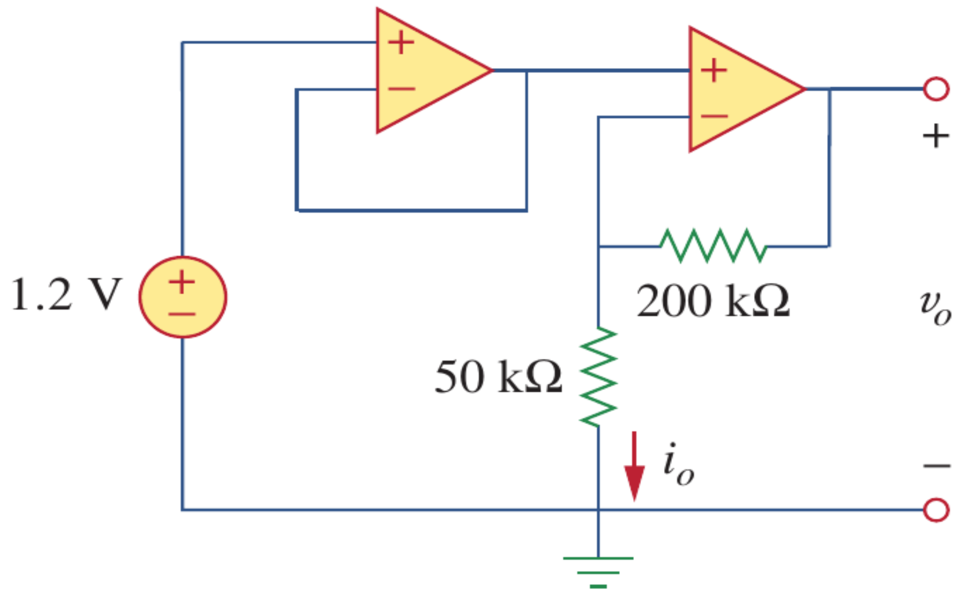
- Example 5.11

Example 5.14: Find the v_o and the i_o in the following circuit.



Solution:

Stage 1 is a voltage follower, $v_{o1} = v_i$,

$$G_1 = 1$$

Stage 2 is a noninverting op-amp, the gain

$$G_2 = 1 + 200k/50k = 5$$

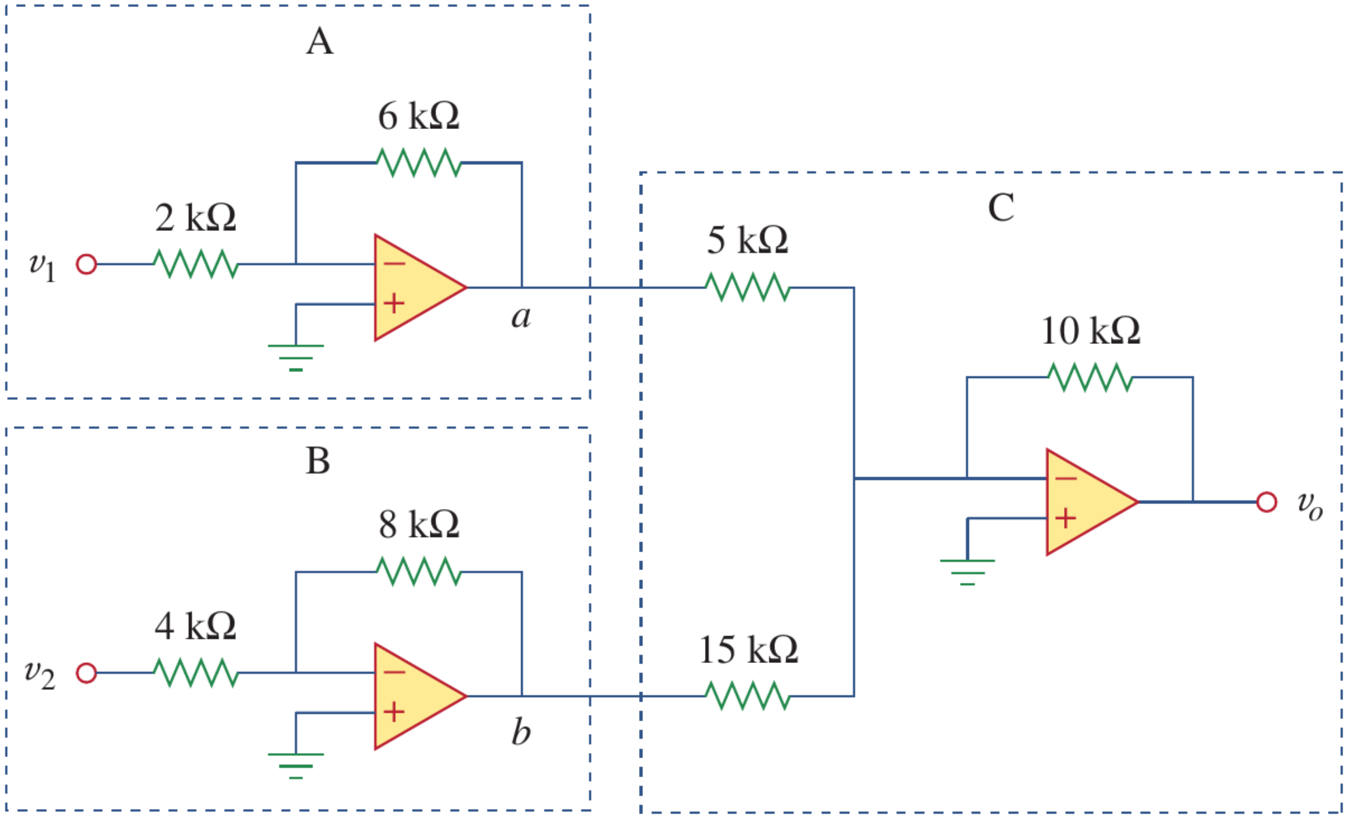
So the total gain, $G_{TOT} = G_1G_2 = 5$, $v_o = 5v_i = 6$ V

The $i_o = (v_o - v_{o1})/200k = (v_o - v_i)/200k = 24\ \mu A$

5.10 Cascaded Op-Amp Circuits (e)

- Example 5.12

Example 5.12: Find the v_o in the following circuit if v_1 = 1 V and v_2 = 2 V.



Solution:

Stage A, inverting amplifiers

$G_A = -3 \qquad v_{o1} = -3v_1 = -3$ V

Stage B, inverting amplifiers

$G_A = -2 \qquad v_{o2} = -2v_2 = -4$ V

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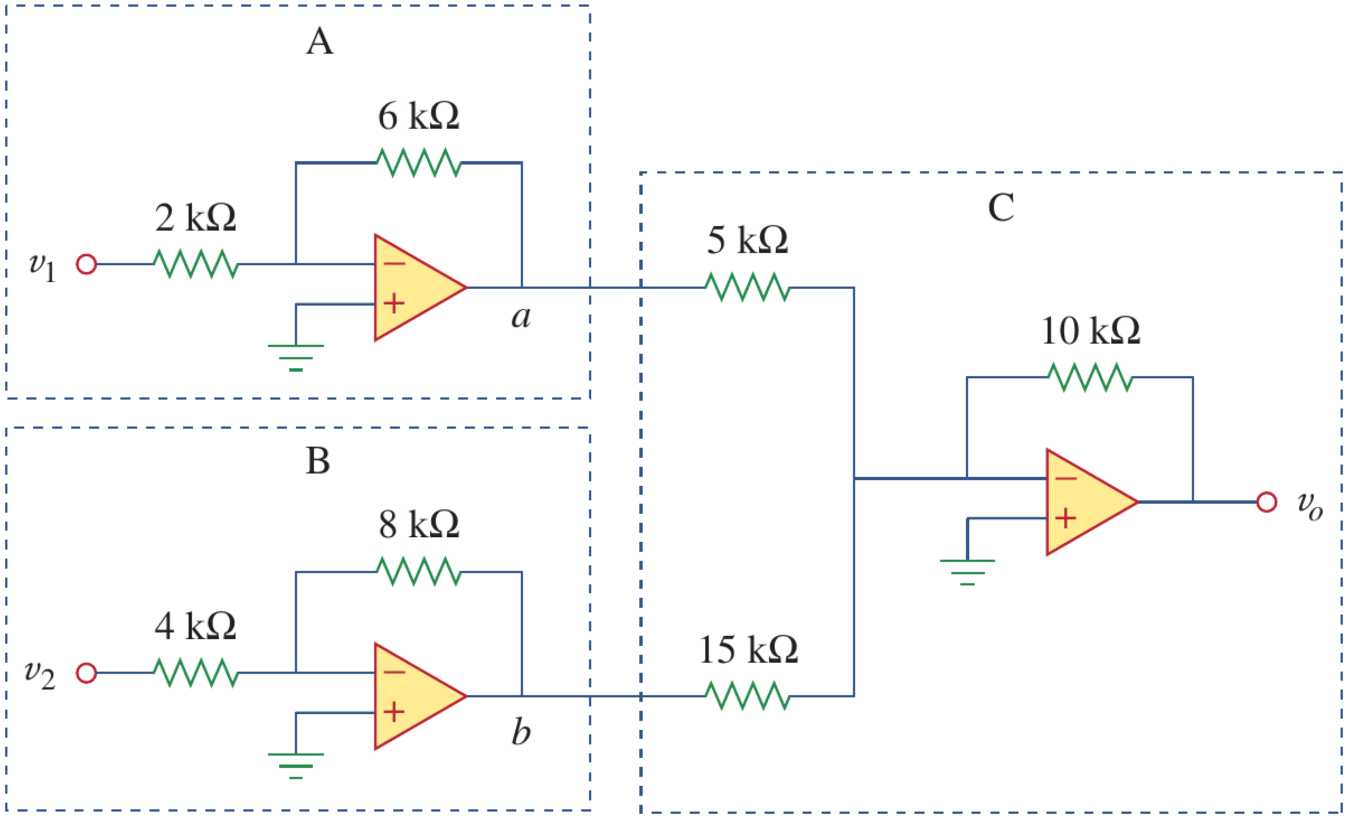
- Example 5.12

Example 5.12: Find the v_o in the following circuit if v_1 = 1 V and v_2 = 2 V.



Solution:

Stage C, summing amplifers

$$v_{oC} = -\frac{10\text{ k}}{5\text{ k}} v_{oA} - \frac{10\text{ k}}{15\text{ k}} v_{oB}$$

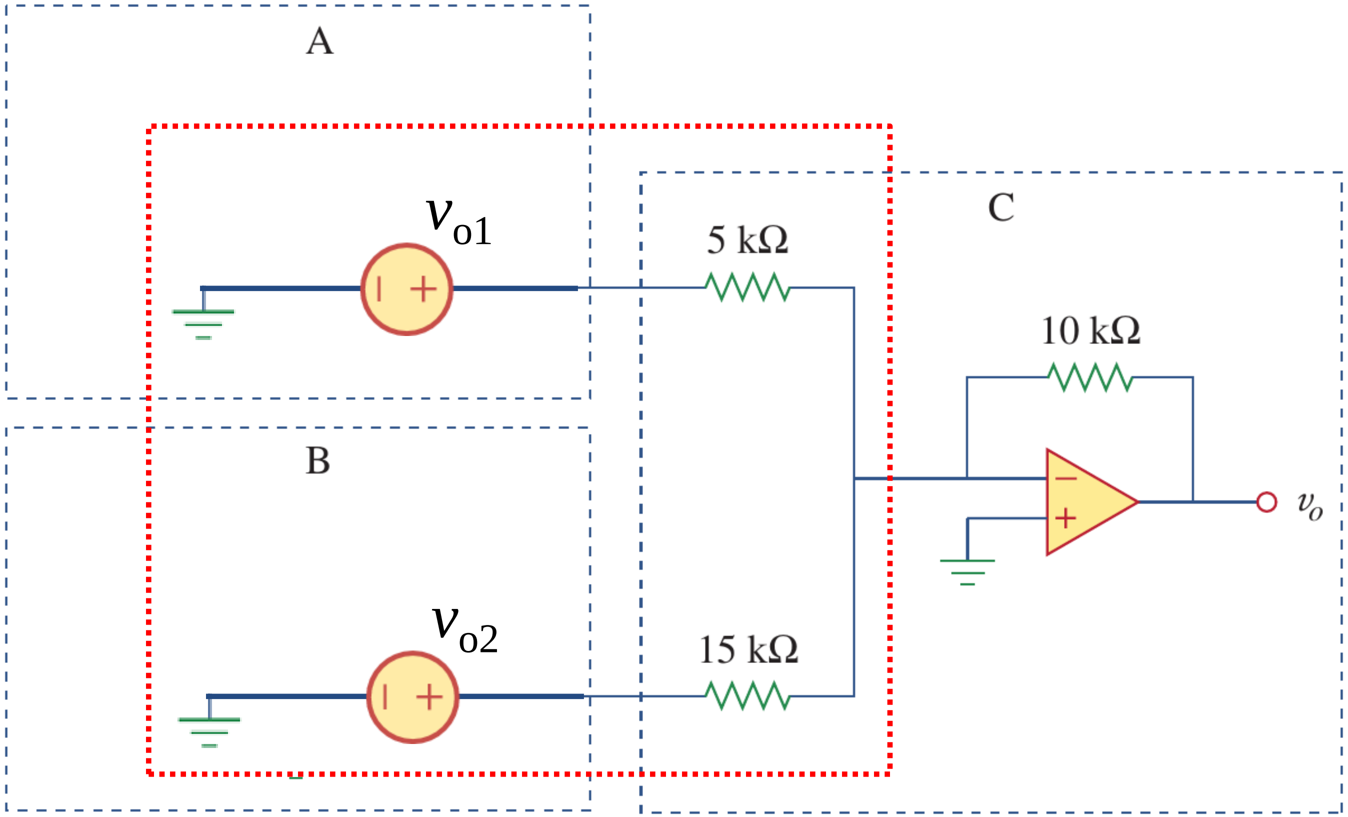
$$= 6 + 2.67$$

$$= 8.67$$

5.10 Cascaded Op-Amp Circuits (g)

- Example 5.12

Example 5.12: Find the v_o in the following circuit if v_1 = 1 V and v_2 = 2 V.



Or Solution:

Stage C, inverting op-amp

Effective input resistance:

$$R_{i3} = 5||15 = 30/8\ \text{k}\Omega$$

Effective input voltage

$$v_{i3} = \frac{\frac{1}{5\text{k}} v_{o1} + \frac{1}{15\text{k}} v_{o2}}{\frac{1}{5\text{k}} + \frac{1}{15\text{k}}} = -\frac{13}{4}\text{V}$$

$G_C = -10/30/8 = -8/3$, $v_o = G_C v_{i3} = 8.67$ V

5.10 Cascaded Op-Amp Circuits (h)

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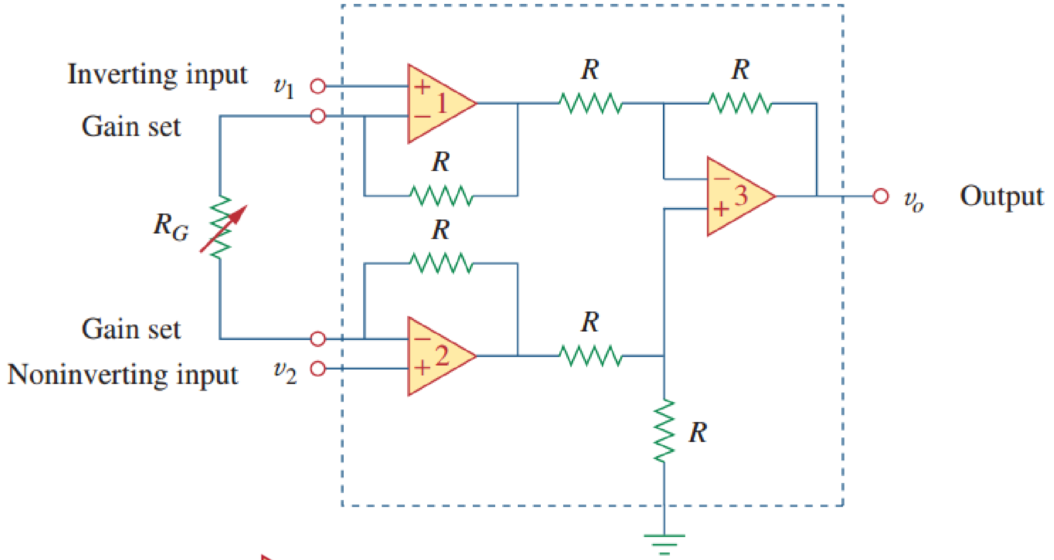
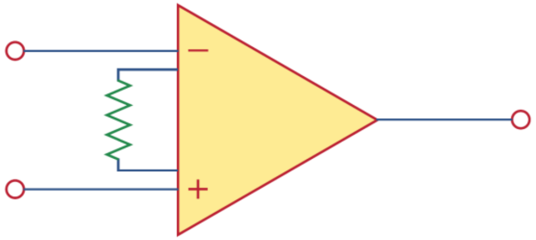
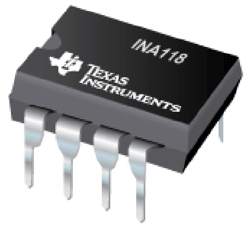
• Instrumentation Amplifier (IA)

- In typical IA (as shown in the right), $R_1 = R_2 = R_3 = R$, $R_4 = R_G$ ('*G*' for gain), hence $v_o=(1 + 2R/R_G)(v_2 - v_1)$. The gain $A_v=(1+2R/R_G)$ can be typically tuned from 1 to 1000 by varying R_G, without affecting the input impedance.
- Instrumentation amplifiers are so useful, they are often packaged as a single chip with the gain resistor being the only external component. They are very effective at extracting a weak differential signal out of a large common-mode signal (共模信号). In circuits exposed to external electrical noise, this is important in order to maintain a high signal-to-noise ratio (高信噪比).







5.10 Cascaded Op-Amp Circuits (i)

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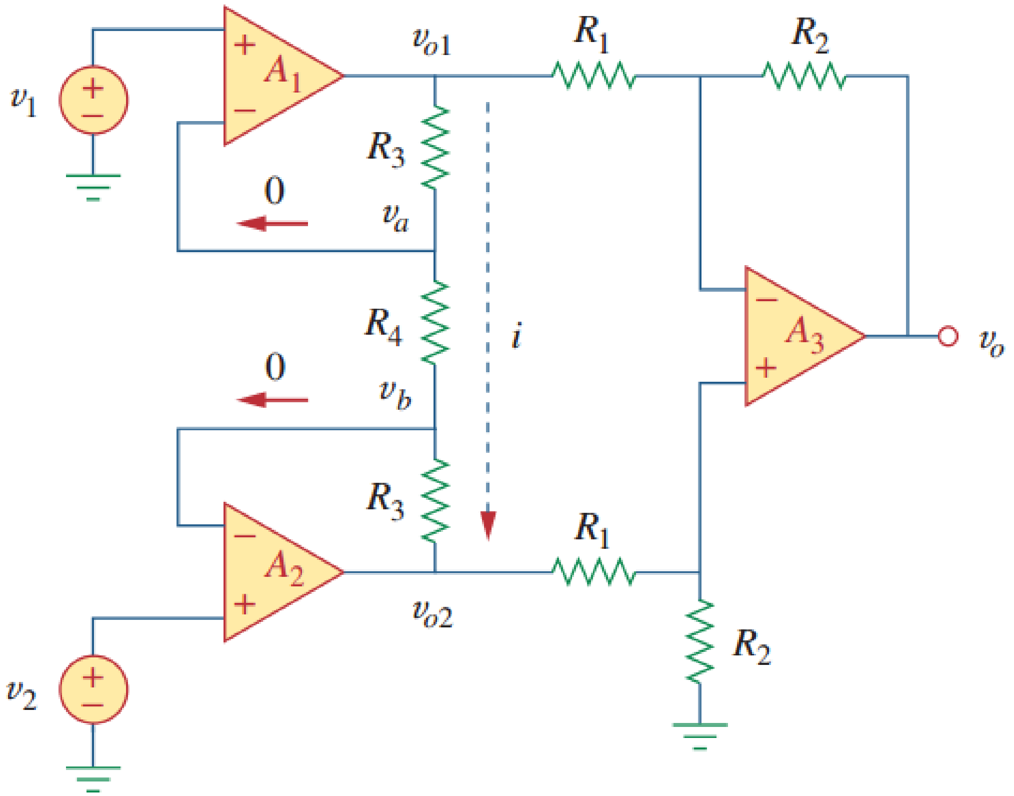
- Instrumentation Amplifier (IA)



Analysis:

$$\begin{cases} \text{For stage 3: } v_{o3} = \dfrac{R_2}{R_1}(v_{o2} - v_{o1}) \\ v_{o1} - v_{o2} = i \cdot (2 \cdot R_3 + R_4) \\ i = (v_a - v_b)/R_4 = (v_1 - v_2)/R_4 \end{cases}$$

$$\boxed{v_o = \frac{R_2}{R_1}\left(1 + \frac{2R_3}{R_4}\right)(v_2 - v_1)}$$

If $R_1 = R_2 = R_3 = R$:

$$v_o = (1 + \frac{2R}{R_4})(v_2 - v_1)$$

Lesson 5 Operational Amplifiers (Op Amp)

5.1	Introduction	Slides 11-19
5.2-5.4	Op amp basics and negative feedback	Slides 21-48
5.5-5.6	Inverting and noninverting amplifiers	Slides 50-60
5.7-5.9	Summing and difference amplifiers	Slides 62-72
5.10	Cascaded Op Amp circuits	Slides 74-82
5.11	**Summary and assignments**	**Slides 84-85**

5.11 Summary and assignments (a)

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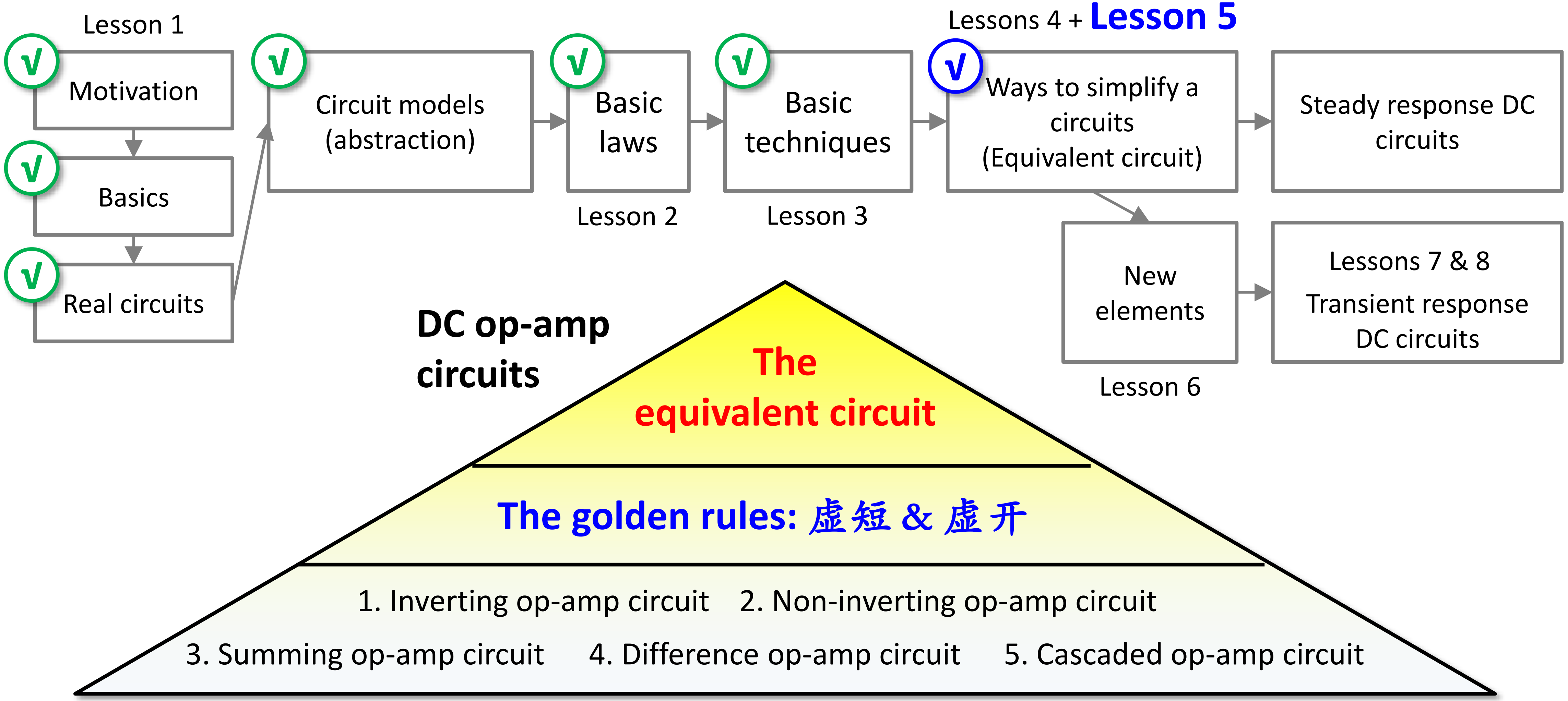
- EE104 roadmap (DC portion)



5.11 Summary and assignments (b)

- Assignments for lesson 5

Chapter 5:

Problems 5.12, 5.14, 5.33, 5.40, 5.45, 5.64

(a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.

(b) There are plenty of problems in the textbook, try to complete as many as you can.

Lenticular clouds

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